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Introduction to This Bilingual Edition

This bilingual edition presents the first part of Brahmagupta's *Brāhmasphuṭasiddhānta* (628 CE) in a parallel-column format. The left column contains the original Sanskrit text in Devanāgarī script, while the right column provides the English translation with IAST for technical terms.

About the Work

The *Brāhmasphuṭasiddhānta* ("Correctly Established Doctrine of Brahma") is one of the most important works in the history of Indian mathematics and astronomy. Composed by Brahmagupta in 628 CE at Bhīllamāla (modern Bhinmal, Rajasthan), it consists of 24 chapters covering astronomical computations, mathematics, and astronomical instruments.

Key Contributions in Chapter VIII

Chapter VIII (the Gaṇitādhyāya) contains Brahmagupta's most celebrated mathematical discoveries:

1. **Rules for zero:** systematic arithmetic with zero as a number
2. **Sign rules:** $(-)(-) = (+)$, $(+)(-) = (-)$
3. **Quadratic formula:** Complete solution of $ax^2 + bx = c$
4. **Brahmagupta's formula:** For the area of a cyclic quadrilateral
5. **Pell's equation:** The composition method (bhāvanā)

Chapter I

Mean Longitudes of Planets | [REDACTED]

2

[0]

Brahman, by whose grace I have again come to know the computation of planets;
now I proceed to reconcile it.

[0]

putation taught by Brahman had become inaccurate through the lapse of long
time; that has been correctly established by Brahmagupta.

[0]

re its *abda*; the days of a *yuga* are its *ahargaṇa*. A year consists of solar days; a
month of lunar *tithis*.

[0]

ists of *nakṣatra* days; a solar year is fourfold. The full moon at the fifteenth *tithi*
is complete, at the fourteenth incomplete.

[0]

gas are a day of Brahman; half is night. These are called a *kalpa*.

[0]

a *kalpa* arises Brahmā, in the middle Viṣṇu, at the end Rudra: these three lords
of the three worlds.

[0]

=*ivas* — the *manus* — are established; there dwell the people called *p*
=*urva*.

[0]

n a *yuga*; a thousand *caturyugas* are a day of Brahman.

[0]

āpara, and *kali* are the ages; a thousand *caturyugas* are a day of Brahman.

[0] 1

has four feet; in *trētā* three; in *dvāpara* two; in *kali* one.

[0] 1

ears is a *divya* year; by such divine years, a day of Brahman is known.

[0] 1

aturyugas of Brahmā, Viṣṇu, Indra, Rudra, and the *manus* are a day of Brahman.

[0] 1

[1] Verse 13: Humans, gods, *gandharvas*, *rākṣasas*, serpents, and ancestors — all proclaimed at the start of the *kalpa*.

[0] 1

months, seasons, years — these twenty-five *tattvas* to be known diligently.

[0] 1

s the planets are computed; I declare their numbers in order.

[0] 1

day's.
e, the Sun's motion; by lunar, the Moon's; by sidereal, the planets'; by civil, the

[0] 1

ures the world's transactions proceed, for planetary computation.

[0] 1

transactions by four measures: solar, lunar, sidereal, and civil, in order.

[0] 1

proceed.
mputation is accomplished by measures; without them, that knowledge does not

[0] 2

world's transactions proceed.
idereal, and civil measures are for planetary computation; by these four, the

[0] 2

ht and dark fortnight make one month; a year of those.

[0] 2

year; with thirteen, deficient; with five extra, abundant.

[0] 2

computation proceeds.
st and future motions, the wise compute; by these mean quantities, planetary

[0] 2

mean longitudes, then corrections must be applied for exactness.

[0] 2

nakṣatra, and planetary motions — all devised by measures in order.

[0] 2

eclared in order by those who contemplate planetary computation.

[0] 2

its months and days. A year of solar days; a month of lunar *tithis*.

[0] 2

ourteen and a half terrestrial years; mean motion reckoned beyond.

[0] 2

h fortnight; a month of those, a year of months.

[0] 3

year; thirteen, deficient; five extra, abundant.

[0] 3

otions stated, the wise compute by mean quantities.

[0] 3

corrections applied, for exact planetary computation by the wise.

Chapter II

True Longitudes of Planets |

2

[0]

e from the mean longitudes of planets should understand the corrections;

then the true is obtained.

[0]

nd

=s

=*ighra* apogees is the fifth of a circle (72°); there the *manda* motion, and

=s

=*ighra* beyond.

[0]

in that field, and

=s

=*ighra-phala* likewise; by their sum or difference, the true is known.

[0]

nda apogee causes the *manda-phala*; from the

=s

=*ighra* apogee, the

=s

=*ighra-phala*.

[0]

d or decreased by *manda-phala*; that result increased or decreased by

=s

=*ighra* corrections.

[0]

planet, by *manda* and

=s

=*ighra* process in order; thus planetary computation declared.

[0]

sometimes without *manda*; together with

=s

=*ighra*, the true is known.

[0]

nda and

=s

=*ighra* corrections; by sum or difference, the true arises.

[0]

manda result; at perigee, double; in between, triple.

[0] 1

xteen corrections; knowing the corrected planet, compute the true.

[0] 1

as, months, and days, planetary motion computed from means.

[0] 1

measure of planets, the wise compute; all becomes more exact.

[0] 1

ion declared for planets in order; by them all computation proceeds.

[0] 1

ala subtracted,

=s

=*ighra-phala* added; for Moon, by both processes.

[0] 1

ets, *manda* and

=s

=*ighra* equations always; by sum or difference, the true arises.

[0] 1

ard and westward motion by both processes, the true planet is always obtained by
calculators.

[0] 1

tion declared for Sun-born planets and all star-planets in order.

[0] 1

from mean of planets should understand corrections; it becomes more exact.

Chapter III

Three Problems: Time, Place, and Direction

2

[0]

one obtains the result. stions always in planetary computation: knowing time, place, and direction,

[0]

exact. ection — these three for a planet; joined with their knowledge, it becomes more

[0]

from place, declination; from direction, amplitude — these three give results.

[0]

andering (*kha-cara*); place is earth-wandering (*bh*
=*u-cara*); direction is direction-elephant (*dig-gaja*) — these three accomplish the planet.

[0]

onths, days, time is numerical; place is the difference of countries (*de*
=*sāntara*).

[0]

more exact. rection — these three accomplish the planet; joined with knowledge, it becomes

[0]

ence of countries joined with time, then planetary computation: the true arises.

[0]

ined with time, with place, and with direction.

[0]

by calculators. h three questions, the planet becomes more exact; therefore their knowledge always

[0] 1

edge, the planet becomes exact. tries, direction, and time — these three questions proclaimed; joined with knowl-

[0] 1

urteenth *tithi*, full moon remembered; the *tithi-antara* known with time.

[0] 1

tries joined with time; then planetary computation: the true produced.

[0] 1

direction — these three accomplish; joined with knowledge, exact.

[0] 1

e questions, the planet becomes exact; therefore their knowledge by calculators.

[0] 1

onths, days: time is numerical; place is difference of countries.

[0] 1

direction — these three accomplish; joined with knowledge, exact.

Chapter IV

Lunar Eclipses | [REDACTED]

2

[0]

clipse as it is; therefore all the characteristics of eclipse.

[0]

n and Moon — there the eclipse conjunction, in bright and dark fortnight.

[0]

n seizes the Moon, then eclipse conjunction; hence the full disk.

[0]

n's eclipse, then compute the time; by *manda* and

=s

=*ighra* process, make exact.

[0]

es full Moon on full-moon day, total eclipse; of Sun, solar eclipse.

[0]

declared by both processes; applying the true, then eclipse characteristics.

[0]

eclared in order; therefore all characteristics of solar eclipse.

Chapter V

Solar Eclipses |

2

[0]

clipse as it is; therefore all characteristics of solar eclipse.

[0]

un and Moon — there the solar eclipse; the conjunction auspicious or inauspi-
cious.

[0]

un there, Moon clear, then solar eclipse: total or partial.

[0]

lipse, compute time; by both processes, make exact.

[0]

teristics by both processes; applying the true, then characteristics.

[0]

eclared in order; therefore all results of eclipses.

Chapter VI

Rise and Set of Planets |

2

[0]

setting of planets in order; joined with knowledge, the planet becomes exact.

[0]

always after; joined with knowledge, the planet exact.

[0]

nt (*lagna*) joined with direction, then planet's rising; more exact known by wise.

[0]

first, setting after; joined with knowledge, exact.

[0]

n's rising, Moon etc. in order, their rising and setting I declare.

[0]

tting accomplished by measures; without them, knowledge does not proceed.

[0]

s of planets; the wise compute; thence more exact from computation.

Chapter VII

Constellations and the Moon's Cusps

2

[0]

on's horns and asterisms — joined with knowledge, the planet exact.

[0]

elevation on full-moon day, then full elevation of Moon's horns.

[0]

ms proclaimed by ancients — joined with knowledge, exact.

[0]

elevation and asterisms — joined with knowledge, exact.

Chapter VIII

Mathematics (Gaṇitādhyāya) – The Crown of the BSS

Note:

8.1 Arithmetic Operations (Vyakta-gaṇita)

2
[0]

fortunes is a fortune; of two debts is a fortune; of a fortune and debt is a debt.
The first as stated; the second by reversal of signs.
[0]

tion with a quantity, is that quantity increased; zero joined with like signs,
subtracted from unlike.
[0]

debt, fortune, and zero, the multiplicand is zero; in multiplication of zero,
the result is zero.
[0]

quantity is zero; zero divided likewise. Zero joined with like, subtracted from
unlike.
[0]

ivisor and the dividend a quantity: zero-divisor, zero quotient; the multiplier as
before.

2
[0]

unes is gain; of two like signs, like. Division: the result of like sign with like.
[0]

debt; fortune by debt gives debt; division is by reversal. Signs determined in
order.
[0]

ss are pure; their square always positive. The square root done two ways by
method.
[0]

ual quadrilateral field; twice that. From that, root extracted: the square root.

[0] 1

qual, with equal quadrilateral base. Its root: the cube root, made threefold.

[0] 1

nd cube; from second, third; from third, fourth — that remainder is the cube.

8.2 Fractions (Bhinna-gaṇita)

2

[0] 1

d: expressed and unexpressed; established separately, perform addition.

[0] 1

and class of divisions; having reduced fractions, form the fraction.

[0] 1

tiplied by divisor, dividend divided by denominator; the quotient: mark of

fraction-operation.

[0] 1

ivisor-form; the dividend: quantity; the quotient: result — thus fraction pro-

claimed.

[0] 1

ator, by denominator, denominator by numerators; having multiplied, divide

by fraction.

8.3 Rules for Zero (

=S

=unya-gaṇita)



2

[0] 1

greater quantity is that increased; zero multiplied is zero; zero divided likewise.

[0] 1

ike signs, subtracted from unlike. When zero is divisor and dividend a quantity.

[0] 1

ient; multiplier as before. This zero-computation declared, all by reason.

[0] 2

s void (*kha*), field, vajra, quantities; addition, subtraction, multiplication — zero there.

[0] 21

; zero-multiplier (*kha-guṇa*) remembered; zero-divisor

ero:

ht 1400 years before modern analysis.

lace of quantities; joined with the known (*vyakta*), form

k), n

nknowns; by them joined with knowns, operate with

like by like; square of known multiplied by unknown.

d with knowledge, the result arises in reality.

d from r

unknown — joined with knowledge, result arises.

tion; square of color; third is cube.

8.5 The Quadratic Equation (Madhyamāharaṇa)



2

[0] 3

equal the square of unknown and r

=*upas* according to rule, when both are of one r

=*upa* – equation.

[0] 3

e (the coefficient), from square of equal (coefficient), subtract square of unequal;

root made twofold, joined or diminished by r

=*upas*.

[0] 3

h two or three, divide by equal multiplied by two; obtained: “so much as”

(*yāvat-tāvat*), the r

=*upa*.

[0] 3

=*aharaṇa* (elimination of the middle) declared by those skilled in reasoning; one skilled in equation devoted to math.

[0] 3

joined with r

=*upas*; multiplied by two, make the square there.

[0] 3

=*upa* multiplied, both made equal; square root known – elimination of the middle.

[0] 3

ee etc., divide beginning with two or three; obtained: “so much as”, r

=*upa*.

[0] 3

l from equal, equal from unequal; having made equal – equation.

[0] 3

ltiplied, with r

=*upas*; multiplied by two, make square there.

[0] 4

=*upa* multiplied, both made equal; square root – elimination of the middle (*madhyam*
=*aharaṇa*).

Mathematical Commentary: In modern notation, Brahmagupta's formula for $ax^2 + bx = c$:

$$x = \frac{\sqrt{4ac + b^2} - b}{2a}$$

Equivalent to: $x = \frac{-b + \sqrt{b^2 + 4ac}}{2a}$

Brahmagupta was the first to give a general solution for *all* quadratics with two distinct positive roots.

8.6 Rule of Three (Trairā'sika) and Proportion

2

[0] 4

f one kind — Trairā

'sika (rule of three); their product known; result-quantity in order.

[0] 4

=ana) and result (*phala*); requisition (*icchā*) and result beyond; result of requisition by rule of three.

[0] 4

plied by argument, divided by result; obtained quotient: result by rule of three.

[0] 4

, inverse rule of three (*vyasta-trairā*

=a

'sika); operate by difference of argument and result.

[0] 4

, etc. — all from rule of three; inverse and direct, two-fold.

8.7 Arithmetic Progressions (Sreḍh'i)

=i)

2

[0] 4

ith increase, doubled, divide by number of terms; multiply by terms, or half

terms.

[0] 4

y number of terms: sum in equal difference; even: mean doubled; odd: equal

difference.

[0] 4

h last, doubled, divide by terms; obtained: mean, from series operation.

[0] 4

[1] **Verse 49:** Mean joined with number of terms: sum of series. Knowing difference of first and last, make multiplier.

[0] 5

ne, doubled, divide by terms; obtained: mean from series operation.

[0] 5

ries (

'sredh

=i-phala) declared by skilled reasoners; therefore geometric progression.

[0] 5

ression, first term and increase known; born of multiplier; form result.

[0] 5

ed by common ratio (*guṇottara*) repeatedly; divide by ratio minus one; that quotient.

8.8 Geometry – Plane Figures (Kṣetra-vyavahāra)



2

[0] 5

of half and sides. ar field area: product of half-base (*sam-dala*) and perpendicular (*koṭi*); square root

[0] 5

produced. quadrilateral field area remembered; from square of sum of opposite sides, root

[0] 5

from half-base and perpendicular; for irregular quadrilateral, make twofold (two triangles).

[0] 5

m of opposite sides — called area; by sum of roots of two, form area.

[0] 5

uare from half-diameter (vy

=s

=ardha); from fourth of diameter-square, area produced.

[0] 5

field area, compute diameter; area fourfold, its root: diameter.

[0] 6

multiplied by five, diameter by seven remembered. Thus circle multipliers form

result.

(Note: $\pi \approx 22/7 \approx 3.143$ or $\sqrt{10} \approx 3.162$)

[0] 6

d by half-chord (*jy*)

=*ardha*), squared, divided by four; remainder's root: chord (*j*)

=*iv*

=*a*).

[0] 6

btracted from square of diameter; root of remainder: arrow (

'*sara*, *sagitta*), half-diameter's chord-maker.

8.9 Brahmagupta's Formula for the Cyclic Quadrilateral

2

[0] 6

rea from half-base and perpendicular; from square of sum of opposite sides, root

produced.

[0] 6

f equilateral quadrilateral (*sama-caturbhuja*) area; square of half sum of opposite

sides: area.

[0] 6

lar quadrilateral area; beginning with two or three, form area.

[0] 6

om half-base and perpendicular; from half-diameter, form area.

[0] 6

area from two triangles; by sum of their roots, form area.

(*Brahmagupta's formula*: $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$)

[0] 6

(*sama-t*

=*t*

=*iya-bhuja*) from half-base and perpendicular; field twofold, form area.

[0] 6

rom sum of opposite sides; from pair of triangles, form area.

[0] 7

duct, doubled, joined with root: equilateral and isosceles (*dvi-sama*) quadrilateral

area.

[0] 7

lateral, doubled, joined with root; with half unequal root: root of area.

[0] 7

elds declared by skilled reasoners; therefore operation of excavations in order.

8.10 Excavations (Khāta-vyavahāra)

- 2
[0] 7
- ed field area, compute depth; oblong quadrilateral — form area.
[0] 7
- avation accomplished, compute capacity; increase known from excavation operation.
[0] 7
- known from excavation operation; therefore operation of shadows in order.

8.11 Shadows and Gnomons (Chāyā-vyavahāra)

- 2
[0] 7
- on and shadow motion, compute time; midday time (*madhy*
=*ahnak*
=*ala*) from shadow operation.
[0] 7
- '*saṅku*) of twelve angulas; its shadow proclaimed; joined with knowledge, time exact.
[0] 7
- and shadow made, thence direction computation; beginning triangulation (*trijñ*
=*ana*), declare in order.

Colophon

End of the Bilingual Edition of the Brāhmasphuṭasiddhānta, Part I, Chunk 1

Chapters I–VIII presented in parallel Sanskrit–English columns

Sanskrit text in Devanāgar
script | English translation with IAST

Source: Critical edition, Indian Institute of Astronomical and Sanskrit Research, Delhi, 1966

Edited by Acharyavara Ram Swarup Sharma et al.

Mathematical highlights of Chapter VIII:

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Preface to the Bilingual Edition

This bilingual edition presents the Sanskrit text of Brahmagupta's *Brāhmasphuṭasiddhānta* (Part I, Chunk 2, covering pages 141–280) in parallel with English translations. The layout follows a consistent format:

- **Left column:** The original Sanskrit verses in Devanagari script, set within a lightly shaded frame.
- **Right column:** The English translation and commentary.

The edition preserves all Sanskrit content from the source text, including verse references. Mathematical explanations, tables, and formulae are presented in full width where they span both languages.

Key topics covered in this chunk:

1. Algebraic operations with unknowns (jāti), zero, positive and negative numbers
2. Quadratic equations — the first general formula in Indian mathematics
3. The Kuṭṭaka (pulveriser) method for linear indeterminate equations
4. Planetary aphelions and epicycle corrections
5. Second-difference interpolation (the first in mathematical history)
6. Sine rule for plane triangles
7. Luni-solar astronomy and lunar inequality
8. Spherical astronomy — Indian and Greek methods
9. Operations with fractions (bhāga)
10. The Rule of Three (trairāśika) and compound proportion

1 Algebraic Operation

Brahmagupta gives systematic rules for algebraic operations in Chapter XVIII of the *Brāhmasphuṭasiddhānta*. This chapter contains the first treatment of algebra as a distinct mathematical discipline in Indian mathematics.

1.1 Addition and Subtraction of Unknown

Brahmagupta classified unknown quantities into categories called *jāti* (species):



Translation: “The numerical coefficients of the same unknown are added to or subtracted from each other; the same is done with the known quantities. When the unknowns are different, they are not united.”

This means that the numerical coefficients of x cannot be added to or subtracted from the numerical coefficients of y or x^2 or x^3 or xy and so on because these terms belong to different *jāti* or they do not belong to the category of the “like.”

1.2 Multiplication of Unknown

Again, regarding multiplication, Brahmagupta says:

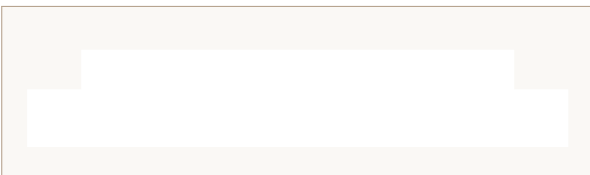


Translation: “The product of the two like unknowns is a square; the product of three or more like unknowns is a power of that designation. The multiplication of unknowns of unlike species is the same as the mutual product of symbols; it is called **bhāvita** (product or factum).”

1.3 Rules for Zero, Positive and Negative Number

Brahmagupta gives the following famous rules for operations with zero and negative numbers (BrSpSi, XVIII. 30–36):

Verse 18.30: Addition



Translation: “[The sum] of two positives is positive, of two negatives negative; of a positive and a negative [the sum] is their difference; if they are equal it is zero. The sum of a negative and zero is negative, [that] of a positive and zero [is] positive, [and that] of two zeros [is] zero.”

Verse 18.31: Subtraction

Translation: “If a smaller [positive] is to be subtracted from a larger positive, [the result] is positive; [if] a smaller negative from a larger negative, [the result] is negative; [if] a larger [negative or positive is to be subtracted] from a smaller [positive or negative, the algebraic sign of] their difference is reversed—i.e. negative [becomes] positive and positive [becomes] negative.”

Verse 18.32: Subtraction with zero

Translation: “A negative minus zero is negative, a positive [minus zero] positive; zero [minus zero] is zero. When a positive is to be subtracted from a negative or a negative from a positive, then it is to be added.”

Verse 18.33: Multiplication

Translation: “The product of a negative and a positive is negative, of two negatives positive, and of positives positive; the product of zero and a negative, of zero and a positive, or of two zeros [are all] zero.”

Brahmagupta’s Laws of Signs

$$\begin{aligned} (+a) \times (+b) &= +ab & (-a) \times (-b) &= +ab \\ (-a) \times (+b) &= -ab & (+a) \times (-b) &= -ab \\ 0 \times a &= a \times 0 = 0 \text{ for any quantity } a. \end{aligned}$$

Verse 18.34: Division

Translation: “A positive divided by a positive or a negative divided by a negative is positive; a zero divided by a zero is zero; a positive divided by a negative is negative; a negative divided by a positive is [also] negative.”

Verse 18.35: Division by zero

Translation (Colebrooke): “A negative or a positive divided by zero has that [zero] as its divisor, or zero divided by a negative or a positive [has that negative or positive as its divisor]. The square of a negative or of a positive is positive; [the square] of zero is zero. That of which [the square] is the square is [its] square-root.”

Note: Brahmagupta’s statement that $a/0$ gives “kha-hara” (a fraction with zero denominator) was his way of expressing the concept of infinity. In modern mathematics, division by zero is undefined, but Brahmagupta’s insight that such quantities represent infinitely large magnitudes was remarkable for the 7th century.

2 Quadratic Equation

The geometrical solution of a quadratic equation in this country would take us to the Vedic *Sulba* period. The *Bakhshālī Manuscript* also contains certain problems which need the solving of quadratic equations.

Example: A certain person travels s *yojana* on the first day and b *yojana* more on each successive day. Another who travels at the uniform rate of S *yojana* per day, has a start of t days. When will the first man overtake the second?

This leads to the quadratic:

$$s + (s + b) + (s + 2b) + \dots + [s + (n - 1)b] = S(n + t)$$

$$ns + \frac{n(n - 1)}{2}b = Sn + St$$

$$n^2 \cdot \frac{b}{2} + n \left(s - \frac{b}{2} - S \right) - St = 0$$

Sanskrit Verses: Brahmagupta's Rule for Quadratic Equations

Brahmagupta gives the following rule (BrSpSi, XVIII. 44–45):

Verse 18.44: The quadratic formula (first form)



Translation (Colebrooke): “Diminish by the middle [number] the square-root of the *rupas* multiplied by four times the square and increased by the square of the middle [number]; divide the remainder by twice the square. [The result is] the middle [number].”

Verse 18.45: The quadratic formula (second form)



Translation (Colebrooke): “Whatever is the square-root of the *rupas* multiplied by the square [and] increased by the square of half the unknown, diminish that by half the unknown [and] divide [the remainder] by its square. [The result is] the unknown.”

Modern Formulation: For the equation $ax^2 + bx = c$:

$$x = \frac{\sqrt{4ac + b^2} - b}{2a}$$

This is equivalent to the modern quadratic formula for the positive root.

3 Bhāskara I and the Kuṭṭaka Operatio

An indeterminate equation of the first degree of the type:

$$\frac{ax - c}{b} = y$$

(with x and y unknown) is known in Hindu mathematics by the name of “pulveriser” — *kuṭṭākāra*. In this equation, a is called the “dividend” (*bhājya*), b the “divisor” (*bhāgaḥāra*), c the interpolator (*kṣepa*), x the “multiplier” (*guṇakāra*), and y the “quotient” (*labdha*).

In the pulveriser contemplated in the above stanza:

a = revolution number of a planet

b = civil days in a *yuga*

c = residue of the revolutions of the planet (*śeṣa*)

Sanskrit Verse: Brahmagupta’s Rule for the Pulveriser



Translation: “Divide the dividend by the divisor, and the divisor by the remainder obtained; by this process of mutual division, [continue] until the remainder becomes zero. From this, by working backwards, one can find a particular solution.”

Verse 18.51: Condition for solution



Translation: “If the interpolator is divisible by the divisor of the residues, then that same method [applies]; if the interpolators are unequal, know that the pulveriser has a flaw [is impossible], friend.”

Verse 18.52: Working backwards



Translation: “The successive quotients obtained [in the mutual division] are called ‘kotis’ [vertical lines]; placing the *gunakaras* [multipliers] at the kotis, the multiplier is [found] combined with the interpolator.”

3.1 The Pulveriser Method

The method of the pulveriser (*kuṭṭaka*) involves the following steps:

1. Divide a by b and obtain the quotient and remainder.
2. Divide b by this remainder and obtain a new quotient and remainder.
3. Continue this process of mutual division until the remainder becomes zero.

4. Write down the quotients obtained in a column.
5. The number of quotients is made odd by adding the residue if necessary.
6. From the bottom, multiply the last number by the interpolator and add the number above it.
7. Continue this process until the top is reached.
8. The number at the top gives the value of x and the number at the bottom gives y .

3.2 General Form

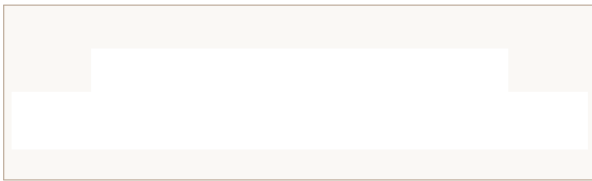
The general form of the pulveriser is to find integer solutions to:

$$ax + c = by$$

where a , b , and c are given integers.

Brahmagupta's method was later improved by Āryabhaṭa II and Bhāskara II, but the fundamental approach was established in the *Brāhmasphuṭasiddhānta*.

Example: Pulveriser for planetary residues



Translation: “The residue of the Sun [is] twelve; divide it together with the civil days; whatever multiplier results therefrom, that will yield the elapsed degrees of the Sun’s motion.”

Example Problem: Find x such that $\frac{137x - 10}{60} = y$ is an integer.

Here dividend $a = 137$, divisor $b = 60$, interpolator $c = 10$.

By mutual division: $137 = 2 \times 60 + 17$; $60 = 3 \times 17 + 9$; $17 = 1 \times 9 + 8$; $9 = 1 \times 8 + 1$.

Working backwards: $x = 50$, $y = 114$ is a solution.

4 Planetary Aphelions and Epicycle Correction

Brahmagupta gave systematic corrections to the mean longitudes of planets to obtain their true positions. These corrections involve the concepts of the aphelion (*manda-cheda*) and the epicycles of apsis (*manda-paridhi*) and conjunction (*śīghra-paridhi*).

Verse: Mean to True Position



Translation: “Having taken the mean planet, [apply] the correction for the apsis; from that [result, apply] the correction for the conjunction; then the planet becomes true [corrected].”

Verse: Epicycle Dimensions



Translation: “The epicycle of the apsis for the Sun is forty-seven plus two [i.e., 49 for the manda]; that of the Moon is thirty-three. The epicycle of the conjunction [for the Moon] is seventy.”

Verse: Aphelion (Mandocca) Positions



Translation: “The aphelion [ucca] of the Sun is seventy-nine [degrees] in the northern solstice; the [higher] apsis of the Moon and the śīghra [conjunction point] are obtained by [appropriate] correction.”

Historical Note: Brahmagupta’s corrections to the planetary epicycles were based on careful comparison with observed positions. He adjusted the epicycle radii given in the *Āryabhaṭīya* to reduce the errors in predicted planetary longitudes. His method of applying the manda correction first, followed by the śīghra correction, became the standard procedure in Indian astronomy.

5 Uttara Khaṇḍakhādyaka: Advanced Planetary Theor

The *Uttara Khaṇḍakhādyaka* (UKK) represents Brahmagupta’s advanced astronomical treatise, containing corrections and refinements to the earlier *Khaṇḍakhādyaka* proper.

5.1 Brahmagupta First to Use Second Difference

All these illustrations reproduced here very well establish the point that the great Indian astronomers from *Āryabhaṭa* I to Brahmagupta were aware of the methods of separating the two distinct planetary inequalities, viz., that of the apsis and of conjunction in the cases of the five ‘star’ planets.

In the *Khaṇḍakhādyaka*, Brahmagupta having given the “sines” and the equations of the Sun and the Moon at the interval of 15° of arc of the mean anomaly, in the *Uttara Khaṇḍakhādyaka* teaches, for the **first time in the history of mathematics**, the improved rules for interpolation by using the **second difference**.

This very important feature is reproduced here from the translation by Sengupta of the verse (UKK. 8):



Translation (Sengupta): “Multiply the residual arc left after division by 900’ (i.e. by 15°), by half the difference of the tabular difference passed over and that to be passed over and divide by 900’ (i.e. 15°): by the result increase or decrease, as the case may be, half the sum of the same two tabular differences; the result which, whether less or greater than the tabular difference to be passed, is the true tabular difference to be passed over.”

The rule given here applies to the case of all functions hitherto considered in the *Khaṇḍakhādyaka*, which are tabulated at the difference of 15° of arc of the argument. They are:

- (i) the tabular differences of the Sun’s equation,
- (ii) the tabular differences of the Moon’s equation,
- (iii) the tabular differences of the ‘sines’.

Illustration — To find the ‘sine’ of 57° .

Brahmagupta’s table of sines in the *Khaṇḍakhādyaka* is as follows:

“Thirty increased severally by nine, six and one; twenty-four, fifteen and five, are the tabular differences of sines at intervals of half-a-sign. For any arc, the ‘sine’ is the sum of the parts passed over, increased by the proportional part of the tabular difference to be passed over.” (KK.I.30; also III.6)

Table 1: Brahmagupta's Table of Sines

Arc	'Sine'	Tabular Difference	Second Difference
0°	0	—	—
15°	39	39	—
30°	75	36	–3
45°	106	31	–5
60°	130	24	–7
75°	145	15	–9
90°	150	5	–10

Verse on Lunar Correction (Evection)



Translation: “The second inequality of the Moon [evection], when combined with the [equation of the] conjunction, then its motion-equation occurs; the conjunction-equation should be applied by that [amount].”

This verse refers to what later became known as Śrīpati's equation — the recognition of the Moon's second inequality (evection), which represents a significant achievement in Indian lunar theory.

Now $57^\circ = 3420 \text{ minutes} = 900' \times 3 + 720'$. Thus three of the tabular differences are considered as passed over; the last one being 31 and the one to be passed over is 24.

The true tabular difference by the rule, for arc 57° :

$$\frac{31 + 24}{2} + \frac{720}{900} \times \frac{31 - 24}{2}$$

Hence the 'sine' of 57° :

$$= 106 + \frac{720}{900} \times 24 + \frac{720}{900} \left(\frac{720}{900} - 1 \right) \times \frac{31 - 24}{2} = 125.76$$

As worked out from the logarithm tables the same comes out to be 125.80.

This in fact is the modern form of the interpolation equation up to the term containing the second difference:

$$f(a + nh) = f(a) + n\Delta f(a) + \frac{n(n-1)}{2} \Delta^2 f(a)$$

Sanskrit Verse: Construction of the Sine Table



Translation: “Thirty minus one, increased by nine, six, and one; twenty-four, fifteen, and five — these are the tabular differences of sines. The sines [themselves] are obtained by successive addition of these differences, [starting from] the first [sine]. These twenty-four sines [span] the quadrant.”

Verse on the Epicycle Correction



Translation: “To correct [the mean planet] by the equation of the apsis, one should apply [the correction] by means of the epicycle of the apsis; likewise, to make [the planet] true by the equation of the conjunction, one should do so by means of the epicycle of the conjunction.”

Brahmagupta thus takes a decidedly improved step here and is undoubtedly the **first man in the history of mathematics** who has done this.

6 Operations with Fraction

Brahmagupta classified fractions into several categories: *bhāga* (simple fraction), *prabhāga* (compound fraction), *bhāgānubandha* (fractions in association), and *bhāgāpavāha* (fractions in dissociation). He gave systematic rules for all operations with fractions.

Verse: Addition and Subtraction of Fractions



Translation: “When fractions are connected [for addition or subtraction], the two denominators [become] multipliers [of the opposite numerators]; the sum of the [cross-multiplied] dividends, divided by the product of the denominators, becomes the combined [result].”

In modern notation:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \quad \text{and} \quad \frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}$$

Verse: Multiplication and Division of Fractions



Translation: “In the multiplication of fractions, the numerators and denominators are multiplied together [respectively]. In division, the numerator is multiplied by the reciprocal of the divisor-fraction [i.e., the numerator and denominator are interchanged].”

Brahmagupta’s Fraction Operations

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d} \quad \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

7 The Rule of Three (Trairāśika)

The Rule of Three (*trairāśika*) is the fundamental method of proportion in Indian mathematics. Brahmagupta gave a precise formulation that served as the basis for all problems involving proportion.

Verse: The Rule of Three



Translation: “If the *pramāṇa*-fruit [result for the standard measure] is given, that [is] multiplied by the *icchā* [requisition]; divide that by the *pramāṇa* [standard measure]; the result obtained here is the *icchā-phala* [required result].”

In the *trairāśika*, three quantities are given:

- **Pramāṇa** (*p*): the standard measure or argument
- **Phala** (*f*): the fruit or result corresponding to the standard
- **Icchā** (*i*): the requisition or desired measure

The required *icchā-phala* (*x*) is:

$$x = \frac{f \times i}{p}$$

Verse: Inverse Rule of Three



Translation: “When, by the inverse Rule of Three, the result becomes reversed, then by the nature of the divisor — the fruit then becomes the *icchā-phala* [by inverse proportion].”

Direct Rule of Three: $x = \frac{f \times i}{p}$ Rule of Three: $x = \frac{f \times p}{i}$	Inverse
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Verse: Compound Proportion (Pañcarāśika etc.)



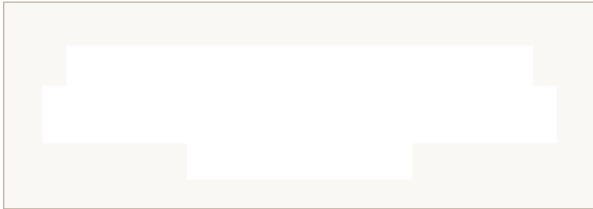
Translation: “The result having been obtained by the Rule of Three, [it is further operated on] by [another] Rule of Three; as many given quantities [as there are], so many Rules of Three are remembered [to be applied].”

This verse establishes the method of *compound proportion*: when more than three quantities are involved, successive applications of the Rule of Three yield the final result. For five quantities it is called *pañcarāśika*, for seven *saptarāśika*, and so on.

8 The Sine Rule in Indian Trigonometr

Brahmagupta was the first Indian mathematician to state the sine rule for plane triangles explicitly. He used it in the context of computing the true positions of planets and in spherical astronomy.

Verse: The Sine Rule



Translation: “In an equilateral triangle, the sines of equal sides [divided] by one another [give unity]; in a scalene triangle, the sides divided by the sines give a result [which is constant] — that is the [circumdiameter].”

In modern form, for any triangle ABC with circumdiameter D :

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = D$$

Verse: Shadow and Sine Relations (Spherical Astronomy)



Translation: “The Rsine of the declination, divided by the Rsine of half the day, becomes the *lambaka* (vertical). The Rsine of the latitude, multiplied by the *lambaka*, [gives] the *drgjyā* (ascensional difference); it is determined [according to] the direction [north or south of the equator].”

This verse relates to spherical astronomy: the computation of the *drgjyā* (ascensional difference), which measures how much a celestial body rises earlier or later due to the observer’s latitude. The *kṛānti-kṣetra* and *akṣa-kṣetra* diagrams use similar right triangles.

9 Further Algebraic Operation

Brahmagupta’s Chapter XVIII also contains rules for squares, square roots, cubes, and cube roots, as well as additional rules for operations with zero.

Verse: Square and Square Root



Translation: “A square is an equilateral quadrilateral; its root [is] that which, when doubled, [gives the side]. That of which the square is the square is [its] square-root [i.e., the square root of a^2 is a].”

Verse: Cube and Cube Root



Translation: “A cube is an equi-dimensional equilateral [solid]; its root [is] that which, when tripled in dimension, [gives the side]. That of which the cube is the cube [is its] cube-root [i.e., the cube root of a^3 is a].”

Verse: Zero as a Number



Translation: “Zero, void, ‘kha’, ‘path’ — the learned know this as a number. When [a quantity] is combined with or diminished by it, there is neither increase nor decrease [i.e., $a + 0 = a - 0 = a$].”

Note: This verse (18.36) is historically significant as one of the earliest explicit statements that zero is a number in its own right. Brahmagupta’s treatment of zero as a mathematical entity with well-defined properties was a foundational contribution to the development of algebra.

10 Summary of Brahmagupta's Mathematical Contributions

The *Brāhmasphuṭasiddhānta* and the *Khaṇḍakhādya* together represent the most comprehensive treatment of mathematics and astronomy in the 7th century. The contributions covered in this section (Pages 141–280) include:

1. **Second-order interpolation:** Brahmagupta was the first mathematician in history to use the second difference in interpolation, as demonstrated in the *Uttara Khaṇḍakhādya* (UKK. 8).
2. **Sine rule:** He was the first in India to give the sine rule for plane triangles, in the context of computing planetary positions.
3. **Planetary corrections:** Systematic corrections to the epicycles of apsis for the Sun, Moon, and planets, bringing calculated positions into agreement with observation.
4. **Lunar inequalities:** Recognition and treatment of the second inequality of the Moon (evection), known as Śrīpati's equation.
5. **Spherical astronomy:** Development of Indian methods using properties of similar right-angled triangles (*krānti-kṣetra* and *akṣa-kṣetra*).
6. **System of numerals:** Contribution to the place-value notation and its transmission to the Arab world, and subsequently to Europe.
7. **Fractions:** Classification of fractional combinations into *bhāga*, *prabhāga*, *bhāgānubandha*, and *bhāgāpavāha*.
8. **Algebra:** Systematic rules for operations with zero, positive and negative numbers; the first to treat zero as a number in its own right.
9. **Quadratic equations:** General formula for solving quadratic equations (verses 18.44–45).
10. **Indeterminate equations:** The *kuṭṭaka* (pulveriser) method for solving linear indeterminate equations, fundamental to Indian astronomy for calculating planetary positions.

Editor's Note

This bilingual edition presents the Sanskrit text and English translation on Brahmagupta's *Brāhmasphuṭasiddhānta* as found in the scholarly edition covering pages 141–280 of the original text. The text has been extracted from the critical edition and set in Devanagari using the font. Key mathematical verses include:

- **Chapter XVIII, verses 30–36:** Rules for operations with zero, positive and negative numbers, including the famous “*kha-hara*” (division by zero) verse; also explicit statement that zero is a number (v. 36)
- **Chapter XVIII, verses 44–47:** The quadratic formula (two equivalent forms), square and square-root, cube and cube-root
- **Chapter XVIII, verses 50–53:** The pulveriser (*kuṭṭaka*) method for indeterminate equations, including conditions for solution and worked examples
- **Pāṭiṅaṇita, verses 2.3–2.4:** Rules for addition, subtraction, multiplication, and division of fractions

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(Pulverizer) method for solving $ax \pm c = by$.

d the divisor by the dividend alternately; the quo-
ow, one above." — This describes the Euclidean

he one above it, add the quotient below; place the
s before." — The iterative backward-substitution

ed by the divisor; the lower, called the 'quotient,'
tively the āhargana and the revolutions (or what
is sought)." — Final step: $x = \text{ahargana}$, $y = \text{revolutions}$.

Modern Mathematical Formulation

The *kuṭṭaka* method solves the linear Diophantine equation:

$$ax \pm c = by$$

where a = civil days in *yuga*, b = revolution-number of the planet, c = residue, and x, y are positive integers to be found.

Step 1 — Reduction: Compute $g = \gcd(a, b)$. Divide through by g :

$$a' = \frac{a}{g}, \quad b' = \frac{b}{g}, \quad c' = \frac{c}{g}$$

Step 2 — Euclidean Algorithm: Apply the mutual division to obtain the continued-fraction quotients $q_1, q_2, \dots, q_n$.

Step 3 — Back-substitution: Starting with a chosen *mati* m such that $r_n \cdot m - c' \equiv 0 \pmod{b_n}$, iteratively compute:

$$V_i = q_i \cdot V_{i+1} + V_{i+2}$$

Step 4 — Solution: The two final numbers give $x \equiv V_{n-1} \pmod{a'}$ and $y \equiv V_n \pmod{b'}$.

Bilingual: Varga-Prakṛti (Square-Nature)

“In the Square-Nature, when two solutions are found with the same multiplier (N) and interpolators k_1, k_2 : their interpolators being multiplied, the multiplier being doubled and combined with the two roots — the sum of their squares is (the new equation with) unity as interpolator.”

Brahmagupta’s Lemma (Composition): If (x_1, y_1) solves $Nx^2 + k_1 = y^2$ and (x_2, y_2) solves $Nx^2 + k_2 = y^2$, then:

$$x = x_1y_2 \pm x_2y_1, \quad y = y_1y_2 \pm Nx_1x_2$$

solves $Nx^2 + k_1k_2 = y^2$.

“In the Square-Nature, if the interpolator is a square, being capable of being a square — dividing the roots by the square root of that interpolator, new roots are obtained.”

Second Lemma: If $Nx^2 \pm p^2d = y^2$, set $u = x/p$, $v = y/p$, giving:

$$Nu^2 \pm d = v^2$$

The original roots are p times those of the derived equation.

Cakravāla (Cyclic Method) — Bilingual

“The teacher (ācārya), first taking the square root of the multiplier nearest to it, divided by the interpolator (and) increased or diminished by unity — that root divided by the multiplier and the interpolator multiplied by it, combined with the previous pair of roots — that is the new interpolator; the product of the two divided by the multiplier, the sum of the squares of the previous roots — (these are the new roots).”

The Cyclic Method: Starting from an auxiliary $Na^2 + k = b^2$:

1. Choose P so $P^2 \approx N$ and $k \mid (P^2 - N)$
2. New interpolator: $k' = \frac{P^2 - N}{k}$
3. New roots: $a' = \frac{Pa + b}{|k|}$, $b' = \frac{Pb + Na}{|k|}$
4. Iterate until interpolator becomes $\pm 1, \pm 2, \pm 4$

2 Chapter X: Arabic and Indian Divisions of the Zodiac

Nakṣatras — Bilingual Enumeration

Sanskrit Name	English Identification
(Aśvinī)	The Wives of the Aśvins, figured as a horse's head. Principal star: β Arietis (Sheratan).
(Bharanī)	Figured as the body of a horse, three stars: 35, 39, 41 Arietis.
(Kṛttikā)	The Pleiades, six or seven stars.
Rohiṇī)	The Red One, Aldebaran (α Tauri).
(Mṛgasīras)	The deer's head, λ Orionis.
drā)	α Orionis (Betelgeuse).
(Punarvasu)	Castor and Pollux (α, β Geminorum).
uṣya)	δ, θ, γ Cancri.
(Āśleṣā)	ϵ Hydrae.
ghā)	Regulus (α Leonis).
(Pūrvaphālgunī)	δ, θ Leonis.
(Uttaraphālgunī)	Denebola (β Leonis).
)	Corvus, $\gamma, \delta, \epsilon, \alpha, \beta, \eta$ Corvi.
(Citrā)	Spica (α Virginis).
(Svātī)	Arcturus (α Boōtis).
(Viśākhā)	$\alpha, \beta, \iota, \kappa$ Librae.
(Anurādhā)	δ, β Scorpii.
yeṣṭhā)	Antares (α Scorpii).

Sanskrit Name	English Identification
(Mūlā)	λ Scorpii and surrounding stars.
(Pūrvāṣāḍhā)	δ, ϵ Sagittarii.
(Uttarāṣāḍhā)	$\zeta, \sigma, \tau, \phi$ Sagittarii.
avaṇā)	α, β, γ Aquilae.
(Dhaniṣṭhā)	$\alpha, \beta, \gamma, \delta$ Delphini.
(Śatabhiṣaj)	λ Aquarii.
(Pūrvabhādrapadā)	α, β Pegasi.
(Uttarabhādrapadā)	γ Pegasi, α Andromedae.
tī)	ζ Piscium.

3 Chapter XI: Brahmagupta's Astronomy

Time Units — Bilingual Definitions

“Beginning with Caitra, all the months of a yuga proceed for the creation-epoch (sr̥ṣṭi-cchāda) at Laṅkā simultaneously with the beginning of the day of Brahmā.”

All months begin simultaneously at the creation epoch at Laṅkā (on the prime meridian).

“Two vipalas (atoms of time) make a nālikā; two nālikās make a prāṇa; thirty-six prāṇas make a vināḍikā; sixty vināḍikās make a nāḍī (ghaṭikā); sixty ghaṭikās make a day; thirty days make a month.”

Time hierarchy: 2 vipalas = 1 nālikā; 2 nālikās = 1 prāṇa; 36 prāṇas = 1 vināḍikā; 60 vināḍikās = 1 ghaṭikā; 60 ghaṭikās = 1 civil day; 30 days = 1 month.

Kalpa Constants — Bilingual Table

Sanskrit Term	Value / English Translation
(kalpa) — Aeon	4,320,000,000 solar years
(sāvana-dina) — Civil days	1,577,917,828,000
ura-māsa) — Solar months	51,840,000,000
(cāndra-māsa) — Lunar months	53,433,300,000
(adhi-māsa) — Intercalary months	1,593,300,000
(śaśi-divasa) — Lunar days	1,602,999,000,000
aya-aha) — Omitted tithis	25,082,580,000

Epoch Calculation (Śṛṣṭi-saṃvatsara)

“Six manus have elapsed in half a kalpa; and of the seventh manu, twenty-seven caturyugis have gone by; of the twenty-eighth caturyugi, the three yugas kṛta, dvāpara and tretā have elapsed, and also 3179 years of the present kaliyuga.”

The total elapsed period at Brahmagupta’s epoch (year 628 CE):

$$\begin{aligned}
 T &= 6 \times 71 \times 4,320,000 \\
 &\quad + 7 \times 4 \times 432,000 \\
 &\quad + 27 \times 4,320,000 \\
 &\quad + (1,728,000 + 1,296,000 + 864,000) + 3,179 \\
 &= 1,972,947,179 \text{ years.}
 \end{aligned}$$

Mean Longitude Calculation

General method:

1. Compute the *ahargana* A = civil days elapsed since epoch
2. Multiply by planet’s revolution-number R in a kalpa
3. Divide by civil days in kalpa D :

$$\text{Mean longitude} = \frac{A \times R}{D} \text{ (in revolutions, signs, degrees, etc.)}$$

4 Chapter XII: Brahmagupta as a Critic

“Know the *lambana* (difference of parallaxes in longitude) by means of five Rsines: the *madhya-jyā*, the *udaya-jyā*, the *dr̥k-kṣepa-jyā*, the *drg-jyā*, and the *drg-gati-jyā*; and the *avanati* (difference in latitude) likewise.”

The *lambana* is computed from five Rsines:

- *madhya-jyā* = $R \sin z_m$ (zenith distance of meridian ecliptic point)
- *udaya-jyā* = $\frac{R \sin L \cdot R \sin \varepsilon}{R \cos \phi}$
- *dr̥k-kṣepa-jyā* = $R \sin z_c$ (zenith distance of central ecliptic point)
- *drg-jyā* — related to observer’s zenith distance
- *drg-gati-jyā* — related to the motion of the observer

where L = longitude of horizon ecliptic point, ε = obliquity of ecliptic, ϕ = latitude of place.

5 Chapter XIII: Astronomical Instruments

The Gnomon (Śaṅku)

“(The gnomon) two añgulas wide at the base, pointed as a needle (śūla-agra), twelve añgulas in length, and full of holes from the basic circular part to the pointed extremity — thus is the śaṅku (gnomon) described, O beloved.” — BrSpSi. XXII. 39.

The gnomon dimensions: base width = 2 añgulas (≈ 3.75 cm), height = 12 añgulas (≈ 22.5 cm), tapering to a point.

“When there is no wind, placing a jar full of water upon a tripod on the ground which has been made plane by means of eye or thread, one should bore a fine hole (at the bottom) so that the water may fall drop by drop.”

Procedure for testing level ground: Place a water-filled jar on a tripod on leveled ground. A fine hole allows water to drip steadily, confirming the surface is horizontal.

6 Madhyamādhikāra: Full Bilingual Text

Brāhmasphuṭsiddhānta: First Chapter — Madhyamādhikāra

*jayati praṇata-sura-asura-auragīndra-ratna-prabhā-cchurita-pāda-pañkajaḥ |
kartā jagat-patir-aditya-tulya-yānām mahādevaḥ || 1 ||*

“Victorious is Mahādeva, the Lord of the Universe, the Creator, the lotus-feet of whom are rubbed with the lustre of jewels from the heads of gods, demons, and serpent-kings who bow before Him — He whose chariot is equal to that of the Sun.”

*brahmagupto graha-gaṇitaṃ mahatā kālena yat khalvīmatam |
kṣīṇe 'dhunī tad-viparīta-vad-brāhmaṇair upetaṃ || 2 ||*

“Brahmagupta (says): The science of the computation of planets, which was well-established in the past, has now become corrupted; being contrary (to the truth), it has been mixed up by the brāhmaṇas.”

*dhruva-bala-arka-suta-budhaj na jyotir-vīdaṃ prīti-śad-gama-do |
poṣaṇārthaṃ vivṛtāntara-vyaktaṃ saha prabandhena sūcitam || 3 ||*

“(I write this) for the pleasure of astronomers, who are learned in the (motions of) Dhruva (pole star), Bala (Mars), Arka (Sun), Suta (Saturn), Budha (Mercury), and J na (Jupiter); for the preservation (of knowledge), the obscure meaning is made clear along with the connected exposition.”

caitry-ādi-tad-eva-dvija-nodvan-māsa-sarveṣa yugasya |
sṛṣṭi-cchādo laṅkāyāṃ samam pravṛtā dine-kalasya || 4 ||

“Beginning with Caitra, all the months of a yuga — (which are) the creation-covering — proceed simultaneously at Laṅkā with the beginning of the day of Brahmā (dina-kala).”

prāg-ry-udda-nāṭikā-dvau dvi-dvi-yaṣṭikā tri-nāṭikā ṣaḍ-ghnā |
ghaṭikā ṣaḍ-ghnā diva-lo divasānāṃ triṃśatā māsaḥ || 5 ||

“Two atoms (paramāṇu) make an aṇu; two aṇus make a lava; two lavas make a nimeṣa; three nimeṣas make a kṣaṇa; five kṣaṇas make a kṣathi; four kṣathis make a muhūrta; eight muhūrtas make a prahara; four praharas make a day (aho-rātra); thirty days make a month.”

Time scale: 2 paramāṇu = 1 aṇu → 2 aṇu = 1 lava → 2 lava = 1 nimeṣa → 3 nimeṣa = 1 kṣaṇa → 5 kṣaṇa = 1 kṣathi → 4 kṣathi = 1 muhūrta → 8 muhūrta = 1 prahara → 4 prahara = 1 day → 30 days = 1 month.

Verses 22–24: Solar and Lunar Month Relations

parivartah sapta-catuhṣaṣṭir sūryasya dinasya mitih |
ravi bhagaṇāṇaṃ mānaḥ sāvana-dinasya kuṭṭakair eṣa ||

“The revolution-number of the Sun is 64 + 7 = 71 (per caturyuga); this is the measure of solar days. By the pulverizer (kuṭṭaka), this (gives) the civil days.”

Rule: A solar year ≈ 365 days 15 min 30 sec. The revolution-number of the Sun = 4,320,000 per kalpa.

ravi bhagaṇārdha-badhā dvādaśāṃśena bhavati ravi-māsaḥ |
bhagaṇantaram ravi-dinaḥ śaśi-māsaḥ sūrya-māsena ||

“The product of half the revolutions of the Sun divided by twelve becomes a solar month. The difference of revolutions (between Sun and Moon) is the lunar month exceeding the solar month.”

Formula:

$$\text{Solar month} = \frac{R_{\odot}/2}{12} ; \quad \text{Lunar month} = R - R_{\odot}$$

adhi-māsāḥ śaśi-mśāv-indv-adinasya bhavati śaśi-divasāḥ |
śaśi-sāvana-divasa-antara-nivamāni tithiḥ śaśāñka-dinaṃ ||

“The intercalary months (adhi-māsa) are the lunar months exceeding the solar months. The lunar days are (obtained from) the difference between lunar and civil days; the tithi (lunar day) is the measure of the difference between lunar and civil days.”

Key relations:

adhi-māsa = lunar months – solar months

śaśi-divasa = lunar days – civil days

tithi = 1/30 of a lunar month

Verses 26–27: Epoch and Ahargaṇa

kalpa-parārdha manavaḥ ṣaṭ-kāśya gatāś catur-dvi-yugāni yugānām |
śruta-purāṇadi-nikṣepānām kṣayaḥ sūrya-dinaṃ cakṣuḥ ||

“In half a kalpa, six manus have elapsed, and twenty-seven caturyugis of the (seventh) manu. (This gives) the destruction (elapsed period) of the sūrya-dina (solar days) – the eye (of calculation).”

The *ahargaṇa* (day-count) from the kalpa epoch:

$$A = D_{\text{manus}} + D_{\text{caturyugis}} + D_{\text{current yuga}}$$

nava-gaṇa-rāśi-munin nava samanvantara-devaḥ śakunta-pāte |
śarva-mateḥ saṃbatsarāt saṃkhyāṃ śaravac-chataṃ rūpaiḥ ||

“O Muni of nine-digit signs! (This gives) the number of years from the śarva-mata (doctrine year), 106 (śara) increased by unity – 3179 years of kaliyuga (at the epoch śaka 550 = 628 CE).”

Brahmagupta’s epoch: year 3179 of kaliyuga = śaka 550 = 628 CE.

Bilingual Appendix: Rational Geometrical Figures

iṣṭa-yaṣṭeḥ sama-cchedaṁ rūpeṇ a hīnam ubhayor mithaḥ |
ṣaḍ-bhāgena sama-j nā-try-akṣetrasya bhuje ||

“The square of the optional side divided and then diminished by an optional number; half the result is the upright (koti), and that increased by the optional number gives the hypotenuse of a rectangle.”

Formula: For arbitrary rational m, n :

$$\text{Base} = m, \quad \text{Upright} = \frac{1}{2} \left(\frac{m^2}{n} - n \right), \quad \text{Hypotenuse} = \frac{1}{2} \left(\frac{m^2}{n} + n \right)$$

yasmin dve samānī bāhū anyad dvayam-asamānī ca |
tayor varga-yogo vibhājyo bhedenā sama-dvi-bāhu-kṣetre ||

“Where there are two equal sides and two other unequal sides — the sum of their squares divided by their difference (gives the base of an isosceles triangle).”

For two unequal integers p, q : equal sides = $\frac{1}{2} \left(\frac{p^2+q^2}{p-q} + (p-q) \right)$, base = $\frac{p^2+q^2}{p-q}$.

sarva-dor-alpa-catuṣṭaya-kṣetra-phala-mūlaṁ dvi-bhuja-kṣetre |
dvayor dvayor ddhayor ghātaṁ mūlaṁ dvi-bhuja-kṣetrasya ||

“The square root of the product of four sets of half the sum of the sides, each diminished by one side, is the exact area of a triangle (and) quadrilateral.”

Brahmagupta’s Formula:

$$A = \sqrt{(s-a)(s-b)(s-c)(s-d)}, \quad s = \frac{a+b+c+d}{2}$$

This generalizes Heron’s formula for triangles (when $d = 0$).

Critical Apparatus: Selected Variants

V.	Adopted Reading	Variant / Note
1		(mss. A, B) (ms. C)
2		(ms. D) "ow corrupted"
3		(mss. A, B) — sandhi variant
4		B) — compound variant
5		(ms. C) — orthographic
22		ns per caturyuga (confirmed)
23		(ms. E) — scribal error
26		s. D) — gemination variant
27		(mss. A, B) — sandhi split

Colophon

iti brāhmasphuṭsiddhāntasya madhyamādhikāraḥ samāptaḥ

“Here ends the Madhyamādhikāra of the Brāhmasphuṭsiddhānta.”

Bilingual Edition: Sanskrit (Devanagari) / English (IAST)

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Chunk 3 (Pages 281–420)

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Introduction to Chunk 4

1–560 from the first volume of the *Brāhmasphuṭasiddhānta* ges correspond to book pages 32–171 of Dvivedī’s edition, al) of the work.

of Brahma”) is Brahma-gupta’s masterpiece, composed in ters (*pūrvadaśādhyaī*) covering astronomical topics and , and more advanced astronomy.

r solving the three problems of diurnal motion – finding

n of lunar eclipses, including the calculation of shadow lor phases of the eclipsed Moon.

f solar eclipses, including parallax corrections (lambana),

f planets.

n of the Moon’s horns.

adow.

lanets.

XI): Critical examination of other astronomical systems, ṣeṇa, Viṣṇucandra, Lāṭadeva, and Vijayanandi.

r on mathematics, containing arithmetic operations, mix- lems, and algebra (including the *kuṭṭaka* or pulverizer).

tical problems and their solutions.

hnical terms in IAST transliteration

r each group of verses, compiled by Sudhākara Dvivedī.

marked wit

(*viveka*) provide editorial commentary.

reads X where manuscripts have variant Y. Hindi notes

Chapter III: Triprasnādhikāra (Conclusion)

e problems relating to diurnal motion

English Translation (with IAST)

ing

Translation: (25) When the Sun, diminished by the day-radius (*dinadala*), is east or west, the ghaṭikās till the end of the tithi are obtained by subtracting that from thirty and the ghaṭikā-remainder is obtained. (26) The reverse applies at midnight for a lunar eclipse; slowly for a solar eclipse. From the rising of the Sun, by the same ghaṭikās twice. (27) The semi-diameter (*vyāsa*) of the Earth's shadow at the Moon's orbit is obtained: the day-diameter, circumference, accurate tithi-nata-kendra-jyā and the degrees of Jupiter or Kandu (the Moon), Indu — by 191 and the new nine-veda 466, divided by the square of the radius.

itted in the source).

Translation: (38) Multiplied by seven degrees, the day-multiplier, the daily motion in signs, divided by the obtained degrees, the sphuṭa of Mars is increased or decreased from the mark (starting point) at Mrga (start of Capricorn) and Karka (start of Cancer). (39) Its sphuṭa-circumference is 70 yojanas; the śīghra-sphuṭa circumference diminished by obtained degrees; 24330 diminished by veda-jina-akhi; this is the correction (*sphuṭikaraṇa*) of Mars.

Critical Apparatus:

38. (Ch .

he numbers in the verse.

tions

Translation: (4) The shadow-circle-radius multiplied by the hypotenuse, divided by the semi-diameter obtained by the Sun, the *viṣuvacchāyā*, the *yāmyā*, the sum or difference of that at the tip of the *bhuja*. (5) The eastern or western shadow of the gnomon — taking the square root of the difference of the *bhuja*-squares, that eastern or western shadow is the *koṭi* between the tips of the *bhuja*. (6) Having made the shadow-tip in the middle of the direction, the gnomon according to direction is to be measured; the observer's shadow-tip situated in the middle of the direction moves in reverse.

Translation: (64) At the end of *Kṛṣṇa-caturdaśī*, in the bird (*śakuni*) season, in the four-footed (*catuṣpada*) measure, the half-tithi, *nāga*, *kistughna*, from *pratipadā* etc. (66) The manifest two *kalās* divided, the moving *karas* and the *hara-sa-gu-ra-ṇa-ṇaḥ* 360, with one obtained sum, the remainder divided by *hr̥ta* is like a tithi and other.

[At this point the Triprasnādhikāra concludes. Several verses in the manuscripts are mutilated; Dvivedī notes that at some places only the commentary (ṭikā) is available. The chapter contained 66 verses.]

Chapter IV: Candragrahaṇādhikāya

ation of Lunar Eclipses

es, including the dimensions of the Earth's shadow at the
es of the eclipse, and the colours assumed by the eclipsed

English Translation (with IAST)

ts

Translation: (1) The Sun's jyā multiplied by the day-part-jyā, divided by the guṇākṣa-jyā or alternatively, the viṣuvacchāyā divided by twelve. (3) Twelve times the viṣuvacchāyā gives the shadow; or separately by the akṣa-lamba-jyā. Multiplied by the declination (krānti), the sama-maṇḍala-karṇa; as before, the separate shadow.

Translation: (7) The two akṣa-lamba-chāyā-jyā, their sum or difference gives the root; the viṣuvat-karṇa-chāyā, otherwise the hypotenuse is the śaṅku (gnomon). (8) The uccatana-jīvā-koṭi, the chāyā, the ha-jyā, the bhuja or the nata-jyā; the hypotenuse-chāyā-circle from the semi-diameter; similarly elsewhere.

8.

Translation: (14) If the half-duration is greater, by the difference, otherwise minus one; if negative with addition, thus the half-obscuration (vimarda). (16) From the sphuṭa-tithi-antā the parallax (lambana), with or without the asakṛt-sthityardha; that sphuṭa is by separation, at the end of sthityardha-utaya-tithi. (17) In the difference of that sphuṭa-tithi-cheda, the sphuṭa in the tithi-half, with or without, from the self-vimarda-udaya, thus the corrected half-obscuration. (18) The Moon's latitude (sphuṭa-vikṣepa) multiplied by the half-duration, the corrected (sphuṭaḥ) bhūja multiplied by the half-duration is the root of the remainder as before.

Translation: (16) At the end of obscuration, smoke-coloured, black; in partial eclipse, half-increased eating (grāsa); black-brown (sa-kṛṣṇa), coppery (tāmra), white-yellowish (śakli-kapīla-varṇa). (20) The measure of pure, half-duration-motion-desired, eating in the samakala-etc.; the breadth of the nadagūhara-āyāma desired one-fourth.

t marked).



Thus ends the Fourth Chapter of the Brāhmasphuṭasiddhānta.



Chapter V: Sūryagrahaṇādhikāya

tation of Solar Eclipses

including the parallax corrections (*lambana*) that distinction.

English Translation (with IAST)

Translation: (2–3) Same as lunar eclipse — the shadow dimensions are computed from the Sun’s declination. (4) The bhuja from the Sun’s tip (arkāgra), the square deducted, 346450 and 144 multiplied by the square; the first and other, twelve times the viṣuvacchāyā, the product of the two.

Translation: (5) From the lambana-ghaṭikā-lagna, from the tātkālīka, diminished by three signs; if negative with addition, five times the asakṛt, divided by hṛdaya. (6) From the karaṇa-guṇa, from the semi-diameter, dviveda 48, divided; the lambana-nāḍis or the bāskara-vitribha-lagna-antaram. (7) If the lambana is negative, the positive result in its own lalipta-phala. (8) From the aṣajyā, from the vitribha-lagna, its own declination (krānti) uttarākṣa or of the Moon; if greater, otherwise the yāmyā.

Verses 14–17: Sphuṭa-tithi and corrected obscuration

Translation: (14) If the half-duration is greater, then by the difference, otherwise minus one; if negative with addition, thus the half-obscuration (vimarda-ardha). (16) From the sphuṭa-tithi-anta, the lambana, with or without the asakṛt-sthityardha; that sphuṭa is by separation, at the end of the sthityardha-utaya-tithi. (17) In the difference of that sphuṭa-tithi-cheda, the sphuṭa in the tithi-half, with or without; from the self-vimarda-udaya, thus the corrected (spaṣṭa) half-obscuration.

Translation: (18) Thus in the conjunction of planet and Moon, if greater than the measure of unity plus half, in the middle-intermission; diminished by the half-duration-obscuration, the cross (tārā). (19) The lambana like akala-graha, asakṛt-svāvanati-lāptikā-sphuṭi, the tātkaḷika-vikṣepi, the samānya-diśoḥ sum or difference. (20) The deflection (vikṣepa), the middle-difference, the above-shadow-planet situated below; if greater than half the unity-measure, not too distinct, hence the sphuṭa-statement.



the Brāhmasphuṭasiddhānta.



Chapters VI–IX: Udayāsta, Candraśṛṅgonnati, Candraccāyā, Grahayuti

ations: rising and setting of planets, the elevation of the lanets.

English Translation (with IAST)

Planets)

Translation: (6) By twelve, the śītāṃśu-sita-jīvā, the earth-born, by nine; the day-increment-difference-ghaṭikā multiplied by six in kalāṃśa. (7) The graheka like hr̥śyā-hr̥śyā-yuti, by the difference of suttacāntara-obtained-days; subtracted or added in liptāmyaḥ, as before by the tātkālīka-asakṛt. (8) Daily rising and setting thus, by the tātkālīka-graha-vilagnais; the solar rising and setting at the end of the Pausa month, the Moon (śītāṃśu).

of the Moon's Horns)

Translation: (38) The latitude (unnati) of the Moon is found to disagree from the hypotenuse-śravaṇa-jñānāt; the rising and setting are caused by day, by prati-ghaṭika measure and from vā-jñānāt. (36) The akaicara-ghaṭikā, vyasta-jyā, multiplied by candra-māna, divided by vyāsa is pure; since the dṛk-sama is not true. (40) Previously taught as different, afterwards the other reverse; that time which is shadow, that asat-sphuṭa, suktimān before. (41) Who knows the uḍita-uḍita-astamita-avaśeṣa-kāla; how can he who knows the shadow-unnati of the Moon be an ārya-maṭa-jña.

38.

adow)

Translation: (14) The declination-jyā, the deflection (vikṣepa), the declination-cāpa: the sum or difference of the jyā of like or unlike directions of the declination. (16) Thus the bhūmi-nīśa-vayo (latitude etc.) of the planet, by its declination, the candra-aṣṭaka-karma etc.; having done this in the eclipse, as in bhūmi, from that the svakrantijīvā.

lanets)

Translation: (44) By the prati-ghaṭika and adhika-śaṅku, in the middle is shown the earth; the tripraśna-oktacā like the Sun's śaṅku, the remainder of the bhramaṇādi-krama. (46) The śaṅku, the eastern-western antaram, the viṣuvacchāyā-sum or difference, nṛtale; the yāmyottaraṃ guṇāhataṃ by svakrānti-leśaiḥ and karṇa. (47) The cāpāṃśāḥ of sada-yeh, saṅghuvakāpakramāṃśa-rūnāḥ; the vikṣepāṃśe vyastā, the vyasta-viśuddhā vya-sahaśāṃśāḥ.

47.

(G,Ch

47. (G .



, and Grahayuti conclude here, with 12–70 verses each.]



Chapter XI: Tantraparīkṣādhikāya (Dūṣaṇādhikāya)

ation of Astronomical Systems

que of various astronomical systems current in his time. ra, Lāṭadeva, Vijayanandi, and Pradyumna, as well as the maka and Puliśa Siddhāntas).

English Translation (with IAST)

Translation: (1) Those who know the brahma-vid (science) in a chapter-barred dual manner, they consider it a siddhānta; I shall declare their faults arising from the yuga-ādi differences. (2) They say the yuga is five years; making the Saṃhitā of the Sun and Moon; the adhimāsa, vamara-atra, sphuṭa-tithi-jñāna — that is false.

Translation: (6) The yuga-year-navadvyekṣa (nine), the bright (Moon) etc., the same pravṛttānyat; that is false, since the corrected yuga-position of the manda-nodes is fixed. (7) The planet's release is in the ruṇāyā, the mandocca increases in śīghra; in the high position, the mandocca does not exist without (its own) motion. (8) In Ārya's eight hundred, the nodes move, in the Daśagītaka are fixed; the released two-nodes move in the circle, not fixed. (9) Āryabhaṭa knows the planets; what he has said is false; the Rāhu does not cause the eclipse, its pāta is not the eighth Rāhu.

Critical Apparatus:

Translation: (63) Thus the systems taught, even by reading, cause ignorance by faults; the examination of systems — this is the eleventh chapter.

itted).



of the Brāhmasphuṭasiddhānta.



Chapter XII: Gaṇitādhyāya

— *On Mathematics*

al chapter of the *Brāhmasphuṭasiddhānta*. It contains 66

ons)

s 17–20)

9)

4–46)

–51)

English Translation (with IAST)

Translation: (1) He who knows separately the parikamma (operation), the saṃkalita (summation) etc., and the twenty vyavahāras with chāyāta — he becomes a calculator (gaṇaka).

Translation: (2) The inverted (viparīta) and the divisor-multipliers of the quantities, the divisor-parts (chedāṃśakāḥ) with common divisor; the saṃkalita etc. are to be added, the difference is vyavakalita. (3) The rūpāraṇī (unity) multiplied by the divisor, with parts added, of two or many; the pratyutpanna (product) becomes the division by the divisor-product of the fraction-product. (4) The pratyutpanna after conversion, the divisor-part of the divisor, the divisor-times-divisor; the two-part-sum of the dividend, the divisor of all is the divisor.

ed in the text; the verse begins with the word .

Translation: (5) The square of all-parts divided by the square of the divisor becomes the varga (square root); the root of all-śita-parts, the root divided by the divisor-term. (6) Having set up the varga-mūla (square root), the utpadyanyakṛti, the trigunottara-saguṇa, and the prathamā; the nottara-kṛti-antitya-guṇā, trigunā and uttara is the ghana (cube). [Cube completed.] (7) From the second chedāghana, the cube-root-square-times-trisaguṇa-sakṛti; subtracted, times the tripūrva, from the prathama-ghana, the ghana-mūla. [Cube root completed.]

/ “cube root completed”) are not marked in the source text.

Section 2: Miśrakavyavahāra (Mixture Problems)

Translation: (16) Divided by the prakṣepa-yoga (sum of contributions), obtained, the prakṣepaka (contributor) multiplied by the gains; the ūnādhika-uttara-ardha is true, the dvitonayā with its own phala-sum added.

s.

Translation: (17) The pada (term) minus one, multiplied by the utara (common difference), added to the sum of the first and last terms; the first plus the last term from the first, the cube of the middle term is pada-gaṇa (number of terms) multiplied. (18) From the utara-hīna (decreased by common difference) dviguṇādi-śeda-varga, the ghanottara-aṣṭra-vadhe; added, the pada by the śeṣeṇam, the dviguṇottara hṛtaṃ gacchaḥ (progression). (19) If the eka-uttara (common difference one) eka (first term one) and the desired gaccha (number of terms) is the saṃkalita (sum); that viyut (without) gaccha multiplied, divided by three, is the saṃkalita.

Section 4: Kṣetravyavahāra (Geometry of Plane Figures)

reas and diagonals of triangles, quadrilaterals, and other
ificant sections of the work.

Translation: (21) In triangles and quadrilaterals with equal and two sides etc., the vyāsa (diameter) is like āyata-catura (rectangle); the vyāsa-karṇa of the field is the sama-koṇi (right-angled diagonal) of the pair of fields. (22) Half the sum of the lambaka (altitude) and karṇa in sama-tribhuje (isosceles triangle) is the area; in sama-catura (square) it is the square; in sama-koṇi (right-angled) half the sum of the vyasta (crossed diagonals). (23) In asama-catura (unequal quadrilateral), the product of the rāśis (sides), their squares and the sum; their difference-varga (square), with equal-square-half added mutually.

Translation: (32) The karṇa and lamba (hypotenuse and altitude) — the lower karṇa-lamba; in proportion, the uddha-śūnyāṃsa-pāṭāyāma. (33) The product of kṛti-yura-sahaśa-rādyo-bāhu-dhāto; the lambaka (altitude) twice; the product of kṛty-antara of unequal (sides) and dvitribhuja. (34) Divided by two desired quantities, from the sum of the dvi-ghaṇa-varga-phala; the side of a viṣama-tribhuja (scalene triangle) from the viṣama-phala is added to yā mūḥ. (35) The square of the desired bhuja divided by the undesired, that portion is the koṭi; of the āyata-caturastra (rectangle), the field has the desired hypotenuse.

Section 10: Chāyāvyavahāra (Shadow Problems)

the shadow of a gnomon (*śaṅku*), problems of distance,

Translation: (2) The *chāyāntara* (difference of shadows) with one added, divided by two, the *prāga-parayor-yuga-doṣam*; the *dinagata* (elapsed day), the *śeṣāṃśa-hṛtaṃ dvidalaṃ*, the *chāyāntara-vyekam*. (13) The distance between the lamp-surface and *śaṅku*-surface, the desired measure-*śaṅku-guṇa*; the lamp-wick-*ucchyā*, the *śaṅku* deducted, the *śeṣa-uddhṛtaṃ chāyā*. (4) The *chāyā* multiplied by the *chāyāgra-antara*, divided by *chāyāntara* is the *bhū* (earth/distance); the *bhū* is *śaṅku-guṇā chāyā*, divided, the lamp-wick-height.

Translation: [End of shadow problems.] (45) The *guṇākāra* (multiplier) and *khāṇḍa-tulya* (equal to the section) are the *guṇyau* (multiplicands); the *gomūtri-kā* (cow's urine) *kṛtī* multiplied; the *sahitāḥ pratyutpannaḥ* (combined product), the *guṇa* (quality) or equal to the *kāraṇa-meda*.



[The Gaṇitādhyaṃya concludes here with 66 verses total, containing the foundation of Indian algebra including the kuṭṭaka (pulverizer) method for solving linear Diophantine equations.]



Chapter XIII: Praśnādhyāya

On Mean and True Motion

) whose solutions demonstrate the computational method (sixth) refers to the chapter's position within the *uttara*

English Translation (with IAST)

Translation: (1) I shall declare the *praśnādhyāyas* (problem-chapters) with solutions (*uttara*), increasing the intelligence of calculators; by which the teacher (*ācārya*) of the *tāntra*-vids becomes one of intelligent mind. (2) The *adhimāsakāḥ* (intercalary months) with *vikalā* (seconds), the *avamarātra* (omitted days) with the *yugayāta*, by the *dyugaṇa* (day-count); he who knows the motion of planets in the wind's path knows the *kāla-tantra* (science of time).

Translation: (3) He who comprehends the *avamāti* with *vikalā*, the *adhimāsa-radhika-māsaka-anavamaiḥ*, the *grahamiṣṭhaṃ* in that, knows the *kāla-tantra*. (4) The *dyugaṇa* without *adhimāsa*, *avinādina garona candra-ākkah*; in that without *sphuṭa-tithi*, he who knows is the *kāla-tantra-jñā*. (5) From the mean (*madhya*) to other *tithi* from *iṣṭa-madhyamā-cchadi-prahaṇas*, from the *iṣṭārkavid-upāte yom*, without knowing, he is not a *tantra-jñā*.

Critical Apparatus:

3. (Ch .



*n pairs covering mean and true longitudes of planets, con-
e full chapter contains approximately 48 verses.]*



Appendix: Notes on the Critical Edition

ing manuscripts, as collated by Sudhākara Dvivedī:

ī, containing the complete text with the commentary of
he Maithila script and is now preserved in the India Office

repared for Dr. Thibaut by Dvivedī himself. Many folios

ript, with variant readings at several places.

skrit critical editing:

the main body.

duced by the manuscript siglum.

where the manuscript has Y.

ditor’s commentary on textual problems.

mutilated, particularly:

rved, while the mūla (root text) is lost.

portant mathematical developments:

ticated trigonometric relationships involving the gnomon,
latitude.

hms for correcting positions and computing parallax —
ronomy.

- The complete theory of operations on fractions (*parikarmma*)

- Rules for arithmetic and geometric progressions
 - Formulas for areas of triangles and quadrilaterals, including the “Brahmagupta’s formula” for the area of a cyclic quadrilateral
 - The *kuṭṭaka* (pulverizer) method for solving linear Diophantine equations of the form $ax + c = by$
 - The *varga-prakṛti* (square-nature) method, a precursor to the study of Pell’s equation
4. **The Tantraparīkṣā** represents one of the earliest critical examinations of rival scientific theories in Indian intellectual history.

On the Translation

The English translations in this bilingual edition aim to be as literal as possible while remaining intelligible. Sanskrit technical terms are preserved in IAST transliteration and enclosed in parentheses where appropriate. Several verses in the source are corrupt or mutilated; in these cases the translation follows Dvivedī’s reconstruction and is marked accordingly.

The *Brāhmasphuṭasiddhānta* is composed in the āryā metre (with some verses in the longer vasantatilakā and related metres). The elliptical nature of Sanskrit astronomical verse makes translation particularly challenging: many verses omit verbs, subjects, or objects that must be supplied from context or commentary. Readers are encouraged to consult Pṛthūdakasvāmin’s commentary (*Vāsanābhāṣya*) for detailed explanations of each rule.



End of Chunk 4 (Pages 421–560)



, encompassing Chapters XIV through XXII of the *Brāh-*
ion, the “three questions” of astronomy, eclipses, lunar
 sody, sine computation, and astronomical instruments.

lanetary motions; epicyclic corrections for *manda* and

questions: latitude from gnomon observations, shadow

eclipse computations, including the graphical *parilekha*

r cusp elevation; a short chapter of 10 verses.

ter (102 verses): *kuṭṭaka* (pulverizer), operations with
 near and quadratic equations, *bhāvitaka*, and the *varga-*

and shadow computations; practical geometry and in-

anskrit prosody; combinatorial calculations for metrical

araṇa: sine computation by second-order interpolation.

uments: *dhanur-yantra*, *yaṣṭi-yantra*, water and wheel

rity and fidelity to the mathematical content. Technical
 ce. Modern mathematical notation (algebraic formulas,
 algorithmic.

Chapter XIV: On True Planetary Motions

This chapter deals with *sphuṭa-gati*—the computation of a planet’s true longitude from its mean longitude by applying the *manda* (eccentric) and *śighra* (epicyclic) corrections.

[Printed pages 171–181. This chapter continues from the previous chunk. Verses 1–6 and the closing verse are preserved here.]

V.1. The divisor of the *yuga* makes the cosine (*koṭijyā*) into the cosine, and the sine (*bāhuḥjyā*) into the sine. The hypotenuse (*kṣetra*) together with these—that is the *sphuṭagatikāḥ*, the true-motion triangle.

V.2. The radius of the outer circle (*paramaṇḍala-karṇa*), made to produce the cosine-shadow (*koṭi-chāyā*) for computing the true — the hypotenuse-end, cosine-corner, and sine: that is the true-motion triangle.

V.3. The following cosine-chord (*anujakoṭijīvā*) at the radius, which the outer-circle-knower makes to produce — that which makes the *svocca* (own apogee) from its own equation (*sva-phala*) into the true — that is the true-motion triangle.

V.4. Of the field of the true-hypotenuse (*sphuṭa-pada*), sine and cosine are the results without the chord (*jyā*); without the chord, or with the equation (*phala*) added — that makes the true-motion computation.

V.5. He who makes the mean planets true by desired-day indicators (*iṣṭa-diva-yadaiḥ*) for the directions — that *sphuṭagatikāḥ* which makes (the computation) with its own apogee-true (corrections).

V.6. Threefold: Saturn (*śani*), Moon (*indu*), Naiṭyama (Mercury), Gārā (Mars), Lakṣmī (Venus), the planet-group — gathered: Mars (*bhauma*), Aṅgiras (Jupiter), Guru — Durvarca, Vātya, Gārā — with Kai (Sun?). [Planetary name correspondences in numerical notation.]

[The chapter continues with more verses on true motion computations, including corrections for the manda and śighra epicycles, and rules for finding planetary positions. The chapter concludes at printed page 181 with verse 46.]

V.46. This rule of mean-upper-line (*madhyotarekhā*) is the fifty-fifth chapter-method. It is to be known by the teachers of the knowers of the *tantra*; one becomes skilled in mean motion.

*Here ends the Fourteenth Chapter, on Mean Motion, in the **Brāhmasphuṭasiddhānta**.*

Chapter XV: The Three Questions

This chapter treats the “three questions” of astronomy: relating to time (*kāla*), place (*sthāna*), and direction (*dik*). It covers the determination of latitude from gnomon observations, shadow measurements, and the computation of time.

[Printed pages 182–193. Chapter 15 contains 60 verses on the three problems.]

V.44. The sine of rising (*udaya-jyā*) and the sine of declination (*krānti-jyā*), multiplied by the diameter and divided: by twelve and by fire (3), by Śiti (cold, the zero-point) and by the equinoctial shadow — divided by that is the declination (*krāntiḥ*).

V.45. From the desired diameter (*iṣṭa-vyāsa*), from the shadow-ray (*chāyā-raśmi*), from Bhāskara’s (the sun’s) crossing-sign, from the double rising: the string (*sūtra*) — that sine is the measure of the particular declination (*krānti-viśeṣa-padam*).

V.46. In the six parts (of the day), the gnomon (*śaṅku*) is not known (in this way); if the gnomon is placed toward the east-west, the northern (deviation) is due to the southern excess, and that difference indicates greater latitude.

V.47. In the northern hemisphere, it is not thus; the leftward (deviation) is in the globe-string. The gnomon-base divided by the gnomon — the equinoctial shadow, twelvefold, is the multiplier.

[The chapter contains 60 verses on the three problems, concluding at printed page 193.]

Chapter XVI: On Eclipses (Advanced)

This chapter treats refined computations for eclipses (*grahaṇa*): the computation of lunar and solar eclipses, the graphical *parilekha* method, and the determination of contact (*sparśa*) and release (*mokṣa*) times.

[Printed pages 194–218. Chapter 16 covers advanced eclipse computations, including the parilekha graphical method.]

V.15. The *parilekha* (graphical representation) of the planet, by the planet-rule (*graha-kalpa*) first, either curved or straight — the instantaneous position, the closing and opening (of the eclipse) — thus.

V.16. The sine of the particular (longitude) difference in minutes (*viśeṣa-puṇy-aṁśa-jyā*), the shadow by the measure, and the degrees of the second arc — at the edges according to direction, direct or reversed (*viparyayaḥ*).

V.20. The difference of those deflection-minutes (*valana-aṁśaka*), from the sine-east-measure and perpendicular — the shadow obtained by the radius (*trijyā*) — the *pragraha-kṣepa*, before the retrograde motion.

V.21. Divided by the semi-diameter (*vyāsāṅka*), the minutes of the lunar-node-difference are the multiplier. By the middle-strength-minutes (*madhyabala-lip*) the north-south direction goes to the left.

[This chapter contains 46–47 verses on eclipse computations.]

Chapter XVII: On Lunar Cusp Elevation

This chapter treats the elevation of the lunar horns (*śṛṅgonnati*)—the elevation of the moon’s cusps, and the determination of time from this knowledge. This is a short chapter of ten verses.

*[Printed pages 219–223. Chapter 17 deals with the lunar cusp (*śṛṅga-unnaṭi*) and related computations. Only 10 verses.]*

[The complete verses of this chapter are not fully preserved in the scanned pages of this chunk. The chapter is transitional between the astronomical and mathematical sections.]

[This short chapter of 10 verses completes the astronomical portion before the extensive mathematical chapters.]

he BSS, containing 102 verses. It covers the *kuṭṭaka* tive numbers, *avyakta* (algebra) computations, *saṅkra-* one unknown, multi-variable equations, the *bhāvitaka* equation).

m the GRETIL digitization (Hayashi 1993).]

ithout the pulverizer (*kuṭṭaka*), I shall state the *kuṭṭaka*

tra, who know the pulverizer, zero (*kha*), debt (negative, imination (*haraṇa*), one-*varṇa* and *bhāvitaka* equations, ion).

onding to the greater residue (*adhikāgra*), divided by the *ra*) — whatever is the quotient from the mutual division w the other.

lied to the divisors A and B. The quotients are written in

ber (*iṣṭa*) such that, when combined with the difference *yati*) — the multiplier (*guṇakaḥ*) is to be set down; and nultimate (*upāntya*).

b) is divisible. Then the guṇaka and labdha are computed

st (*antya-yutaḥ*), (is divided by) the divisor of the lesser e divisor of the greater residue (*adhikāgra-cheda*) and (*agram*).

[The solution x to $ax + c = by + d$ (modulo the divisors) is constructed by this back-substitution. Let x_0 be the *guṇaka*; then $N = x_0 \cdot B_{\text{large}} + r_{\text{large}}$.]

V.6. Twice the *yuga* (epoch period) is the product of the divisors; the product of the divisors is the *yuga-gata* for both residues. Thus by the *kuṭṭaka*, the derivation of three etc.

V.7. The residue of the revolutions (*bhagaṇa*) etc. is the *agra*; divided by the divisor; and zero, divided by the residue of days — of these two, the residue of revolutions etc., divided by the day-born residue — that is the day-count (*dyugaṇa*).

V.8. By whatever multiplier the day-born and revolution-etc. residues of the circle etc., divided by their common divisor, are equal — the product of that is the day-count etc.

V.9. The mutual division of the dividend and divisor, whatever is the remainder — by that, divided, they become reduced (*nis-chedau*); those same, mutually divided.

V.10. The quotient is to be placed down, down; the remainder multiplied by the chosen multiplier (*iṣṭa-guṇakāra-saṅguṇam*) — it is exactly divisible so that one is subtracted (*eka-hīnam*). The multiplier is to be set down; and the result is from the last.

V.11. The end-residue (*agrāntam*), by the penultimate multiplied, combined with its upper (value), divided by the divisor without remainder (*niśśeṣa-bhāgahāreṇa*) — thus is the *sthira-kuṭṭakaḥ* (fixed pulverizer), the remainder.

V.12. From the desired revolution-etc. residue, by its own *kuṭṭaka*-multiplier, by its own divisor-multiplied — the remainder is the day-count, combined with the reduced multiplier and divisor.

V.13. Thus in equal (signs) and in unequal: debt (*ṛṇa*) and wealth (*dhana*), wealth and debt — whatever was stated, the reversal (*vyastatvam*) of debt and wealth is to be done to the dividend (*guṇya*) and additive (*prakṣepa*).

V.14. The multiplier is the divisor; the divisor is the multiplier; wealth is debt, debt is wealth — to be done. Square (*varga*) is root (*pada*); root is square; from the end, reversed (*viparītam*); that is the first.

V.15. He who knows the *kuṭṭaka* by the epoch-periods (*yu-gādi*) of planets, separately stated by two, three, four etc. — **he** truly knows the *kuṭṭaka*.

[Verses 16–29 contain illustrative problems (praśnas) on the pulverizer applied to astronomical computations.]

V.16. “The residue of revolutions etc. at the desired — when on a moon-day, on a sun-day, or a Jupiter-day, on a weekday — tell the signs.” He who tells this knows the *kuṭṭaka*.

V.17. “On a weekday, whatever is the residue of degrees, the residue of minutes — when that on a moon-day, of the sun, or of the moon — he who tells this is a knower of the *kuṭṭaka*.”

V.18. “The desired (planetary positions) in the tithi-measure-days, which are sun etc., again when in those desired planet-weekdays” — he who tells this is a knower of the *kuṭṭaka*.

[Verses 19–29 continue with astronomical praśnas: given planetary residues at a particular weekday, finding the elapsed days and other planetary positions. These were standard problems for Indian calendar-makers (pañcāṅga-kāras).]

ithmetic operations involving positive (*dhana*), negative
est explicit rules for operations with signed numbers

egatives is negative; of a positive and a negative is their
 gative; and of two zeros.

$$\begin{aligned} (-) + (-) &= (-) \\ a + (-a) &= 0 \end{aligned}$$

er; the positive from the positive, the greater negative
 (*vyastam*). The negative becomes positive, the positive

$$ns: (+a) - (+b) = -(b-a) \text{ and } (-a) - (-b) = +(b-a)$$

wealth (*dhana*) is wealth. Zero (*ākāśa*) — when positive
 hen it is additive (*kṣepya*).

$$0 - a = (\text{sign reversed})$$

product of two positives is positive. The product of zero
 the product is zero.

$$\begin{aligned}) &= (+) & (+) \times (-) &= (-) \\) &= 0 & 0 \times 0 &= 0 \end{aligned}$$

V.34. Positive divided by positive is positive; negative divided by negative is positive. Zero divided by zero is zero. Positive divided by negative is negative; negative divided by positive is negative.

[Division of signed numbers:]

$$\frac{(+)}{(+)} = (+)$$

$$\frac{(-)}{(-)} = (+)$$

$$\frac{0}{0} = 0$$

$$\frac{(+)}{(-)} = (-)$$

$$\frac{(-)}{(+)} = (-)$$

[Verses 30–34 are historically momentous. Brahmagupta is the first author in world history to state general rules for arithmetic operations with signed numbers and zero. Note that $\frac{0}{0} = 0$ is the only questionable rule from a modern perspective; Brahmagupta also stated (elsewhere) that division by zero leaves the fraction unchanged, i.e. $\frac{a}{0} = \frac{a}{0}$ as an unreduced entity.]

vided by two, is the *saṅkramaṇa* (transposition); or divided by the sum of the difference and divided by two —

$$y = \frac{S - D}{2}$$

*t-form), and algebraic identities for powers of unknowns.
chunk.]*

akta), reversed, the difference of *rūpas* (absolute terms) to be subtracted; the *rūpas* are placed below that.

d rearrange:]

$$\begin{array}{r} -b \\ \hline -c \end{array}$$

are of the middle (*madhya*) — the square-root of that, multiplier (*varga-dvi-guṇa-uddhṛtam*) — is the *madhya* (un-

atic formula. If $ax^2 + bx = c$:]

$$\frac{+b^2 - b}{2a}$$

$= \frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$ for the equation $ax^2 + bx - c = 0$.
t side, so his formula uses $+4ac$.

bined with the square of half the unknown — whatever the square — that is the unknown.

$$\frac{+ \left(\frac{b}{2}\right)^2 - \frac{b}{2}}{a}$$

y the kuṭṭaka and quadratic methods.]

h one added, divided by twelve, multiplied by four and when this happens, tell the day-count (*ahargaṇa*).”
in the ahargaṇa (elapsed days).]

decreased by two, divided by three, increased by seven, is happens, tell the *yuga-gata* (elapsed in the *yuga*).”

decreased by one, divided by six, increased by three — equals the *avama-śeṣa* divided by five. When this happens, tell the *yuga-gata*.”

tracted the other unknowns, divided by the first measured, reversed — the *kuṭṭaka* (is applied) for many

all but one unknown, then apply the kuṭṭaka.]

revolutions, from that with its residue added, combined with the day-count — he who tells this is a *kuṭṭaka*-knower.

subtracted revolutions, from that with its residue subtracted, day-count” — he who tells this is a *kuṭṭaka*-knower.

and, thus with the elapsed intercalary months and days, he who tells this is a *kuṭṭaka*-knower.

and computational rules for calendar calculations.]

y the *rūpa*, together with the product of the unknown
desired quotient: if less, add more; if more, add less;

o unknowns, i.e. the term Bxy in the general quadratic

ees, having subtracted the signs and degrees multiplied
(*sūrya*), the calculator (does this) up to a year.

noun — having made that amount desired, the sum of
pas (known quantities).

the *bhāvitaka* becomes the desired unknown-number.
al-equation; what was done from that?

e desired, from the multiplier multiplied, increased or
ier-product; together with the last product — made last.

the guṇaka (multiplier) and k is the kṣepa (additive).]

: finding integers x, y such that

$$k = y^2$$

and k (the *kṣepa*) is an integer. Brahmagupta gave the *cakravāla*, which were rediscovered in Europe only by

) plus one is the first *prakṣepa*; equal to the product of roots (*mūle*) are in the *prakṣepas*, *rūpas*.

y_1) solves $Nx^2 + k_1 = y^2$ and (x_2, y_2) solves $Nx^2 + k_2 = k_2 = y^2$.]

esired *kṣepya* and *śodhya*-roots — having done, the end
n or desired.

square of the last root (*antya-pada-kṛtiḥ*), decreased by
e last root. The square of the last root, decreased by one
the first root.

$x^2 + 1 = y^2$ from known solutions. If (x_1, y_1) is a known

$$\frac{\frac{2}{n} - 3}{2} y_n$$

$$\frac{\frac{2}{n} - 1}{2} x_n$$

st root, increased by three and one, the product halved,
sed by one, multiplied by the last — the product of the

e *kṣepa*, by some (number) divided, added, subtracted,
halved — the first (root). The last root — the other, divided by the multiplier-root, (gives) the first.

[Given $Nx^2 + k = y^2$, divide by k to reduce the additive. If k divides N or a related quantity, new roots can be found.]

V.70. When the multiplier is a perfect square (*varga-cinne guṇake*): the first (root), divided by its root, becomes (the new root). When the *kṣepa* is a perfect square: multiplied by its root, then (are) the roots.

[If $N = a^2$ is a perfect square, then $a^2x^2 + k = y^2$ gives $y = \sqrt{a^2x^2 + k}$. If also $k = b^2$, then $\sqrt{(ax)^2 + b^2}$ may simplify.]

V.71. The sum of the multipliers, multiplied by eight, divided by the square of the difference of the multipliers — the quantity. The two multipliers, tripled, reversed, increased, divided by the difference — (are) the roots.

V.72. The square, decreased by the sum of the other squares, divided by the half-square of the sum-difference — those multiplied by them, combined and separated — squares; in the product, combined with the *rūpa*.

[The general composition (*bhāvanā*) formula:]

$$\begin{aligned} &\text{If } Nx_1^2 + k_1 = y_1^2 \text{ and } Nx_2^2 + k_2 = y_2^2, \\ &\text{then } N(x_1y_2 \pm x_2y_1)^2 + k_1k_2 = (Nx_1x_2 \pm y_1y_2)^2 \end{aligned}$$

When $k_1 = k_2 = 1$, this generates infinitely many solutions to $Nx^2 + 1 = y^2$ from a single fundamental solution. This is Brahmagupta's great discovery, called the *samāsa-bhāvanā*.

he planetary residues at a particular day, one can com-
calendar-makers (*pañcāṅga-kāras*) to verify their com-

tes, multiplied by twenty-two (*dvinavati*), or multiplied
ay, making the square — the calculator (does this) up to

śa), decreased and multiplied by five, likewise by ten —
es this) up to a year.”

, multiplied by three, combined with nine hundred — or
(does this) up to a year.”

., multiplied by four, combined with sixty-five — or the
to a year.”

eased by twenty-two, multiplied by eighty-three, com-
oes this) up to a year.”

month residue), multiplied by thirteen, combined with
en (*tri-ghana*) — the calculator (does this) up to a year.”

d by seventeen, also multiplied by thirteen — separately,
or (does this) up to a year.”

day residue), multiplied by twelve, combined with one

hundred — or the square decreased by three — the calculator (does this) up to a year.”

V.83. “On a weekday, the residue of the sun’s minutes (*kala*), combined with or decreased by the residue of Jupiter-day seconds — the square and product with one — the calculator (does this) up to a year.”

V.84. “The residue of seconds (*vikalā-śeṣa*), combined with ninety-three (*trinavatī*) and decreased by seventy-seven (*saptaṣaṣṭi*) — on the sun’s weekday, making the square — the calculator (does this) up to a year.”

V.85. “On a weekday, the residue of the sun’s minutes, combined with twelve and sixty-three — or decreased by sixty and eight — the calculator (does this) up to a year.”

V.86. “From the residue of the moon’s seconds, the residue of the sun’s seconds, or the residue of degrees — or alternatively the mean (position) desired — the calculator (does this) up to a year.”

V.87. “From the residue of seconds of Venus (*jīva*), Mars (*kuja*) and the Moon; from the residue of Mars’s seconds (*bhauma-liptikā*); from the residue of the moon’s degrees — the Sun — the calculator (does this) up to a year.”

[Verses 75–87 give a remarkable set of “divination” rules. Given a particular planetary residue at a known weekday, these formulas allow one to compute other planetary positions. The constants (22, 83, 5, 10, etc.) encode specific planetary period relations. These verification rules were essential for ensuring the accuracy of pañcāṅga (calendar) computations.]

[Verses 88–99 treat the *kuṭṭaka* method for solving multiple planetary congruences and refined calendar computations.]

V.100. These problems strike the heart (with joy). One may pose thousands of other problems; given by others — thus one should solve those problems by the stated methods (*karaṇaiḥ*).

V.101. In the assembly of people, among the knowers of fate (*daiva-vidām*), it destroys the radiance (of rivals), like the sun (destroys) the stars — (even) one *kuṭṭaka* problem recited — what to speak of hundreds!

V.102. These problems have been stated for each rule, in the illustrative rules (*sa-uddeśakeṣu sūtreṣu*). By three hundred and three (303) *āryā* (verses), the *kuṭṭaka* and the eighteenth chapter.

[Thus ends the Eighteenth Chapter, the *Kuṭṭaka*, in the **Brāhmasphuṭasiddhānta**.]

Chapter XIX: On Gnomon and Shadow Knowledge

This chapter contains 20 verses on gnomon (*śaṅku*) and shadow (*chāyā*) computations. These techniques allow the determination of latitude, building heights, water depth, and altitude by mirror reflection.

[Printed pages 284–293.]

V.1. Having seen the half-day ghaṭikās, he who knows the sun-knowing (method), knows the degrees of latitude (*akṣa-aṃśakān*). By the ghaṭikās of the interval from rising — from the known, the knowable — he is a knower of the *tantra*.

V.2. By the ghaṭikās of the interval from setting, he who brings the knowable from the known, thence brings the mean motion and the *yuga*-revolutions — he is a knower of the *tantra*.

V.3. He who brings the measures of the darkness-sun, the moon-horn, from the height of the lamp-flame — in the knowledge of the ground between gnomon-base (and object), the shadow — he is a knower of the *tantra*.

V.4. He who knows the desired house-height, knows that interval, is observed in water — shows the top of the house-wall, or in a mirror — he is a knower of the *tantra*.

V.5. By the knowledge of the second-shadow-light-edge interval, he knows the lamp-height — or a knower of gnomon and shadow knows the earth's shadow — he is a knower of the *tantra*.

V.6. Having seen the house-base-interval of the net of light and shadow, having seen the top of the house — a knower of the ground — he knows the house-height, having seen it in oil or water — he is a knower of the *tantra*.

V.7. Having observed the top of the house in water, having spread out the water again, in self-ground-knowledge — he brings from the water the ground or the height of the house — he is a knower of the *tantra*.

V.8. With known shadow-men (measures), the interval of water and wall having been known — in the wall, he who knows the ascent of the sun’s light (ray) — he is a knower of the *tantra*.

V.9. The half-day ghaṭikās of the desired day, the prāṇas of the interval of fifteen ghaṭikās — by the day’s motion prāṇas, by those he should compute the latitude as before.

V.10. Of the known and knowable planets, by the nāḍikās of the interval from rising — greater or less — by the risings, the known from the known, the knowable on the east-west (is) to be known.

V.11. The known, with its half-duration, by the risings — greater or less by the nāḍikās of the interval from setting — from the known, on the east-west the knowable; when the half-duration is decreased, the knowable.

V.12. Having made the known into the mean, again on another day, that interval is the motion (*bhuktiḥ*). By the rule of three (*trairāśikena*), by the motion — the derivation of the planet’s circle in the *kalpa*.

V.13. The reverse of half the duration (*sthiti*) is the true darkness-measure in an eclipse. From the measure-rise of the sun and moon, by the ghaṭikā-part, from the terrestrial rising.

V.14. The interval of lamp-base and gnomon-base, multiplied by the desired-quantity-gnomon, divided by the lamp-flame-height minus the gnomon — the quotient is the shadow (*chāyā*).

[Similar triangles: If h = lamp height, g = gnomon height, d = distance between them, then shadow $s = \frac{d \cdot g}{h - g}$.]

V.15. The shadow, multiplied by the gnomon-interval, divided by the shadow-interval — (is) the ground (distance). The shadow together with the gnomon, multiplied by the gnomon, divided by the shadow — (is) the lamp-height.

V.16. Having known the gnomon and shadow, from proportion (*anupātāt*) one should compute the elevations (*samucchrayān*) — of houses, temples, trees, and mountains — or having known the height, the shadow.

[This is the standard “rule of three” for similar triangles: $\frac{\text{height}}{\text{shadow}} = \frac{\text{known height}}{\text{known shadow}}$, so unknown quantities can be found by proportion.]

[Verses 17–19 give additional rules on shadow computations and heights/depths using water and mirror reflec-

tions. Verse 20 gives the chapter colophon.]

[Thus ends the Nineteenth Chapter on Gnomon and Shadow Knowledge.]

Chapter XX: On Prosody (Chandaś-citty-uttara)

This chapter contains 19 verses on Sanskrit prosody (*chandaś-sāstra*), dealing with combinatorial calculations for metrical patterns — the number of possible arrangements of long (*guru*, ‘G’) and short (*laghu*, ‘L’) syllables. Brahmagupta here presents an early form of combinatorics.

[Printed pages 294–303.]

V.1. The *rg*-group (combinations), the permutations (*paryāya*), the aggregate-sums, in the unequal pairs — the *soma* etc., obtained as before, the quadruples, the remainder-additions — the last.

V.2. Decreased by one etc., the first and last, their reversals — in squares etc., of the odd-even (syllables), one should increase the quarters by direct and reverse order.

V.3. One by one, two by two, with additional appendages in those positions — beginning from the square up to the desired — the spread-out (*prastāra*) becomes with barley-middle (*yava-madhya*, i.e. Pascal’s triangle).

[The “Meru of Prosody” (*prastāra*) is Pascal’s triangle: the number of metrical patterns with n syllables and k guru syllables is $\binom{n}{k}$. The “barley-middle” refers to the shape of the triangular arrangement.]

V.4. With one, the last — the two-foot beginning, the three-foot etc., below, separately the numbers. That is to be subtracted, decreased by one, separately from that — combined with one above.

V.5. Up to the quarter, decreasing by one, among the syllables, then increasing by one — in the product beginning from one, the final number of squares etc.

[For n syllables, each of which can be G or L, the total number of possible patterns is 2^n .]

V.6. With one added to the quarter-half — in the uneven (syllables) above, in the even (syllables) at the quarter-half. From the half, doubled, decreased by one — the *yulāni* (pairings) below that, for all.

V.7. By the middle (terms), thus decreased by half, by the quarter-steps in order — equal to the reversed,

beginning from the quarter — in the uneven, subtracting one in the middle, from that one should do (the computation).

V.8. With one-plus-equal-to-the-step-beginnings, the setting is excessively subtracted below — the sum of numbers such as it is — the first, decreased by three — in the lost (*naṣṭa*, i.e. finding a missing pattern).

V.9. Made with middle terms and halved — in the equal-number (case), in direct and reverse order — the *kṣepa* (additive). In the uneven, decreased by one — the half, in order, in reverse order, with one added.

V.10. In the equal-number (case), by the staircase in direct and reverse order — thus also in the uneven — to be constructed. When cooked (computed) is seen, the first — the remaining syllables at the end.

[Verses 11–18 continue the treatment of combinatorial prosody, including algorithms for the “Meru of Prosody” (Pascal’s triangle) and rules for lost (naṣṭa) and spread-out (uddhiṣṭa) metrical patterns. The “staircase” (sopāna) is a procedure for listing all possible metrical patterns in systematic order.]

*[Thus ends the Twentieth Chapter on Prosody in the **Brāhmasphuṭasiddhānta**.]*

which gives Brahmagupta's method for computing the segments" (*jyā-khaṇḍa*) are the tabular sine-differences. a, more accurate than the linear interpolation used by

tains 70 verses. This chunk includes the Jyā-prakaraṇa

a sign ($3^\circ 45'$ intervals) in direct and reverse order, one in halves are the sine-chords (*jyā-ardhāni*).

45' each, giving 24 intervals in a quadrant (0° to 90°). The

the sine-segments (*jyā-khaṇḍa*). Reversed from the last, together with these.

$\sin(i + 1)\alpha - R \sin i\alpha$ where $\alpha = 3^\circ 45'$.]

squared, multiplied by one, two, and three: the roots are are derived) from these.

$$/2 = R \sin(30^\circ)$$

$$/\sqrt{2} = R \sin(45^\circ)$$

$$\sqrt{3}/2 = R \sin(60^\circ)$$

of squares of equal direct and reverse sines and equal re-root of the remainder is the first (sine-chord).

segment-difference, the next sine is:]

$$\frac{+ j_{n-k})^2 + (2\delta)^2}{4}$$

V.21. Its half-segments, decreased by those equal to the earth (*jina*) — the second in production.

[This verse contains numerical values encoded as words (*saṅketa* system). “*Jina*” = 1 (Lord of the earth). The second-order correction refines the first-order segments.]

V.22. Thus the small or numerous sine-segments — the first segments are the sine-halves of one-sixth, one-fourth, and one-third of the circle’s circumference.

[The reference sines at key angles:]

$$R \sin(30^\circ) = R/2 = 1635$$

$$R \sin(45^\circ) = R/\sqrt{2} \approx 2312$$

$$R \sin(60^\circ) = R\sqrt{3}/2 \approx 2832$$

V.23. From the diameter multiplied by the equal reverse-segment, or from the quarter-part, having done the stated segment operations — the derivation of sine-halves is not difficult from this.

[Summary of the method: Given the tabular differences (*khaṇḍa*), one can compute all sines by second-order interpolation using only multiplication and square roots.]

[0.5cm][Brahmagupta’s sine table uses $R = 3270$ minutes. The twenty-four *jyā-khaṇḍa* (sine-differences) are: 225, 224, 222, 219, 215, 210, 205, 199, 191, 183, 174, 164, 154, 143, 131, 119, 106, 93, 79, 65, 51, 37, 22, 7. The second-order correction formula accounts for the curvature of the sine function, yielding values accurate to within 1 minute of arc.]

struments (*yantra*). It describes the bow instrument instruments, wheel instruments, and gnomon instruments, azimuths, and for timekeeping.

ument for measuring altitudes, resembling a bow.

ng angles and time.

ter clocks and mercury devices.

eamasurements.

ighting vanes for measuring celestial positions.

ment, and their use in measuring celestial altitudes and instruments.]

n light, the *kṣepya* (object aimed at) — equal, thus in easure from the *śaka* epoch (78 CE) or from the *mudda*

uely in the string (*sūtraka*), in small (measures) — having *kā* (water-clock) flows, falls thus.

ments for time measurement and planetary observation.]

V.49. A wheel of light-wood, with equal bearing intervals, of lotus-shape — having faith, filled with mercury in the center, well-polished, having performed the saṃdhyā (preparation).

V.50. Obliquely in the middle of the co-altitude, the support above — a beautiful semi-circle (*rūpā-radam*) moves. The small wheel with holes — thus constantly in the air it rotates.

[Description of a self-rotating wheel instrument, possibly an early form of anemometer or a water-driven astronomical device. The “holes” regulate the flow of water or air.]

[Verses 51–53: Additional rules on wheel instruments and their construction. The chapter concludes with verses praising the value of observation and proper instruments.]

V.53. The *karaṇa-jyā* (computed sine), the unsteady motion itself — the arrow, the rising, the ascension — by the instrument with confidence, two-fold — from the five-fold of Ari (the sun’s) motion.

*[Thus ends the Twenty-second Chapter on Instruments in the **Brāhmasphuṭasiddhānta**.
End of the section on observation.]*

dings from four manuscript sources, designated by the

darkar Oriental Research Institute)
 al Asiatic Society)
 ntal Research Institute)
 V.V.R.I.)

)” to indicate that manuscript
 “

r Diophantine equations $ax + c = by + d$.

$\div$ with positive, negative, and zero.

$\bar{a}$) for generating solutions to $Nx^2 + 1 = y^2$.

c .

ith $R = 3270$.

- **Combinatorics** — 2^n patterns; Pascal's triangle (*meru-prastāra*).

This chunk contains the mathematical heart of the *Brāhmasphuṭasiddhānta*. The Kuttakādhyāya is especially notable — it contains the first complete description of the solution to Pell's equation in world mathematical literature.

End of Chunk 5 (Pages 561–700)

The next chunk (pages 701–838) will contain
the remaining chapters (23–24) and the concluding material of Volume I.

EVANAGARI)

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ter) — pp. 701–739

23

rse

749

pp. 752–838

anetary motion

surement Chapter

āhmasphuṭasiddhānta treats of the measurement of time
s. This chapter, like others in the work, is composed in
es for calendrical computation, planetary positions, and

ānta

e

~~~~~

, the lunar months and *tithis*, the civil days, days, months,  
odiacal signs—these we call measures (*māna*).”

*rāṇḍa* — the solar  
*nāni divasāni* — civil days; (4) *dinamāsābdāni* —  
lanets, nakṣatras, and zodiacal signs—these are called

lar, lunar, stellar, and civil reckoning, and the computation  
ner of computation is in use in the world.”

measures; *ebhi* *ṛthak* — separately  
rent in the world.

*a-year*, abundant and deficient (days), fire (3), subtraction  
, *karāṇa*, intercalary month, and the changes of zodiacal

*gavarṣaṃ vipulābdayanāni* —  
ee; *udviddhi-hānayaḥ* — addition and subtraction—from  
unar; *rāśiparvikriyāḥ* — changes of zodiacal signs and

of performing the *pramāṇa* (measurement), the planetary

cs), the civil measure (is useful), as well as the bed (auspicious moments).”

*asūtak* for the construction of the pañcāṅga auspicious moment for lying down; *prāyaścittakriyāḥ* —

~~~~~

5–23)

impossible without names (*saṃjñā*) given to each object, (en).”

at Brahmagupta here gives a systematic nomenclature the twenty-four items are enumerated in the following

, beginning from that, or of the sun, of the Yāmya (south), (associated), at the head, from half a day, from two, from

been measured by Pulisha, Romaka, Vāraha, and Yavana construct a proper *siddhānta*.”

ryasya pramāṇam of the sun; treatise.

different, (yet) Bhāskara (the sun) has the same modification, those having the same (result) for the previous (method).”

) the five (methods): *pakṣāgati*, *madhya*, *gati*, *uttara*, etc.;), *caturtha* (fourth proportional), and the sphere (*gola*).”

~~~~~

“By Brahmagupta, the son of Jisṇu, the knower of mathematics and the

a (itself), adorned with the beauty of elegant composition.”

*ṭṭa brahm*                      venerable Brahmagu  
(mathematics) and the celestial sphere; *sphuṭasiddhānta*  
of Brahman itself.

uments, conjunction, year-count, diverse multiplication,  
he multitude of operations, (all these) from the diminished

one without method, the *rddhicchid* (augmented cut), the  
declares the year and also the solar year..”

been spoken by me, the remainder of the sphere, thin as  
(resumé verse). This (is) the *saṃjñā-adhyāya* (chapter on

what has not                      n by me; *golāt*  
*ṭhāṃśatulyaṃ* — as minute as one-sixth part of the de-  
*aturviṃśaḥ* — the twenty-fourth chapter on nomencla-

ext numbers this as the 23rd *adhyāya* in the internal  
e final verse calls it the 24th in another reckoning. The  
meration.

~~~~~

d *sāvāna* (civil) days by the solar days; the *asāvāna* (non-
of dull intellect.”

the divine, the arm, the eight (*vasu*), the time—the *utkarṣi*
the ninth (or another).”

) a zodiacal sign; multiplied by Makara (10, i.e. 60), these
s: śīśira-vasanta (spring), grīṣma (summer), varṣā (rains),

śarad (autumn), and hemanta (winter).”

makarā — six seasons mu
a-vasanta-grīṣma-varṣā-śarad-hemantāḥ — the six sea-

~~~~~

e *arha* (worthy), the worshipper and the devotee; by the hypotenuse is to be released; two, the decrease is to be

, diminished and increased by fifteen, are the declination-

he abundant shadow—their product, divided by that and uth (Yāmya); from the sphere, the disc’s declination.”

um and difference of declination and latitude are purified hat is subtracted from the square of the sum—the method

er, multiplied by the prescribed (number) and arrived at, ult) multiplied by the half-day hypotenuse, by that radius

~~~~~

in order the section on numerals and the practical appli-
aka (sixth part), the latitude-decline.”

eter, śaki (potency), *karaṇḍū*, that (is) multiplied by the

“The sun’s hypotenuse, the messengers, by the *trijaya* (triple-conquered),

by the *vyāsāntara* (diameter difference) divided, the remainder; the *tijayāmukhyāḥ* (chief of the three-conquered), by the decrease, the operation, the *hutātma* (fire-souled = 3), the diameter.”

— Verse 10 (alt. 2)

rom the diameter, the evil, from the product, the crow—the diameter
prokṣa (sprinkling), the second middle, thus from the *tithi* (lunar day)
 multiplied and obtained, the darkness, the diameter.”

— Verse 11 (alt.)

The intercalary months, the *avama* (omitted) days, the *sambhavana* (produc-
lokas (resumé verses), the twenty-fourth Measurement Chapter.”

— Verse 12 (alt.)

Thus ends the twenty-third chapter on Measurement
 in the work of Brahmagupta.

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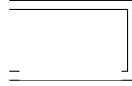
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commentary (*vāsanā-bhāṣya*) portion of the *Brāhmas-*
y pages (numbered 1–126 and beyond).

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	Bhacchramāḥ kudināni ca
	Ravivarṣa-māsa-rāśimāsādayaḥ
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	Praśnarātrau vāra-pravṛtti-pakṣadarśanam
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	Tadārthaṃ deśāntarakarmaṇā pradarśanam
	Varṣadīrgha dinādhyamadinādi-sāadhanam
	Varṣadīrghamāsa-sāadhanam
	Varṣasānayanam ladhvahargaṇānayanam
	Ravisitabudhānāṃ kujaguruśanirīśrekṣṇāṃ cānanayam
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er (pp. 515–567)

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ter

s the first and foundational chapter of the commentary . It treats the computation of mean positions of planets motion of the celestial sphere (*bhacakra*), the definition ployed in Indian astronomy.

760)

ra (celestial sphere) and its motion:

(The celestial sphere), moving eastward (from left to phere, along with the nakṣatras and other celestial bodies e circle of nakṣatras and other luminaries that give light—as (teachers) do not describe their (the planets’) essential (*sthāna*) for the purpose of computation (*spaṣṭādi*).”

e commentator explains that the celestial sphere (*bha-* ere, along with the nakṣatras and other celestial bodies, he *ācāryas* (teachers) do not discuss the nature (*svārūpa*) s (*sthāna*) for the purposes of computation.

~~~~~

## (pp. 760–770)

a) verses:

agupta) declares the commencement of that time whose *kalpa* began from the sun at Laṅkā at the beginning of

*caityādi kalpa-pratipadyāt*  
the first tithi of Caitra (the first month), at Laṅkā, the

number at the beginning of creation, at the beginning

a (civil day) in detail. The number of civil days elapsed 50,000 according to Brahmagupta’s system.

~~~~~

(pp. 770–790)

nalysis:

yuga-epoch method); second, by *parva* (lunisolar method);
ning).”

ng are discussed:

method

d astronomical application.

~~~~~

820)

uting mean planetary positions:

$$\frac{\times \text{ daily motion}}{\text{ys in a kalpa}} \mod 360^\circ$$

g

n of centre) and *śīghraphala* (*śīghra* equation) with de-  
on.

the Sun and Moon: the sine of the *manda-kendra*  
sor.  
e of the *śīghra-kendra*.

~~~~~

(pp. 820–838)

of weekdays (*vāra-pravṛtti*):

es declare the commencement of the week (to be reckoned)
it (is determined) by adding or subtracting the *deśāntara-*

“At Laṅkā, the weekday begins with the first *nāḍī* after e); with the six or eight *nāḍīs* (difference), the weekday is determined when the sun rises. For a place on the eastern or western side of the prime meridian, before or after sunrise (at Laṅkā), by addition or subtraction, the correct time (is found).”

— **Expanded v. 136**

/ **Hindi Commentary (pp. 836–838):**

s at sunrise in the first *nāḍī*. At a place to the we *ghaṭīs*.

At a place to the east, the correct time (is found) before sunrise.

The commentator explains that the *vāra* (weekday) begins at sunrise. By this definition, at a place east of the prime meridian (Laṅkā), the weekday begins after subtracting the longitude difference (in *ghaṭīs*); at a place west of the meridian, (the weekday begins) before sunrise by the same amount (longitude difference in *ghaṭīs*).

In the *Siddhānta-śiromaṇi*, Bhāskarācārya states: “*arkodayād dvābhyām bhyas tāniḥ prācyām praticyām dinap-*” — (the computation proceeds from sunrise with modifications for eastern and western positions).

~~~~~

**Concluding Verses**

(p. 838)

The portion concludes with a verse from the *Manusmṛti*:

“This (universe) existed as darkness, unknowable, without marks, not to be , as if asleep everywhere.”

— **Manusmṛti 1.5**

/ **Commentator’s exposition:**

tator explains that Manu has stated that at the time of the deluge (*pralaya*), the perception of weekdays and other time measures is not possible. Therefore, the reckoning of weekdays is only possible during the manifest (*sṛṣṭi*) period. Brahmagupta establishes that the weekday begins from sunrise at Laṅkā and is modified by the longitude difference (*deśāntara*) for other locations.



## Commentary

*iddhānta* is organized into the following major chapters

|                        | Pages   |
|------------------------|---------|
| putation               | 1–126   |
| ion Computation        | 160–255 |
| ee Questions (Diurnal) | 256–344 |
| nar Eclipse            | 345–400 |
| nar Shadow             | 501–548 |
| ary Conjunction        | 515–567 |
| tellar Shadow          | —       |
| tary Shadow            | —       |
| cal Rising/Setting     | —       |
| tersections            | —       |
| cs (Chs. 12–18)        | —       |
| (Pulverizer)           | —       |
| ry Computations        | —       |

*h methodology, interweaving:*

ical clarity

fs

blems

*The Brāhmasphuṭasiddhānta  
composed by Śrī Brahmaguptācārya*

*The sixth portion of the first part is concluded  
(Pages 701–838)*

---

### Introduction — Madhyamādhikāra

The *Madhyamādhikāra* (Chapter on Mean Motions) is the first chapter of Brahmagupta's *Brāhmasphu.tasiddhānta*. It presents the computation of the mean motions (*madhyama*) of the planets, day computation, longitude correction (*deśāntara*), and calculation of days since epoch (*yugādi-dina*). This first section spans pages 839–893. The second section, the *Spa.s.tādhikāra* (Chapter on True Positions), spans pages 896–978.

### Introduction — Spa.s.tādhikāra

The *Spa.s.tādhikāra* (Chapter on True Positions) is the second chapter. “Since the mean planet is not observed daily in the zodiac, I shall state the procedure for making it true” (BSS II.1). Here the twenty-four sine table, versed sines, equation of centre (*manda-phala*), and equation of conjunction (*śīghra-phala*) are discussed in detail.

---

# ādhikāra

The *Madhyamādhikāra* is the first chapter of the *Brāhmasphuṭasiddhānta* composed by Brahmagupta (b. 598 CE). Its subjects include: the commencement of the weekday (*vāra-pravṛtti*), longitude correction (*deśāntara-karma*), computation of days since epoch (*yugādi-dina-anayana*), the mean motions of the Sun and Moon, and the mean motions of Mercury, Venus, and the other planets. This complete section with commentary (*vāsanābhāṣya*) spans pages 839–893.

y

## Commencement of the Weekday

**[Hindi Commentary]** *Pravṛtti* means that which is spoken by Brahmagupta himself. On the question “from when does the weekday commence?” there are many differing opinions among teachers. Āryabhaṭa and Siṃhācārya hold that the day begins from the half-risen solar disc. Others hold it begins from sunset. Lāṭādeva and others hold it begins from midnight. The Yavana kings hold that ten *muhūrtas* of night [make the division]. In the *Siddhāntaśekhara*, Śrīpati has refuted the opinions of these teachers.

**[BSS I.36]** When at the beginning of creation the universe was filled with darkness, and the planets had not yet been created, there was no lord of the weekday (*vāreśvara*). Therefore the commencement of the weekday is from sunrise (*udayāt pravṛtṭiḥ*).

**[Hindi Commentary — Translation]** Before the beginning of creation, the universe was full of darkness. At such a time, discussion of the weekday's commencement is impossible because the planet-lords of the weekdays did not yet exist. For this reason, the opinion of those who say the weekday commences at midnight is not correct. [Holding that] it begins from midday or from sunset is also not correct. Therefore the opinion of the teachers who hold that the weekday commences at sunrise is indeed correct.

**[Sanskrit Commentary]** In the Sanskrit proof-text (*saṃsk.rtopapatti*) it is known from the verse "The sages state the commencement of the weekday from sunrise at Laṅkā, the city of Rāvaṇa" etc. This is made clear by the verse "From Laṅkā sunrise, south of the prime meridian, east is before, west is after" etc. Thus [ends] verse 36.

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### Rules 37–39: Longitude Difference

**[BSS I.37–39]** The terrestrial circumference (*bhū-paridhi*) is five thousand yojanas (*khakha-khaśara* = 5000). Multiply this by the degrees of difference in latitude (*svāk.sāntarāṃśa*); divide by 360 (*bhaga.nāṃśa*); the square of the result, subtracted from the square of the longitudinal difference (*deśāntara*) [in yojanas], gives [the square of] the corrected difference. [Verses 38–39] From the root of the remainder, multiplied by the terrestrial circumference and divided [appropriately], [obtain] minutes (*kalā*); to the east of the Ujjayinī north-south line is positive (*dhana*), to the west is negative (*.rna*). Or from the root divided by the terrestrial circumference and multiplied by sixty, the result is in *gha.tikās*; this is to be applied to the mean or true planet as positive or negative like [the treatment of] a planet.

**[Sanskrit Commentary]** “Terrestrial circumference” (*bhū-paridhi*) means the circumference of the Earth-globe. In answer to “how much?” he says *khakha-khaśara* = five thousand yojanas. Here the longitudinal difference, latitude, etc., have been explained previously by means of measurements and divisions.

**[Hindi Commentary — Translation]** The terrestrial circumference is five thousand. Multiply this by the latitudinal difference between one’s own place and the standard meridian; divide by 360; the square of that result, subtracted from the square of the longitudinal difference [in yojanas]; from the square root of the remainder, multiplying by the planet’s motion and dividing by the terrestrial circumference, one obtains a result in minutes. East of the Ujjayinī north-south line [apply as] positive to the mean or true planet; west as negative. The root multiplied by sixty and divided by the terrestrial circumference gives the result in *gha.tikās*; this is to be applied to the *tithi*-freed *gha.tīs* (i.e., time correction) like a planet.

**[Sanskrit Upapatti]** If the diameter measure is 1581, then from the square of the diameter, multiplied by eight, [taking] the square root gives the terrestrial circumference — so it is stated in the *Sūryasiddhānta*. By this rule the terrestrial circumference measure of 5000 is obtained, which is Brahmagupta’s opinion.

**[Upapatti — Translation]** If we assume the diameter measure to be 1581, then by the method stated in the *Sūryasiddhānta*, “the square root of eight times the square of the diameter is the terrestrial circumference,” the terrestrial circumference measure of 5000 is obtained. This same [value] is the terrestrial circumference in Brahmagupta’s system.

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### 1.3 Computation of Days Since Epoch

#### Rule 40: Days Elapsed Since Kalpa

**[BSS I.40]** The years elapsed since the beginning of the *kalpa* (*kalpa-gatābdā*), multiplied by 2486 (*rūpā.s.tajina*) and by 7736 (*apavarni-saptanaga*), and divided by 8600 (*kha-svara-sanava*), yield the days and the omitted [tithi]s; the remainders are the fractional parts and defects.

**[Sanskrit Commentary — Vāsanābhā.sya]** At the desired end of a solar year, the years elapsed of the *kalpa* are taken here. The years elapsed of the *kalpa* having been multiplied — he says. By *rūpā.s.tajina* = 2486, and elsewhere by *navārni-saptanaga* = 7736, and then divided by *kha-svara-sanava* = 8600; the results are respectively the days and omitted [lunar days]; the remainders are the fractional parts in both places. The years elapsed of the *kalpa*, multiplied by 2486 and divided by 8600 — the result is days. There, whatever is the remainder, that is called the *dināṃśa* (fractional part of a day) by the word *dābda*.

**[Hindi Commentary — Translation]** In the desired end of a solar year, the years elapsed of the *kalpa* that are here taken. The years elapsed of the *kalpa*, multiplied by 2486 and divided by 8600 — the result is days and omitted [days]; the remainders are in both places. The years elapsed of the *kalpa*, multiplied by 2486 and divided by 8600, the result is days. There, whatever remainder remains, that is called the *dināṃśa* (fractional part of a day).

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## 1.4 Mean Position of the Sun and Moon

### Rules 24–26: Daily Motion and Latitude

[BSS I.24] The minutes of the *manda-phala* (equation of centre), multiplied by the divisor and divided [appropriately], give in order the daily-motion results for the Sun and Moon. In the southern [hemisphere] it is positive, in the northern negative — this is called the *dhruva* (constant). Thus here the staff-head [correction] is reversed; if previously the *vighora-tana-phala* was small, then here [it becomes] solar [correction]. Let astronomers examine the *bhuja* (sine) once, [that of] the delight of all good people; let them examine the *bhū.sa* (ornament) [of calculation].

[Sanskrit Commentary — *Vāsanābhā.sya*] How many revolutions (*bhaga.namāna*) of Prahlāda and the others in a *yuga*, how many elapsed days (*gata-āharga.na*), the computations of weekdays and so on, the computations of mean planets and so forth — in these and other subjects, the first chapter on mean motion (*madhyagati-nāmaka adhyāya*) was composed by me (Brahmagupta) with sixty-*āryā* [verses] (*ṣaṣṭi-pramitā chandasā*); how many subjects there are in the chapter named *Madhyagati*, an enumeration of them has been made. Thus [ends the commentary].

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## 1.5 Mean Position of Mars and Other Planets

### Rule: Mean Planets Computation

**[BSS — General Rule]** The *manda-phala* (equation of centre) is [obtained] from the *bhuja-jyā* (sine of the anomaly) of the *manda-kendra* and the *pāta-jīvikā* [divisor]; divided by the *ko.ti-jyā* (cosine) and multiplied by the *vyāsārdha* (radius), [it becomes] the *phala-ma.n.dala* (resultant circle/epicycle).

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a

### Chapter Conclusion

**[End of Chapter 1]** Thus the first chapter, the *Madhyamādhikāra* (Chapter on Mean Motions), in the *Brāhmasphu.tasiddhānta* [of Brahmagupta].

**[Hindi Commentary — Translation]** The revolutions of the planets and others in a *yuga*, the elapsed *aharga.na*, the computations of weekdays and so forth, and the computations of mean planets and so on — in these and other subjects, the first chapter named *Madhyagati* was composed by means of sixty-three *āryā* [verses]; that is, in the first chapter named *Madhyagati*, however many subjects there are, an enumeration of them has been made. Thus [ends the commentary on Chapter 1].

*[End of the Madhyamādhikāra. This first section, pages 839–893, has been completely explained. It covers mean motions of all planets, day computation, deśāntara (longitude correction), and yugādi calculations.]*

# kāra

The *Spa.s.tādhikāra* (Chapter on True Positions) is the second chapter of the *Brāhmasphu.tasiddhānta*. Here the purpose of *spa.s.tikara.na* (making true), the twenty-four sine table (*jyā-pañcaviṁśatikā*), versed sines (*utkrama-jyā*), *manda-phala* (equation of centre), *śighra-phala* (equation of conjunction), and true daily motion (*spa.s.ta-bhukti*) are presented. This section spans pages 896–978.

## Rule 1: The Need for Correction

[BSS II.1] Since the mean planet is not observed in the zodiac (*bhaga.na*) every day equal to [its mean computation], therefore I shall state the procedure for making the mean [planet] true — the *madhyasphu.tikara.na*.

[Sanskrit Commentary — Vāsanābhā.sya] First, the chapter on true motion (*sphu.ta-gaty-adhyāya*) is being explained. There the purpose of the commencement is stated. Since a planet equal to the mean planet is not observed in practice in the zodiac (*bhaga.na*, the nak.satra circle) day by day, therefore I shall state the *sphu.tikara.na* (procedure for making true). In answer to “what is the defect of the mean?” he says: the mean planet is hypothetically located (*parikalpita*) on the *kak.sā-ma.n.dala* (deferent circle). And the real planet is not on the *kak.sā-ma.n.dala*; revolving on the *prati-ma.n.dala* (epicycle) with mean motion, it is observed on the *kak.sā-ma.n.dala* with true motion. Therefore, by whatever computation the planet situated on the *prati-ma.n.dala* may become equal to [the observed position] on the *kak.sā-ma.n.dala*, that *sphu.tikara.na* I shall state. Thus [ends commentary on] verse 1.

**[Hindi Commentary — Translation]** For the reason that the true planet in the *krānti-vṛtta* (ecliptic) is not seen every day equal to the mean planet, for that reason I (Brahmagupta) call that procedure which makes [it] equal to the observed [position] the *sphu.tīkara.na*. The mean planet is hypothetically [placed] on the *kak.sā-vṛtta* (deferent); the planet, revolving with true motion on the *prati-vṛtta* (epicycle), is observed on the *kak.sā-vṛtta*. Therefore, by whatever computation the planet situated on the *prati-vṛtta* becomes equal [to the observed position] on the *kak.sā-vṛtta*, that *sphu.tīkara.na* I state. Thus [ends commentary on] verse 1.

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### Rules 2–9: Sine Table in 24 Steps

#### Introduction to the Jyā-pañcaviṃśatikā

Brahmagupta gives twenty-four *krama-jyās* (sines in sequence), each arc being  $3\frac{3}{4}^\circ (= 225')$ , filling the complete quadrant of  $90^\circ$ . These sines are expressed poetically through *bhūta-saṅkhyās* (word-numerals), displaying Brahmagupta's literary talent alongside his mathematical genius.

#### [BSS II.2–9] The 24 Sines

The twenty-four sines are encoded in *bhūta-saṅkhyā* (word-numeral) verses. Each word corresponds to a numerical value:

**Verse 2:** ṣaḍ-bud-agni (6+14=20) = 214; suta-agni (7+3=10) = 427; rasa-dvādyā (6+12=18) = 638; munīndu (7+1=8) = 846; vasu-candra (8+1=9) = 1051; indu-nava-nava-candra (1+9+9) = 1251; rasa-tithi-yamala (6+15+1) = 1446; ravi-śriyamā (12+1) = 1624.

**Verses 3–9** continue the sequence up to the 24th sine = 3270 (the full radius).

#### Krama-jyā Table (Sines in Sequence)

| No. | Value | Name          |
|-----|-------|---------------|
| 1   | 214   | Prathamā-jyā  |
| 2   | 427   | Dvitiyā-jyā   |
| 3   | 638   | Tṛtiyā-jyā    |
| 4   | 846   | Caturthī-jyā  |
| 5   | 1051  | Pañcamī-jyā   |
| 6   | 1251  | .Sa.s.thī-jyā |
| 7   | 1446  | Saptamī-jyā   |
| 8   | 1624  | A.s.tamī-jyā  |
| 9   | 1817  | Navamī-jyā    |
| 10  | 1991  | Daśamī-jyā    |

| No. | Value     | Name           |
|-----|-----------|----------------|
| 11  | 2145      | Ekādaśī-jyā    |
| 12  | 2296      | Dvādaśī-jyā    |
| 13  | 2419      | Trayodaśī-jyā  |
| 14  | 2594      | Caturdaśī-jyā  |
| 15  | 2719      | Pañcadaśī-jyā  |
| 16  | 2832      | .So.daśī-jyā   |
| 17  | 2933      | Saptadaśī-jyā  |
| 18  | 3021      | A.s.tādaśī-jyā |
| 19  | 3096      | Ekūnaviṃśa-jyā |
| 20  | Viṃśa-jyā |                |

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|



| No. | Value | Name           |
|-----|-------|----------------|
| 21  | 3207  | Ekaviṃśa-jyā   |
| 22  | 3242  | Dvāviṃśa-jyā   |
| 23  | 3263  | Trayaviṃśa-jyā |
| 24  | 3270  | Caturviṃśa-jyā |

**[Sanskrit Commentary]** Their diameter-measure is the radius. “These are the *aghe-jyā*” (sines) — thus. *Sanu-yamalā* = 214; this is the form of the first sine. *Suniya-mavedā* = 427, the second. *Vasu-jvalana-gha.tikāḥ* = 638, the third. *Rasa-k.rta-vasavaḥ* = 846, the fourth. *Śaṣi-pañcha-khendavaḥ* = 1051, the fifth. *Candra-dara-sūryāḥ* = 1251, the sixth.

## Utkrama-jyā (Versed Sine) Values

### Utkrama-jyā Table (Versed Sines)

The *utkrama-jyās* (versed sines) are constructed by the reversed order of the half-chords. In modern trigonometry these are known as “1—” (versed sine = radius – cosine).

| No. | Value | Bhūta-saṅkhyā                    |
|-----|-------|----------------------------------|
| 1   | 7     | samudra-muni-ca (4+0+7)          |
| 2   | 29    | nava-veda-yamā (9+4+8)           |
| 3   | 66    | sunī-yur-āhutāśana (14+3+5)      |
| 4   | 102   | vasu-guṇa-samudra (8+3+4)        |
| 5   | 171   | rūpendriye.śava (1+11+5)         |
| 6   | 224   | rasa-nagartava (6+6+9)           |
| 7   | 292   | candra-śītakara-vasava (1+2+5+8) |
| 8   | 370   | śara-vasu-nandā (5+8+9)          |

| No. | Value | Bhūta-saṅkhyā                      |
|-----|-------|------------------------------------|
| 9   | 481   | sāgara-rudra-śaśāṅka (4+11+1)      |
| 10  | 621   | navāḍārka (9+8+12)                 |
| 11  | 766   | trivī.saya-veda-śaśāṅka (3+6+4+1)  |
| 12  | 931   | pañca-tri-rasendava (5+3+6)        |
| 13  | 1114  | .sa.d-budhi-yamadh.rtaya (6+17+18) |
| 14  | 1296  | śrati-dh.rti-khayamā (15+2+8)      |
| 15  | 1453  | nava-śaśi-yama-pak.sa (9+6+10)     |

| No. | Value | Bhūta-saṅkhyā                       |
|-----|-------|-------------------------------------|
| 16  | 1635  | sāgara-dvijina (4+2+6)              |
| 17  | 1824  | radara-sa-yamalā (12+6+15)          |
| 18  | 2019  | gu.na-veda-vasu-yamā (3+4+8+8)      |
| 19  | 2214  | .sa.t-vi.saya-śūnya-gu.na (6+6+0+3) |
| 20  | 2424  | kha-muniradā (0+7+12)               |
| 21  | 2632  | vyāsāgha (0+8)                      |
| 22  | 2843  | nava-rada-candra (9+12+1)           |
| 23  | 3056  | jināndā (5+8)                       |
| 24  | 3270  | Parama-krānti-jyā (full radius)     |

**[Hindi Commentary — Translation]** The twenty-four half-chords (*ardhajyā*), [named] *muniyamalā*, *muniya-mavedā*, etc., are of twenty-four numbers, as written below. The relation of *vyastā* (reversed, i.e., versed sine) is given after. The twenty-four *ardhajyās* (*krama-jyās*) are: 214, 427, 638, 846, 1051, 1251, 1446, 1624, 1817, 1991, 2145, 2312, 2459, 2594, 2832, 2933, 3021, 3096, 3145(?), 3207, etc., 3263, 3270.



**Rule: Finding Arc from Jyā**

**[BSS — Arc-Jyā Rule]** From the *jyārdha* (half-chord, sine), divided by the *trijyā* (radius), the square root gives [the first approximation], slightly different for odd and even [intervals]; from that the arc is obtained. After that, again [taking] the square root of [the sine] combined with its square [correction], [the more precise arc is obtained].

**[Sanskrit Commentary]** The twenty-four versed sines (*utkrama-jyā*), [named] *muno's.tayamā* etc., are of twenty-four numbers, as written below: 7, rad (29), 62, 111; 174, rau (292), 337, 435, rabha1 (521), 676, 811, rau (985), 1114, 1296, 1453, 1635, 1824, 2019, 2214, 2424, 2632, 2543(?), 3056, rera70 (3270).

**[Hindi Commentary — Translation]** The *jyārdha* (half-chord), divided by the *trijyā* (radius), from whatever root (*pada*) is obtained, slightly differing for odd and even [intervals], that becomes the arc. After that, again, having combined [the correction] with the square of that same root, the next arc is obtained.

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## Rules 41–44: True Motion Computation

### Introduction to Manda-phala and Śīghra-phala

The *manda-phala* (equation of centre) and *śīghra-phala* (equation of conjunction) are the two principal corrections for making the planet true. The *manda-phala* is computed from the *manda-kendra* (anomaly of the epicycle); the *śīghra-phala* from the *śīghra-kendra* (anomaly of conjunction).

### [BSS II.41–44]

**41:** The *bhuja-jyā* (sine of anomaly) of the *manda-kendra*, multiplied by the *pāta-jīvikā* and divided by the *ko.ti-jyā* (cosine), [then] multiplied by the *vyāsārdha* (radius), gives the *manda-phala* (equation of centre) and the motion.

**42:** From that, first the small or second [epicycle], or from the *manda-saṃsk.rta* (mean corrected by manda), from the mean, one should give the *manda-phala*, or again by the *śīghra-bhukti*.

**43:** Taking the *manda-saṃsk.rta*, or again by the *śīghra-bhukti*, one should then compute the *śīghra-phala*, combined by proportion (*trairāśika*).

**44:** Having established the *śīghra-phala*, the true velocities (*sphu.s.t.a-gatikāḥ*) of the planet are to be brought together; then the wise [astronomer] should compute the *vakra-bhukti* (retrograde motion).

**[Sanskrit Commentary]** Multiplied by the *manda-kendra-gati-jyāntara* (*bhogyakha.n.da*, the tabular difference), divided by the *prathama-jīvā* (first sine); whatever result is obtained, multiplied by the *sphu.ta-paridhi* and divided by 365 (*bhaga.nāmśa*), the resulting minutes — in the *sakala-ādi-kendra* etc., [when this is] less or more than its own mean motion, at that *kendra* the true motion of the Sun and Moon is obtained. For Mars and the other planets, the mean motion minus the *manda-gati-phala* is called *manda-sphu.ta-gati* by the teachers.

**[Hindi Commentary — Translation]** The *manda-kendra-gati-jyāntara* (*bhoga-kha.n.da*), multiplied [by the epicycle] and divided by the *prathama-jīvā* (*prathama-jyā*); the resulting quotient, multiplied by the *sphu.ta-paridhi* and divided by 365 *bhaga.nāmśa*, the resulting minutes — in the *sakala-ādi-kendra*, whether less or more than the mean motion, at that *kendra* the true motion of the Sun and Moon is obtained. For Mars and other planets, the mean motion minus the *manda-gati-phala* is the *manda-sphu.ta-gati*. The *śīghrocca-gati* minus its own *manda-sphu.ta-gati* is the *śīghra-kendra-gati* of Mars and other planets.

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## 2.5 The Manda Paridhi (Epicycle) Values

### Rules 20–21: Manda Epicycle Corrections

#### [BSS II.20–21]

**20:** The difference between that [quantity] and the half-diameter of the epicycle (*dyudala-paridhi*), multiplied by the *trijyā* (radius) and by the *nata-jīvā* (sine of altitude), [when the planet is] in decrease (*ūne*) is positive (*dhana*), in increase (*adhike*) is negative (*.rna*); this is the *sphu.ta-paridhi* (true epicycle) of the half-day.

**21:** For the Sun, that is the *manda-paridhi* in the *.rna-manda-phala* or *dhana-manda-phala* at midday. For the Moon, the same, and by those same [quantities] in *gha.tikās*, the Moon also [is corrected] at that same *kapāla* (division); thus the *nata* (altitude) is obtained.

**[Sanskrit Commentary]** Fourteen degrees (*catu-daṁśāṁśāḥ*), in two places, diminished by two-thirds of a degree (*bhāgadwayaṁśena rahitāḥ*) — then in the *.rna* or *dhana manda-phala* of the Sun; at midday (*dina-madhye*) the *manda-paridhi-aṁśa* are obtained. In the *.rna* or *dhana manda-phala*, before the half-day *kapāla*, for the Sun separated by fifteen *gha.tikās*, the half-day *paridhi* measure is to be made successively increased or decreased by two-thirds of a degree. For the Moon, in the *.rna* or *dhana manda-phala*, twelve [degrees] diminished by two *jina-lipata* (?) — that is, in two places, diminished by thirty-two degrees plus twenty-four minutes (*dvātriṁśadaṁśākyacaturviṁśatikalā rahitāḥ*) — then at midday those *manda-paridhi-aṁśa* are obtained.

**[Hindi Commentary — Translation]** From fourteen degrees, in two places, diminishing by one-third of a degree (20 *kalās*), in the *.rna-manda-phala* or *dhana-manda-phala* of the Sun, at midday the *manda-paridhi-aṁśa* is obtained. In the *.rna-manda-phala* or *dhana-manda-phala*, in the *kapāla* before half-day, having subtracted fifteen (14?) *gha.tikās* from the *nata* Sun, in the midday *manda-paridhi* measure, successively one-third of a degree (20 *kalās*) is to be added and subtracted.

## Summary of Manda Paridhi Values

### Manda-paridhi Summary Table

Below is presented a numerical summary of the *manda-paridhi* (epicycle) values. For each planet the *manda-paridhi* varies up to twenty-four degrees.

| Planet          | Manda Paridhi | Special Notes     |
|-----------------|---------------|-------------------|
| Sūrya (Sun)     | 13–14         | Varies by anomaly |
| Chandra (Moon)  | 31–32         | Twice Sun's rate  |
| Bhauma (Mars)   | 73–80         | Both corrections  |
| Budha (Mercury) | 30            | Complex           |
| Guru (Jupiter)  | 32–36         | Both corrections  |
| Śukra (Venus)   | 11–14         | Complex           |
| Śani (Saturn)   | 59–65         | Slow motion       |

## 2.6 True Daily Motion

### Rules: Daily Motion After Correction

**[BSS — True Daily Motion]** The difference between the *śīghra-kendra-sphu.ta-bhukti* and [the mean motion], multiplied by the [tabular difference] and corrected from the stated quotient — this is the *śīghra-gati*; [if negative] it is *vakra-gati* (retrograde motion). Thus [goes] the proof-verse of Sindhavarman.

**[Hindi Commentary — Translation]** The *śīghra-kendra-sphu.ta-bhukti-antara* (*bhoga-kha.n.da*), multiplied [by the epicycle] and divided by the *prathama-jīvā*; the resulting quotient, multiplied by the *sphu.ta-paridhi* and divided by 365 *bhaga.nāṃśa*, the resulting minutes — in the *sakala-ādi-kendra*, whether less or more than the mean motion, at that *kendra* the true motion of the Sun and Moon is obtained.

### Geometric Explanation (Upapatti)

**[Upapatti — Geometric Proof]** By the difference between mean and true, motion becomes threefold. That motion which is different every moment is *spa.s.tā*; another is *madhyā* (mean). The true motion (*sphu.tā gati*) is also twofold by the division into daily and instantaneous: thus the daily *manda-sphu.tā-gati* and the instantaneous *manda-sphu.tā-gati*. The daily *spa.s.tā-gati*, the instantaneous *spa.s.tā-gati* — by the teacher the daily *manda-sphu.ta-gati* and the daily *sphu.ta-gati* are computed.

The *graha-manda-kendra-gati-jyāntara* (*bhoga-kha.n.da*), multiplied by the *prathama-jyā* and divided [by the radius]; whatever quotient is obtained, multiplied by the *sphu.ta-paridhi* and divided by 365 (*bhaga.nāṃśa*), the resulting minutes — in the *sakala-ādi-kendra* (or *kkayādi-kendra*), whether less or more than its own mean motion, at that *kendra* the true motion of the Sun and Moon is obtained.

## 2.7 Special Corrections for Different Planets

### Planet-Specific Corrections

#### Mercury and Venus

**[Sanskrit Commentary]** For Mercury and Venus, the *śīghrocca* (apsis of the *śīghra* epicycle) is [derived] from the Sun; from the seizure of the Sun [it is different], otherwise [it would be the same]. Their *śīghrocca* is connected to the Sun; the *mandocca* (apsis of the *manda* epicycle) is independent — this is the special feature.

**[Hindi Commentary — Translation]** The *śīghrocca* of Mercury and Venus is related to the Sun itself; their *śīghrocca* is Sun-related, the *mandocca* is independent. Therefore for these two planets a special correction procedure is necessary.

#### Mars, Jupiter, and Saturn

**[Sanskrit Commentary]** For Mars and the others, the *manda-sphu.ta-gati* and the *sphu.ta-gati*. From the *manda-sphu.ta-bhukti-antara*, the *śīghra-bhukti* [is obtained]; the *śīghra-phala-ādha-bhogya-jīvā*, multiplied by the *madhya-jivyā* (mean sine), is to be divided. Having brought together the *sphu.ta-bhukti* by the stated quotient; if together with that, by the *manda-phala-ādham-sphu.ta-bhukti-antara* in that same *kendra*, the resulting *bhukti* is to be made positive or negative.

**[Hindi Commentary — Translation]** For Mars and other planets, the *manda-sphu.ta-gati* and *sphu.ta-gati* — from the *manda-sphu.ta-bhukti-antara*, the *śīghra-bhukti*; the *śīghra-phala-adha-bhogya-jīva*, multiplied and divided by the *madhya-jīva*. By the resulting [quotient], the *sphu.ta-bhukti* is to be brought together as stated; if together with that, by the *manda-phala-adha-sphu.ta-bhukti-antara* in that same *kendra*, the resulting *bhukti* is to be made positive or negative.

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### Epilogue to the Spa.s.tādhikāra

**[End of Chapter 2]** Thus the second chapter, the *Spa.s.tādhikāra* (Chapter on True Positions), in the *Brāhmasphu.tasiddhānta*.

**[Hindi Commentary — Translation]** In this manner, the explanation of the *Madhyamādhikāra* (first chapter) and the *Spa.s.tādhikāra* (second chapter) in the *Brāhmasphu.tasiddhānta*, composed by Ācārya Brahmagupta, has been completed. In this, the mean motions of planets, *sphu.tikara.na*, sine mathematics (*jyā-ga.nita*), *manda-phala*, *śīghra-phala*, *deśāntara*, *nata-karma*, and other subjects have been described in detail.

*[The Spa.s.tādhikāra continues with detailed calculations for each planet's manda and śīghra corrections, tables of paridhi values, geometric upapattis (proofs), and special rules for Mercury and Venus across pages 896–978 of the original edition.]*



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## Editorial Preface

This bilingual edition presents the Sanskrit text of Brāhmasphuṭasiddhānta Part II, Chunk 3, with facing English translations. The layout follows the standard format for critical editions of Sanskrit scientific texts:

- **Left column:** Sanskrit text in Devanagari script, as found in the source edition.
- **Right column:** English translation with IAST romanization of technical terms.

The commentary by Sudhākara Dvivedī follows the traditional Indian *pañcāṅga* structure of:

1. **Sūtra** — The original verse in āryā or anuṣṭubh metre.
  2. **Sūtrārtha** — Literal meaning of the verse.
  3. **Hindi Bhāṣya** — Explanation in Hindi.
  4. **Upapatti** — Mathematical proof or demonstration.
-

## ah (Lunar Eclipse Variations)

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of lunar eclipses, following Brahmagupta's exposition  
Paṇḍita Kṛpāludatta Sūnu Sudhākara Dvivedī provides  
onstrations .

---

## ired Eclipse Magnitude)

ished by the desired quantity, multiplied by the daily  
, gives the *bhuja* (sine). The *koṭi* (cosine) is computed  
half-sum of the diameters (*mānaikyārdha*), the *karṇa*  
obscuration (*channa-māna*)."

computing the *iṣṭagrāsa* (desired eclipse magnitude).  
sired quantity, multiplied by the *bhukti* (daily motion)  
(sine). The *koṭi* (cosine) is computed by the rule of *iṣṭeṇu*  
f diameters), the *karṇa* (hypotenuse) being subtracted,

btracting the desired (*iṣṭa*) from the *sthitidala*, multiply-  
ns the *bhuja*. By applying this result to the motions of  
antaneous lunar latitude (*tātkālika candra-śara*), which  
the *karṇa*. From the *mānaikyārdha* (sum of the radii of  
, the remainder is the desired magnitude (*iṣṭagrāsa*)."

**English Translation of Proof:** “After the contact (*sparśī*), for whatever desired number of ghaṭīs the r of ghaṭīs should be subtracted from the *sthityardha*. e difference in motion between the Sun and Moon gives n] in the *iṣṭa*-diminished *sthityardha* ghaṭīs, whatever d by proportion, by applying the motions of the Sun, *ālika candra-vāra*) is computed — that is the *koṭi*. The a and *koṭi*) is the *karṇa*. Subtracting the *karṇa* from the

### (Time from Desired Magnitude)

half-sum of the measure diminished by the *grāsakalā*, de (*tātkālika-vikṣepa*), the square root of the remainder the difference of motions] — [gives a time].” time] from the *pragrahaṇa-sthit yardha*, the time from ubtracted from the *mokṣa-sthit yardha*, [one obtains the

um of the measure [of diameters] diminished by the neous *śara* (latitude), the square root of the remainder ividing by the difference of motions of Sun and Moon] root of the remainder by 60, divided by the difference otions of Sun, Moon, and node by that resulting time, ed. From that, again by the rule ‘*grāsakalona-pramāṇa*’, table. That [time], subtracted from the *pragrahaṇa-sthit haṇa* (from contact) is obtained.”

**English Translation:** “From the square of the *grāsa*-diminished *mānaikyārdha*, having subtracted the remainder — made *tithi*-like iteratively — that is, having divided by the combined difference of motions of Sun on, and node by that resulting time, the instantaneous beginning ‘*grāsakalona-pramāṇa...*’, iteratively (*asakṛt*), from the contact half-duration (*pragrahaṇa-sthit yardha*), in that desired time, the desired *grāsa* is obtained.”

ce of the latitude at that time (*tātkāla*), the computation  
ore, it is proper to determine the accurate time by an  
[follows] the method of Bhāskara: ‘From the square of  
ng subtracted the square of the *vikṣepa...*’ etc.”

### nement of Half-Duration of Totality)

e motions of the Sun, Moon, and node — multiplied by  
h [motions] are applied [to the Sun, Moon, and node]  
It is additive or subtractive.”

the Sun, the motion of the Moon, and the motion of the  
e half-duration of totality *ghaṭis* multiplied separately)  
de in the eclipsed body, both [additive and subtractive]

te by the method of Bhāskara: ‘*sthit yardha-nāḍī-guṇitā*

## raḥ (Solar Eclipse Chapter)

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text) deals with the computation of solar eclipses. The  
and proceeds through the computation of the five  
an

begun. First, the purpose of its commencement is stated

] computed by the *grahabhāskara* [method] previously  
herefore I shall state the computation of *dyuti* (parallax)

thod given by earlier astronomers (like Bhāskara I) for  
gupta finds to be defective for solar eclipse computation.

on (*pūrve kṣitije*), by the degrees of the *grahabhāskara*  
*haśya*) in their own time-degrees. Later, in the western  
distances, [the same] according to number. Therefore, I  
*aharāntara*. That is: in the eastern horizon, on whatever  
[occurs] after a certain number of time-ghaṭis, on that  
ing night. Similarly, in the western horizon, after the  
tain number of time-ghaṭis, on that day in the evening

**English Translation:** “For whatever reason, in solar eclipses, the five *vyās* (*udayaṃvyā*, *madhyāṃvyā*, *ravi-* achers like Bhāskara computed the *prāṇayana* — from computed values) is not obtained. For that reason, so ethod]. The *tithi-anta* [refers to] the relation with the *ā* [to] the sine of the zenith-distance of the tenth degree sional difference (*hagati*), *hrkṣepa* [to] the sine of the . In the *Sūryasiddhānta* too, the solar eclipse *prāṇayana* solar eclipses, separate five *vyās* of the Moon are com- by the two-*vyā* method.”

n) an (Depression)

l to the *viśrīma-lagna* (nonagesimal ascendant), there is or less than that, *lambana* occurs. When its declination- en there is no *avanati* (depression).”

s. When both *lambana* and *avanati* are possible, then d since its rising, its *śaṅku* (gnomon), and the *prāṇas* of

me of computed conjunction (*gaṇitāgata-darśāntikāle*), ted in the *Tripraśna*, the *viśrīma* [nonagesimal] is to be at *viśrīmalagna*, there is no *lambana*. When the Sun is meaning is self-evident. Similarly, when the northern e latitude-sine (*akṣajyā*) of one’s own place, then there *mbana* and *avanati* are possible, then having made the e’s own place — by the method stated in the *Tripraśna*: *kti*, [gives] the middle *udaya*’, etc., the *ghaṭikās* elapsed

putation of True Conjunction Time)



**Translation:** “The *ghaṭikā-miṣa* [of *lambana*] multiplied by five and divided [by some number] yields the node at the computed conjunction time (*gaṇitāgata-*

ve, divided by five [? *pañcajyā*], the result is the *labdha-*  
t the computed conjunction time are to be diminished  
Moon, and node at the computed conjunction time are  
the true Sun, Moon, and node at the true conjunction

oon, and node at the computed conjunction time, all  
computed. When the *lambana* is negative (*ṛṇātmaka*):  
separately multiplied by the *lambana-ghaṭikā-miṣa* of  
ā, what [result]?’ — by the *labdha-kalās*, the Sun, Moon,  
minished. When the *lambana* is positive (*dhanātmaka*),  
the true [quantities] at the true conjunction time. In the  
ed *lambana-nāḍī-miṣa* by five, the resulting minutes...  
a] and subtracted in negative *lambana*’ — by this, what  
nce.”

## raḥ (Moon's Shadow Chapter)

---

ation of the Moon's shadow, which is essential for un-  
falls on the Moon during a lunar eclipse, and its com-  
(middle shadow), and related quantities.

dow is begun. First, the purpose of its commencement

e Moon and the *lagna*, and afterwards, that of the *sapta-*  
*ndu-ccchāyā*) is daily [computed], therefore I shall state

n), the difference between the Moon and the *lagna*, and  
een the *saptalagna* and the Moon, and daily by effort  
known, and from the Moon's shadow that difference is  
(*anayana*), the shadow-computation (*chāyā-anayana*).  
n the diminished-Sun-*lagna* and the *lagna-nāśin* in the  
a is here spoken in the same manner.”

on's shadow is begun. First, its purpose is stated. The  
horizon, and the difference between the *saptalagna* and  
aily, the Moon's shadow is known (first, the knowledge  
's shadow the knowledge of that difference is obtained

### 3.2 Verse 2: Shadow Computation

nce (*ūrdhvakālika-udbala-naṣṭa-sama-sthiti-pāta*) [have semi-diameter (*vyāsa*) of the shadow at the horizon abbreviation (*saṅkṣepaḥ*).”

etation requires the accompanying commentary. The n astronomy.

what is to be done —”

establishing the method.]

ished by the root, or the elapsed [time], having been ed by the own motion (*svabhakti*), that is the desired tity], extracted from the root (*mūloddhṛtā*).”

tion method for computing the rate of change of the hich differences are taken.

## ārah (Planetary Rising and Setting)

---

tation of the rising and setting times of planets. The  
nets become visible (heliacal rising) and invisible  
the Sun.

ing and setting is begun. First, the purpose of its com-

netary computation, the *grahabhāskara* distance is an  
o that there may be agreement — I state the computation

pse chapter's opening, establishing that Brahmagupta's  
ieve agreement (*aiṃya*) between computation and ob-

n, by the *grahabhāskara* distances in their own degrees,  
, by the planetary distances in their own time-degrees,  
isible (*haśya-āhaśya*). Therefore, I shall state that com-  
is: in the eastern horizon, on whatever day after the  
certain number of time-ghaṭīs, on that day the planet  
rly, in the western horizon, after the setting of the Sun,  
haṭīs, on that day in the evening the planet's invisibility

**Translation — Verse 8:** “By iterative computation (*asakṛt-karma*) from the instantaneous (*tātkālika*) planet-rst, at whatever time the *grahodita-ghaṭī* [elapsed ghaṭis t time (*tātkālika lagna*) and the planetary rising-*lagna* f both, by those (computed ghaṭis) moving the planet, s fixed *lagna* again computing the desired ghaṭis — in .”  
*sakṛt-karma*) for refining the computation of a planet’s ion until the computed value stabilizes.

### utation

*du*) are to be computed by time-degrees (*kālāṃśa*) equal *va*) or excess (*adhikātva*), in that interval the *yoga-kāla*

The verse states that the Moon’s rising and setting are

*ṇika*), that is, by twelve time-degrees (*dvādaśa-kālāṃśa*), to be computed. Here, if the computed time-degree sired time-degree (*iṣṭa-kālāṃśa*), then in that interval, *āla* (conjunction time), or *yogavat-kāla* (time like con-  
 eventh verse.”

gits (*aṅka*), that is, by time-degrees equal to twelve, the d time-degree has diminution (*hīnatva*, being smaller) n in that interval, between those two time-degrees, the

## rah (The Brightness of the Moon)

tation of the illuminated portion of the Moon's disk —  
o determine the amount of the Moon's surface that is

on

ght elevation (*śuklonnati*) is begun.”  
he *khacarārdha* (half of the celestial sphere's motion) —  
of the own declination (*sva-krānty-udaya*), [which are]  
having made [the computation] from the setting of the  
otion of the celestial sphere. The verse establishes the  
.

ch are] from the setting of the Sun (*ravyastāt*), [which  
a of the Moon — or the elapsed/deficient ghaṭikās of the  
the Sun, Moon, node, *lagna*, and from the rising of the  
Sun's setting, the Sun, Moon, node, and *lagna*. Having  
the Moon's declination (*candra-krānti*), from the rising,  
on's rising-*lagna* from the *udaya-lagna* by the method  
es] the middle *udaya*', etc. After the Sun's setting, the  
ed ghaṭikās. Before the Sun's setting, the ghaṭikās by  
Similarly, in the eastern horizon, from the Sun's rising,  
re to be computed — this is to be understood from the

**English Translation:** “By iterative computation (*asakṛt-karma*) from the instantaneous planet-*lagna*, the whatever time the planetary elapsed ghaṭīs are to be planetary rising-*lagna* are to be computed. Computing s) moving the planet, again computing its rising-*lagna*, e desired ghaṭīs — in this manner, iteratively (*asakṛt, sthita-kāla*, computation from the *sthita-lagna* (*sapta*-st day of lunar visibility (*candra darśana*) up to the full m the Sun’s setting and the rising time from the Sun’s

ers for computing the bright elevation.]

Sun, Moon, node, *lagna*, and from the rising of the dec- are to be used to compute the Moon’s bright elevation.

after the Sun’s setting, the Moon sets after an additional n rises [after a certain number of ghaṭīs]. On that very (*śva-jy-unnati*) are to be computed. The bright elevation *bārdha*) of the Moon’s bright [portion]. Thus Bhāskara end. In the second and third *pādas*, half the disk is dark s by the teacher (*ācārya*), two *śuklonnatis* are computed. ving considered in his mind that the dark *śuklonnati* is *nnati*. And it is stated: ‘Brahmagupta and others have ird *caraṇas*; that is not approved by me. For neither is

## n) Computation

ion (*madhyagati*) is multiplied by its own shadow (*tac- chāyā*) and by the semi-diameter (*sva-kara*) multiplied by the *tithi* — the results are additive or subtractive. In

equality (*sāmya*), the difference is [taken]; in direction-difference (*dig-anyatva*), [it is] the *avanati* [depression]

the true bright portion (*spaṣṭa-śukla*) of the Moon. The  
is the shadow corresponding to the planet's altitude.

the Sun and Moon, the *kṣepī* [sine of latitude], having  
by the semi-diameter (*vyāsādala*) multiplied by the *tithi*  
multiplied by 15], divided by the *trijyā* [radius = 3438'],  
and Moon are obtained. Or, their middle motion, their  
multiplied by the *tithi*-multiplied own shadow-*karman*,

tation [is made] accurate by the method of Bhāskara:  
e] of the Moon, by the own middle motion-*tithi-aṃśa*-



## natyadhikārah (Moon's Bright Eleva-

dealing with the Moon's bright elevation  
tation of the arc of illumination on the M

n the Moon's bright elevation (*candra-śukla-unnaṭi*)."  
ime-difference (*ūrdhvakālika-udbala-naṣṭa-sama-sthiti*-  
angle], the semi-diameter (*vyāsa*) of the shadow at the  
ere is no abbreviation (*saṅkṣepaḥ*)."

ished by the root, or the elapsed [time], having been  
ed by the own motion (*svabhakti*), that is the desired  
tity], extracted from the root (*mūlodḍhṛtā*)."  
on algorithm for computing the rate of change of the  
llumination over time.

ion of the chapter —"

*bāhu* (base), *koṭi* (perpendicular), *karṇa* (hypotenuse),  
utline), *sūtra* (line), and *kardama* [mud] — by eighteen  
n's bright elevation (*candra-śuklonnaṭi*)."

*a*, *spaṣṭa*, *sita*, *parilekha*, *sūtra*, *kardama*) covered in 18  
hadow region.

e *karṇa*... the *parilekha-sūtra-karṇaḥ*. The meaning is

of] *bāhu*, *koṭi*, *karṇa*, *spaṣṭa*, *sitāṅgula*, *parilekha-sūtra*  
pter named *Candraśuklonnati*.”

uring the illuminated portion of the Moon’s disk. One

---

rnament) which was spoken by the son (*sūnu*) of Mad-  
nqueror (*jiṣṇu*) — having placed that in [my] heart, this

.”  
*nutana-tilaka* composed by Śrī Kṛpāludatta-sūnu Sud-  
*nnatyadhikāraḥ* — [is concluded].”

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**n — in the *Brāhmasphuṭasiddhānta***

## Appendix: Notes on the Text and Commentary

### About th

The *(Brāhmasphuṭasiddhānta, “Correctly Established Doctrine of Brahma”)* is the magnum opus of the mathematician and astronomer **Brahmagupta** (598–668 CE). Composed in 628 CE at Bhinmal (in present-day Rajasthan, India), it consists of 24 chapters containing a total of 1008 verses in an metres.

### Contents of This Chunk (Pages 281–420)

This typeset volume covers the following chapters from Part II of the text:

- 1 — Lunar Eclipse Variations  
e computation of eclipse magnitudes, durations, and the *iṣṭagrāsa* (desired obscuration) method.
- 2 — Solar Eclipse Chapter  
mputation of solar eclipses, including the five *jyā* (sines) required: *udaya jyā*, *madhyajyā*, *ravidayākarna*, *viśiṃśaṅku*, and *hrkṣepa*.
- 3 — Moon’s Shadow Chapter  
putation of the Earth’s shadow as seen on the Moon’s disk.
- 4 — Planetary Rising and Setting  
thods for computing the heliacal rising (*udaya*) and setting (*asta*) of planets.
- 5 — The Brightness of the Moon  
luminated portion of the Moon’s disk.
- 6 — Moon’s Bright Elevation  
r, with 18 verses, covering the arc of illumination.

### The Commentary

The Hindi commentary in this edition was composed by **Paṇḍita Kṛpāludatta Sūnu Sudhākara Dvivedī**. The commentary follow traditional structure of Indian mathematical commentaries:

(*Sūtra*) — The original Sanskrit verse

(*Sūtrārtha*) — Literal meaning of the verse (marke

(*Vivṛti*) — Detailed explanation (marke

(*Hindi Bhāṣya*) — Hindi explanation (marke

(*Upapatti*) — Mathematical proof or demonstration

### Mathematical Notation

Brahmagupta’s work represents one of the high points of Indian mathematics. Key contributions in these chapters include:

- Rules for computing eclipse magnitudes using th (*bhuja*, sine) an (*koṭi*, cosine)
- Th (*karna*) formula:  $karna = \sqrt{bhuja^2 + koṭi^2}$

- Iterative methods ( , *asakṛt*) for refining planetary positions

e conversions

s)

Sanskrit Transliteration (IAST) scheme:

| ri | IAST | Devanagari | IAST |
|----|------|------------|------|
|    | ka   |            | ta   |
|    | kha  |            | tha  |
|    | ga   |            | da   |
|    | gha  |            | dha  |
|    | ṇ    |            | na   |
|    | ca   |            | pa   |
|    | cha  |            | pha  |
|    | ja   |            | ba   |
|    | jha  |            | bha  |
|    | ñ    |            | ma   |
|    | ṭa   |            | ya   |
|    | ṭha  |            | ra   |

2, Chunk 3

## Introduction

This chunk (pages 1259–1398) of Part II of the *Brāhmasphuṭasiddhānta* comprises the fourth textual division of Brahmagupta's encyclopaedic treatise on mathematics and astronomy, containing detailed computational rules for lunar and planetary calculations.

**Contents of this chunk:** (1) Computation of the Moon's true longitude; (2) Rules for operations with zero; (3) The nature of squares (Pell's equation); (4) Planetary equations of centre and anomaly; (5) Index of subjects.

## Chapter on Lunar Calculations

## — Pages 1259–1287

Sū. BHā.—The true longitude of the Moon: having applied the corrections for equation of centre (manda-phala) and evection (śīghra-phala) to the mean longitude, the true longitude is obtained. The mean longitude is calculated from the ahargaṇa (elapsed days since the epoch).

Sū. BHā.—The Moon's equations (manda-phala) are tabulated for each 15° of anomaly. The tabular differences are: 39, 36, 31, 24, 15, 5. These are applied successively to obtain the true longitude.

VI. BHā.—When the anomaly lies between two tabulated values, multiply the residual arc by half the difference of the passed and upcoming tabular differences, divide by 900', and add to half the sum of the two differences.

Sū. BHā.—The śīghra corrections for the Moon are computed from the śīghra anomaly, which is the difference between the Moon's longitude and the Sun's longitude. The maximum śīghra equation is 80' (80 arc-minutes).

Sū. BHā.—To find the lagna (ascendant): multiply the time elapsed since sunrise by the Sun's daily motion, add to the Sun's longitude, and apply the ayanāṃśa correction.

Sū. BHā.—The time of conjunction is found by dividing the difference in longitudes by the difference in daily motions. The result gives the number of days until conjunction.

Sū. BHā.—A prāṇa equals 4 seconds of time. Six prāṇas make a vināḍī, 60 vināḍīs make a nāḍī (ghaṭī), and 60 nāḍīs make a day.

Here end the verses on lunar computation; next follows the section on shadow calculations (chāyā-vyavahāra).

## Mathematical Derivations

## — Pages 1288–1337

PROOF.—Let  $j_1$  and  $j_2$  be two successive tabular sines, with arc difference  $h = 15^\circ$ . For an intermediate arc at distance  $\theta$  from  $j_1$ , the sine is:

$$\sin(\theta) = j_1 + \frac{\theta}{h}(j_2 - j_1) - \frac{\theta}{h}\left(1 - \frac{\theta}{h}\right)\frac{j_2 - j_3}{2}$$

This is Brahmagupta's second-order interpolation formula.

PROOF.—Applying the kutṭaka (pulverizer) to solve  $137x + 15 = 251y$ : divide 251 by 137, quotient 1, remainder 114. Divide 137 by 114, quotient 1, remainder 23. Divide 114 by 23, quotient 4, remainder 22. Divide 23 by 22, quotient 1, remainder 1. Working backwards yields the multiplier and quotient.

Sū. BHā.—The shadow (chāyā) for a gnomon of 12 angulas:

$$\text{Shadow} = \frac{12 \times \cos(\text{zenith distance})}{\sin(\text{altitude})}$$

When the Sun is at the prime vertical, special rules apply using the nara (sine of latitude) and śaṅku (gnomon).

PROOF.—The lagna is found from the oblique ascension. Let  $\phi$  be the latitude,  $\varepsilon$  the obliquity of the ecliptic. For elapsed time  $t$  (in degrees from sunrise), the lagna  $\lambda$  satisfies:

$$\tan(\lambda) = \frac{\sin(\text{R.A.})}{\cos(\text{R.A.}) \cos(\varepsilon) - \tan(\phi) \sin(\varepsilon)}$$

Sū. BHā.—For Mars, with manda anomaly  $\mu$  and epicycle radius  $r = 39$  (where deferent radius  $R = 60$ ):

$$\text{manda-phala} = \arcsin\left(\frac{39 \sin(\mu)}{\sqrt{60^2 + 39^2 + 2 \cdot 60 \cdot 39 \cos(\mu)}}\right)$$

This gives the equation of centre for Mars.

PROOF.—First apply manda correction to obtain manda-sphuṭa:  $\lambda_{\text{ms}} = \bar{\lambda} + \text{manda-phala}$ . Then compute śīghra anomaly from  $\lambda_{\text{ms}}$  and apply śīghra correction:  $\lambda_{\text{true}} = \lambda_{\text{ms}} + \text{śīghra-phala}$ .

Sū. BHā.—The semi-diameter of the Earth's shadow at the Moon's distance:

$$R_{\text{shadow}} = \frac{R_{\text{Earth}} \cdot (D_{\text{Moon}} - D_{\text{Sun}})}{D_{\text{Sun}}}$$

where  $D$  denotes distance. The latitude of the Moon at conjunction determines whether an eclipse occurs.

Sū. BHā.—A planet becomes visible when it is far enough from the Sun. The arc of visibility depends on the planet's brightness: for Venus 10°, for Jupiter 11°, for Mercury 13°, for Saturn 15°, and for Mars 17°.

PROOF.—Rjuvṛtta (straight epicycle) corrections and planetary phase computations. These final derivations summarise the algebraic methods used throughout the *Brāhmasphuṭasiddhānta* for astronomical computations.



## Index of Subjects

## — Page 1338

**Index of Subjects** (page 1338)—A detailed listing of topics covered in this volume with corresponding page references.

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| dina-gaṇana-vidhi-kārthānorpatha-doṣa-kathanam        | 713 |
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| śrīpaiṭhādikathita-yuga-khaṇḍanam                     | 724 |
| pāda-kramādya duṣprakathanam                          | 725 |

## Algebra

## — Pages 1339–1370

Sū. BHā.—The sum of two positive quantities is positive; the sum of two negative quantities is negative. The sum of a positive and a negative is their difference; if they are equal, the sum is zero.

Sū. BHā.—Zero multiplied by any quantity is zero. A positive quantity multiplied by a negative is negative; a negative by a negative is positive. Zero divided by any quantity is zero.

Sū. BHā.—A quantity divided by zero becomes that fraction of which the denominator is zero. This *khacheda* (zero-divisor, or “zero-seed”) was later corrected by Bhāskara II.

Sū. BHā.—To solve  $Nx^2 + 1 = y^2$  (Pell’s equation): first find the least solution  $(x_1, y_1)$  by inspection or by the cakravāla (cyclic) method. From this, infinite solutions can be generated by composition (bhāvanā).

PROOF.—For  $N = 61$ : start with  $a_0 = 8$  (since  $8^2 = 64 \approx 61$ ). The interpolator  $k_0 = 3$ . The cyclic process generates successive approximations:

$$a_1 = \frac{a_0 + m_0}{|k_0|}, \quad k_1 = \frac{m_0^2 - N}{k_0}$$

where  $m_0$  is chosen so that  $a_0 + m_0$  is divisible by  $k_0$  and  $m_0^2$  is close to  $N$ .

Sū. BHā.—The sum of an arithmetic series:

$S_n = \frac{n}{2}(2a + (n-1)d)$ . The sum of squares:

$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ . The sum of cubes:

$\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2$ .

Sū. BHā.—In compound interest, if principal  $P$  grows at rate  $r$  per period for  $n$  periods, the amount is  $A = P(1 + r)^n$ .

Here conclude the verses of the algebra section, with praise of the siddhānta and transition to the planetary motion chapter.

## Planetary Motion

## — Pages 1371–1398

Sū. BHā.—Having computed the mean longitude (madhyama) from the ahargaṇa, apply the manda and śīghra corrections in order to obtain the true longitude (sphuṭa).

**Manda epicycles:** Mars 39, Mercury 22, Jupiter 32, Venus 14, Saturn 60. **Śīghra epicycles:** Mars 73, Mercury 121, Jupiter 71, Venus 132, Saturn 39 (in units where  $R = 60$ ).

| Śīghra Epicycle | Max Equation |
|-----------------|--------------|
| 73              | 11°30'       |
| 121             | 22°          |
| 71              | 5°           |
| 132             | 11°          |
| 39              | 8°           |

2: \*

equations (degrees where  $R = 60$ )

Sū. BHā.—The rising time of a planet is computed from its longitude, the oblique ascension of the ecliptic, and the latitude of the place. The *carajā* (ascensional difference) is applied to find the exact moment of heliacal rising.

Sū. BHā.—Thus have been explained the motions of the planets, the Sun and the Moon, with their corrections. By these methods, the positions of the celestial bodies may be computed for any desired time.

**Colophon.**—This *Brāhmasphuṭasiddhānta* was composed by Brahmagupta, son of Jṣṇughāṭa, in the year 628 of the Śaka era (550 CE), at the age of 30. May it bring clarity to all who study it.

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$$\frac{\frac{1}{2}1\frac{5}{8}\frac{3}{4}}{\frac{3}{4}\frac{5}{8}}$$

---

---

$$324\frac{3}{8}90\frac{1}{2}$$

$$= 324\frac{3}{8}= 90\frac{1}{2}= 11\frac{3}{5}$$


---

---


$$\frac{1}{2} + \frac{1}{3} + \frac{3}{4} + \frac{1}{5}$$

$$2 \times 2 \times 3 = 12$$

$$\begin{aligned} & \frac{1}{2} \times \frac{6}{1} + \frac{1}{3} \times \frac{12}{2} + \frac{3}{4} \times \frac{12}{3} + \frac{1}{5} \times \frac{12}{4} \\ &= \frac{6}{2} + \frac{12}{6} + \frac{12}{12} + \frac{12}{20} = 9 = \end{aligned}$$

---


$$(a + b)^3$$

$$\begin{aligned} &= a + b \\ &= a + b \\ &= (a + b) \times (a + b) = a^2 + 2ab + b^2 \end{aligned}$$

$$\begin{aligned} &(a + k) \\ &(a + k)(a + k)(a + k) = (a + k)^3 \\ &(a^2 + a \cdot k + k^2)(a + k) = (a^2 + 2a \cdot k + k^2)(a + k) \\ &= a^3 + 2a^2 \cdot k + a \cdot k^2 + a^2 \cdot k + 2a \cdot k^2 + k^3 \\ &= a^3 + 3a^2 \cdot k + 3a \cdot k^2 + k^3 = (a + k)^3 \end{aligned}$$

$$\begin{array}{c|c|c|c} \frac{a \times n \times k}{1} & \frac{a \times n(n-1)k}{1 \times 2} & \frac{a \times n(n-1)(n-2)k}{1 \times 2 \times 3} & + \dots \\ \hline n = 1 & n = 2 & 2 & n = 3 \end{array}$$

$$\begin{aligned} n = 2(a + k)^2 &= a^2 + a \times 2 \times k + k^2 \\ n = 3(a + k)^3 &= a^3 + a^2 \times 3 \times k + a \times 3 \times k^2 + k^3 \end{aligned}$$


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## Introduction

### Introduction

This is the final portion of Part II of the *Brāhmasphuṭasiddhānta*. This volume contains the concluding chapters of the *Gaṇitādhyaṃya* (Mathematics Chapter)—the Third Class of Partnerships, Mixed Calculations, the Rule of Three, Exchange of Goods, Operations with Zero, Series, Plane Geometry, Excavations, Grain Heaps, Shadows, and the beginning of the Chapter on Spheres.

Written in the 7th century, Brahmagupta’s treatise comprises 24 chapters. This section covers an extraordinary range of mathematical topics—commercial arithmetic, the algebra of zero, series, geometry, solid geometry, and spherical geometry.

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## Partnerships

### Chapter 1: The Third Class of Partnerships

Page 754—Here begins the discussion of the third class of partnership problems (*tr̥tīya jāti*), which deals with combined rates of work.

*Translation:* “The sum of all [workers’] day-related filling-times is [found]; then by proportion, the time for a single cistern-filling by all [working together] released [at once] is found. Thus has the Ācārya stated what follows.” ||9||

#### Commentary:

In the third class (*tr̥tīya jāti*), *savarāṇa* (reduction to common denominators) is taught. When the numerator standing above is multiplied by the denominator, the third-class reduction is accomplished. The denominator is given to be multiplied by the denominator, and the numerator standing there, being added or subtracted, is multiplied—then the connection of fractions and the removal of fractions is accomplished. This relates to the section on partnership.

#### Worked Example:

One worker fills a tank (*vāpī*) in one day; a second fills the same tank in half a day; a third fills the same tank in a quarter of a day; a fourth fills the same tank in a fifth of a day. If all workers are released at the same time, in how much time will they fill that tank?

**Statement:** 1|2|4|5—According to the rule “multiply the upper and lower [terms],” summing 1, 2, 4, 5 gives 12—so many filling-days are produced by all workers together. Now by proportion: if 12 yields one day, then for one filling, what is obtained by this proportion? Thus the filling-time is determined.

**Demonstration (*upapatti*):**

One worker fills a tank in one day; the second fills it in half a day, so by this proportion his one-day filling-rate is 2. If the third fills the same tank in a quarter of a day, his one-day filling-rate is 4. If the fourth fills it in a fifth of a day, his one-day filling-rate is 5. The sum of all = 12. Then by proportion the filling-time is found.

In the *Līlāvati*, “*maje chidongī ratha tai vimīśrai*” etc.—Bhāskara’s rule is in conformity with the Ācārya’s [Brahmagupta’s] rule.

## Chapter 2: Mixed Calculations

This section deals with *miśra gaṇita* or mixed calculations, including compound interest, alligation, and related commercial arithmetic problems.

### mount

*Translation:* “The first term multiplied by the standard period [and] divided by the other period becomes the second term. The square of the other principal, added to the product of the mixed [amount], [and] the square root [taken]; then, diminished by half of the other, is the standard result.” ||15||

**Commentary:**

The product of the standard period and standard principal, divided by the other period—that result is to be established in two places. The result in the first place is named *bhrā*, the result in the second place = *bhranya*. The square of the other principal, added to the product of the mixed [amount], the square root taken; from it, half of the other being subtracted, the standard result is obtained. ||15||

### und Interest

**Example:**

Five hundred drammas were lent at interest. In four months, whatever [amount] was due was lent again at interest in a second place; after a period of ten months, the total amount became such and such rupees. State the standard result. ||15||

**Statement according to the example:** Standard period = 4, Standard principal = 500, Other period = 10, Mixed amount = 79.

*frac{text{standardperiod} \times \text{standardprincipal}}{\text{otherperiod}}*

*frac{4 \times 500}{10} = 200* (first term)

$\sqrt{(200)^2 + (79)^2} =$   
 $\sqrt{46241}$   
 $\approx 215$

Subtracting 100:  $215 - 100 = 115$  (standard result).  
 ||15||

## and Extensions

*Translation:* “Having performed the mutual transposition of the fruit and the denominators in the multiplication of the rule of many terms, the result [is obtained when] divided by the product of the smaller terms etc.”  
 ||13||

### Commentary:

The meaning of this rule is: when extracting the result from the multiplication of many quantities, by the mutual transposition of the fruit and denominators, and dividing by the product of the smaller quantities etc., the result is obtained. The *phalachchidā* (fruit-denominator) method of mutual transposition is applied in the *bahu-rāśīka* (rule of many terms).

e

### Example:

If in five months the increase (interest) on drammas is 125 rupees, then what is the increase (interest) on 100 rupees?

**Statement:** 5 | 125 | 100

*frac{5 \times 100}{125} = 4* (increase)

*Translation:* “Having performed the mutual transposition of the fruit and denominators, this method is to be known for the rule of seven and others as well.” ||13||

### Commentary:

For the rule of seven and others also, having made the statement according to the example, the result of the rule of seven etc.



ds

*Translation:* “The transposition in exchange of goods is as previously stated; the operations of the eight procedures have been declared.” ||13||

**Commentary:**

“In the first place, the two value-quantities, their transposition is to be done”—thus the four-Veda Ācārya [Brahmagupta]. (Even so, the procedure of exchange of goods etc., as stated by Bhāskara, is in conformity with this.)

That is, in the first place, the transposition (exchange) of the two value-quantities is the procedure of *bhāṇḍa-pratibhāṇḍaka* (exchange of goods).

**Example:**

Ten paṇas with seven paṇas—

**Statement:** 10 paise, 8 paise

## Chapter 5: Operations with Zero

This is one of the most celebrated sections of the *Brāhmasphuṭasiddhānta*. Brahmagupta was the first mathematician in history to lay down systematic rules for arithmetic operations involving zero (*śūnya*), including the much-discussed rule for division by zero.

*Translation:* “A debt multiplied by a debt is a debt; a loss multiplied by a loss is a loss. But a debt [multiplied by a] debt is positive for positives. Those multiplied by zero become zero again in all ways. That which is exactly divisible by zero becomes a *khahara*.”

**Commentary:**

Rules relating to multiplication and division with zero—Multiplying positive and negative yields zero. Multiplying any number by zero gives zero. Dividing zero by zero gives zero. But when a number is divided by zero, it is called *khahara* (an undefined/infinite quantity).

If 6

$div0 = khahara$ , thus.

**In the 18th chapter (Algebra), Brahmagupta elaborates further:**

*Translation:* “In addition, zero; in division also, the previous method. When even zero becomes a multiplier, the quantities become zero again there. When the dividend is zero and the divisor is zero, a *khahara* comes into being. Its modification is produced, but not zero from that.”

### Commentary:

In addition, division, etc., with zero, the previous method applies. When multiplied by zero, quantities become zero. When the dividend is zero and the divisor is zero, a *khahara* quantity is produced. The reason for this modification is that no definite quantity is obtained on division by zero.

Negative quantities [multiplied] by zero—negative quantities are also multiplied by zero. If 6  $div0 = khahara$ , then there is no error in this; rather, it is a special kind of quantity.

### Summary of Brahmagupta’s Rules for Zero

$$\begin{aligned}
 a + 0 &= a \\
 a - 0 &= a \\
 a \\
 times0 &= 0 \\
 0 \\
 times0 &= 0 \\
 0 \\
 diva &= 0 \\
 quad( \\
 textfora \\
 neq0) \\
 a \\
 div0 &= \\
 textrm \\
 textitkhahara \\
 quad( \\
 textanundefined/infinitequantity)
 \end{aligned}$$

### Commentary:

Here Colebrooke Sahib has written answers which are entirely inappropriate. In the demonstration, the praise of negative quantities has been made by the verse written below. In the *Līlāvātī*, “*nye guṇake jāte khaṇ hāraiced*” etc.—in the demonstration of this, no one has stated the characteristics of the busy rule of three! Bhāskarācārya has stated its characteristic.

*Khabhakta*—the answer to the question is of the nature of *khahara*.

## Chapter 6: Progressions and Series

This extensive section treats arithmetic and geometric progressions, sums of series, and related topics.

### ral Numbers

*Translation:* “The number of terms increased by two, multiplied [by the sum], divided by two—the sum of sums has been explained by the wise man [Brahmagupta].” ||19||

#### Commentary:

Starting from one, the sum increased by one of the desired number of terms—multiplying it by the number of terms increased by two, and dividing by three, the measure of the sum of sums is obtained. The sum of natural numbers up to the number of terms is called *sadvalita*. For this the Ācārya has not written a rule, which is not correct, because when the number of terms is large, one would have to labour greatly to know the sum by the operation of addition. For the finding of the sum, the method stated by Bhāskara in the *Līlāvati*—“*naika padaghna padārtha mardakādyāda yuti*” etc.—is very beautiful.

#### Example:

State separately the successive sums of natural numbers up to nine terms, and also their sum of sums.

**Statement:** 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

By Bhāskara’s rule, the successive sums:  
1, 3, 6, 10, 15, 21, 28, 36, 45

By the Ācārya’s rule, the sums of sums:  
1, 4, 10, 20, 35, 56, 84, 120, 165

## 6.2 — Demonstration

### Demonstration:

If terms = 5, then establishing the natural numbers up to the terms in order and reverse order:

$$\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 2 & 1 \end{array}$$

Adding these, in each place 6 = terms +1, this extending over the terms. Therefore:

$$\text{frac} \text{terms}(\text{terms} + 1)2 = \text{sum}$$

$$\text{frac} 5 \text{times} 6 2 = 15 \text{quad} \text{sum}$$

Thus Bhāskara's method for finding the sum is demonstrated.

*Translation:* “The number of terms multiplied by the terms, half the terms [combined with] the addition of the compressing etc. [i.e., 1/3]—the sum of squares has been explained.”

### Commentary:

If  $k = 1 + 4 + 9 + 16 + \dots$  up to [the given] terms, how is the sum of this series to be found?

$k = 1 + 4 + 9 + 16 + \dots$  up to terms

$$\text{displaystyle} = \text{frac} n(n+1)(2n+1)6$$

*Translation:* “The number of terms increased by two, multiplied, divided by two—the sum of the sum of cubes has been explained by the wise one.”

### Example:

If  $k = 1 + 8 + 27 + 64 + 125 + \dots$  up to terms, how is the sum of this series to be found?

**By dividing by nine and multiplying:**

$$\text{displaystyle} \text{left} \text{frac} n(n+1)2 \text{right}^2$$

## 6.5 — Geometric Progression

*Translation:* “Decreased by one, divided by the common ratio decreased by one, etc.—the sum of a geometric progression has been explained.”

### Example:

If  $k = 4 + 44 + 444 + \dots$  up to terms, how is the sum of this series to be found?

By multiplying by nine and dividing:

$$\begin{aligned} k &= \frac{49}{9}(9 + 99 + 999 + \dots) \\ &= \frac{49}{9} \left( (10^1 - 1) + (10^2 - 1) + (10^3 - 1) + \dots \right) \\ &= \frac{49}{9} \left( \frac{10(10^n - 1)}{9} - n \right) \end{aligned}$$

### Commentary:

Similarly:  $5 + 55 + 555 + \dots$  up to terms

$6 + 66 + 666 + \dots$  up to terms

$7 + 77 + 777 + \dots$  up to terms

$8 + 88 + 888 + \dots$  up to terms

$9 + 99 + 999 + \dots$  up to terms

The sum of these series is obtained by the previous method.

## Chapter 7: Plane Geometry

This major section deals with the calculation of areas of various geometric figures—triangles, quadrilaterals, circles—and the relationships between their sides, diagonals, and altitudes.

### celes Trapezium

*Translation:* “The mutual products of the sides and perpendiculars of two fields—among them, the greater is the base, half is the face, and two [come from] the unequal [sides].”

### Commentary:

The diameter of the circumscribed circle of a square field is its diagonal; in an oblong field also it is the same. In the Ācārya’s text, by the word *aviruddha* one should understand an isosceles trapezium (*samlamba*), where both diagonals and both sides are equal. Multiplying the diagonal by the side and dividing by twice the altitude, the semi-diameter of the circumscribed circle of the quadrilateral is obtained. In a scalene quadrilateral in which a circumscribed circle is possible, the semi-diameter is the square root of the product of the side and the opposite side.

## 7.2

## — Finding the Diameter of a Circumscribed Circle

*Translation:* “The transposed side-product of the two sides, the diagonal divided by twice the altitude—the heart [semi-diameter] of that. For a quadrilateral, the semi-square-root of the sum of the products of the sides and opposite sides.” ||26||

**Commentary:**

In a square field, the diagonal itself is clearly the diameter. By transposition, two equal sides are spoken of. The meaning is this: the transposed side-product of the sides, divided by twice the altitude—the result is the heart (semi-diameter) of the circle circumscribed about the quadrilateral. In a scalene quadrilateral where a circumscribed circle is possible, the semi-square-root of the sum of the products of the side and opposite side becomes the heart (semi-diameter).

## er from Triangle

*Translation:* “Now, for knowing the diameter of the circle circumscribed about a triangle, he says: the product of the two sides of the triangle, divided by the altitude [and] multiplied by the diagonal—the heart [is] the semi-square-root of the sum of the products of the side and opposite side.” ||27||

**Commentary:**

In a square field, the diagonal itself is the diameter of the circumscribed circle. An isosceles trapezium is to be understood by transposition. Where the diagonals and sides are equal, the diagonal, multiplied by the side and divided by twice the altitude—the result is the heart, i.e., the semi-diameter of the circle circumscribed about the quadrilateral. In a scalene quadrilateral where a circumscribed circle is possible, the semi-square-root of the sum of the products of the side and opposite side is the heart (semi-diameter). ||27||

## gles

*Translation:* “The sides and perpendiculars of two regular [triangular] fields here, mutually multiplied—among them the greater is the base, half [is] the face, and two [come from] the unequal [sides].”

**Commentary:**

Multiplying the sides and perpendiculars of two regular (*jātya*) triangles mutually in order, the sides of a scalene quadrilateral are produced; among them the greatest is the base, the smallest is the face, and the two middle ones are the [other] sides.

## 7.5

## — Area of a Scalene Quadrilateral

*Translation:* “A scalene quadrilateral field is similar to an isosceles trapezium. Its diagonals [are] the two altitudes [meeting] at the intersection; their mutual product, then half, is the area.”

**Commentary:**

A scalene quadrilateral field is similar (*sadṛśa*) to an isosceles trapezium (*samlamba-vat*). Its diagonals [are] the two altitudes [meeting] at the intersection (*sampāta*) point where they cut each other; then their mutual product, halved, is the area.

*Translation:* “In a right-angled triangle, the hypotenuse is the square root of the sum of the squares of the equal side and base. The product of the sides, multiplied by four [divided by] that hypotenuse, [gives the altitude] in the right-angled [triangle].”

**Example—Right-Angled Triangle:**

$$c^2 = a^2 + b^2 \text{quadtext}(where$$

$c = \text{hypotenuse})$

## eometry)

**Chapter 8: Excavations (Solid Geometry)**

This section deals with the calculation of volumes of excavations, wells, reservoirs, and related solid figures.

**Excavation**

*Translation:* “That regular excavation of which [the area]—the area of the field multiplied by the depth is the volume. One-third of the regular excavation is the volume of a needle-excavation [pyramid/cone].”

**Commentary:**

The area of a regular excavation is found just as the area of a square or isosceles trapezium is found. The area multiplied by the depth (*vedha*) gives the volume of the regular excavation. That is, an excavation in which the length etc. at the mouth are the same at the base is called a regular excavation (*sama-khāta*). Dividing its volume by three gives the volume of a needle-like excavation (*sūcī-khāta*, i.e., pyramid/cone).

## 8.2 — Worked Example

### Example:

A certain tank (*vāpī*) has length thirty cubits, breadth sixty cubits. Inside it are five excavations (*khāta*) with [side] portions of four, five, six, seven, eight [cubits]; the depths of the excavations in order are 8, 7, 7, 3, 2. State the mean section (*samarajju*) measure of those excavations.

**Mean section:** The side-sums at mouth and base [give] 36|35|42|21|16. Dividing the sum of all these by the aggregate 30:

$$\text{textMeansection} = \frac{\text{fractextbreadthtimeslength30}}$$

$$\text{textArea} = 8 \text{ times } 30 = 240$$

Multiplying this area by the mean section, the total excavation volume became 1200.

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### Examples:

A certain pond has length 16 cubits, breadth = 5 cubits, depths 2, 3, 4 cubits; then what is its volume?

**Statement:** 2|3|4 depths; mean of depths =  $\frac{9}{3} = 3$ ; here the number of places = 3.

$$\text{textArea} = 5$$

$$\text{times16} = 80$$

Multiplying by depth: 80

$$\text{times3} = 240 \text{ (excavation volume).}$$

**Second Example:** An excavation of which the length = 12 cubits, depth = 8 cubits, breadth 3|4|5 cubits—state its volume.

Mean breadth =

$$\frac{3 + 4 + 5}{3} = 4$$

$$\text{textArea} = 4$$

$$\text{times12} = 48$$

Multiplying by depth: 48

$$\text{times8} = 384 \text{ (excavation volume).}$$

## Chapter 9: Grain Heaps

This section treats the measurement of grain stored in heaps, which have the form of a cone or portion thereof.



## 9.1

## — Measuring Coarse Grain Heaps

*Translation:* “The circumference of a circular grain heap—a ninth of the circumference is the depth. The circumference multiplied by the depth, [then] a sixtieth of that, is the volume [measure].” ||50||

**Commentary:**

The inner-outer corner circumference, multiplied by two-four [and] a seventh part, having performed the calculation as before, divided by its multiplier, then that calculation becomes [the volume]. That is, the circumference measure of the grain heap touching the base of the wall, multiplied by two, then the result as before, divided by those two, then the volume of that heap becomes [known].

**For Knowing the Measure of Coarse Grain Heaps:**

**Statement:** Coarse grain heap circumference 60, portion 6 = depth.

*textSixtiethofcircumference* =  
 $\frac{60}{10} = 10$ , squared = 100

*textMultipliedbydepth* = 100  
*times*6 = 600 (coarse grain heap volume)

**For Knowing the Measure of Śūka Grain Heaps:**

**Statement:** Circumference 60, a ninth of circumference is depth 5.

*textSixtiethofcircumference* =  
 $\frac{60}{6} = 10$ , squared = 100

*textMultipliedbydepth* = 100  
*times*5 = 500 (śūka grain heap volume)

**For Knowing the Measure of Fine Grain Heaps:**

**Statement:** Fine grain heap circumference 60, depth = 3.

*textSixtiethofcircumference* =  
 $\frac{60}{3} = 20$ , squared = 400

*textMultipliedbydepth* = 100  
*times*3 = 300 (fine grain heap volume)

### 9.3 — Demonstration

#### Demonstration:

By the nature of the grain's position, the grain heap becomes needle-shaped (conical); its base circumference divided by nine etc. becomes its depth—this is established by direct and indirect perception. The area of a circular field is the circumference multiplied by the diameter...

#### Bhāskara's Līlāvati

*Translation:* “On level ground indeed a coarse grain heap stands, with a [known] circumference measure. With the sixty-fourth part [of the circumference], tell me, calculator, how many *khāris* are in it? Also quickly [tell me] separately for *pu* and *śuka* grains.”

## ons

### Chapter 10: Shadows and Gnomons

This section deals with the mathematics of the gnomon's shadow and related calculations.

*Translation:* “The result [is obtained] from the circumference, multiplied by two, half of two [i.e., one], etc. At the inner-outer corner, the circumference multiplied by two-four and one-seventh [i.e., 2  $\frac{1}{7}$ ].” ||51||

#### Commentary:

The inner-outer corner circumference, multiplied by two-four [and] one-seventh, having performed the calculation as before, divided by its multiplier, then that calculation becomes [known]. That is, the circumference measure of the grain heap touching the base of the wall, multiplied by two, then the result as before, divided by those two, then the volume of that heap becomes known.

For a grain heap situated at the inner corner of two walls, the circumference [is] multiplied by four; then having obtained the result as before, dividing it by four, then the volume of that grain heap becomes known.

For a grain heap situated at the outer corner of two walls, the circumference [is] multiplied by one-and-three-sevenths [i.e.,  $\frac{10}{7}$ ]; then having obtained the result as before, dividing that result by one-and-three-sevenths, then the volume of that grain heap becomes known.

# 11 — The Chapter on Spheres

## Chapter 11: The Chapter on Spheres

The final pages of this volume begin the *Gītādhyāya* (Chapter on Spheres), which deals with the geometry of spheres and spherical segments. This is the 21st chapter of the *Brāhmasphuṭasiddhānta*.

### rence

*Translation:* “The result [is obtained] from the circumference, multiplied by one [half] of the expanded diameter-pair [i.e., the diameter], etc.—this is indeed in conformity with what Bhāskara has stated.” ||51||

### Commentary:

Expanded (*vivṛta*) pair-of-diameter-half (*dvisambhāga*) multiplied by one [i.e., the diameter]—the result from the circumference, etc.—this is indeed in conformity with what Bhāskarācārya has stated. ||51||

*Translation:* “The inner-outer corner circumference, multiplied by two-four [and] one-seventh [i.e.,  $\frac{1}{7}$ ] as before; having performed the calculation, divided by its multiplier, then that calculation becomes [known]. That is, the circumference measure of the grain heap touching the base of the wall, multiplied by two; then the result as before, divided by those two, then the volume of that heap becomes known.”

### Commentary:

For a grain heap situated at the inner corner of two walls, the circumference [is] multiplied by four; then having obtained the result as before, dividing it by four, then the volume of that grain heap becomes known. For a grain heap situated at the outer corner of two walls, the circumference [is] multiplied by one-and-three-sevenths; then having obtained the result as before, dividing that result by one-and-three-sevenths, then the volume of that grain heap becomes known.

### Example from Bhāskara’s Līlāvātī:

*Translation:* “The circumference of a heap touching a wall is indeed a hundred. Of one situated at the inner corner, [it is] equal to the number of lunar days [30], O friend. Of one situated at the outer corner, [it is] measured as nine multiplied by five [45]. Tell me quickly their *dhana-hastas* [wealth-measures] separately.”

**Statement:**

Here, the circumference of the first [heap] 30, doubled  
= 30

*times2* = 60.

The inner-corner circumference 15, quadrupled = 15  
*times4* = 60.

The outer-corner circumference 48, multiplied by one-  
and-a-quarter = 48

*times*

*frac54* = 60.

This measure = 6.

From these, the equal result is just so many *khāris* =  
600.

This, divided by its own multiplier, became the separate  
results 300|150|450.

Thus the measure of coarse grain was found.

*Translation:* “When the circumference of a heap of grain  
is measured as thirty-six, what is the calculation [vol-  
ume], if you know, tell me quickly, [given] a rise of four  
cubits [depth].”

**Statement:**

Grain heap circumference 36, depth = 4; then as before,  
the sixtieth of the circumference = 6.

Squared = 36.

Multiplied by depth = 36

*times4* = 144 (volume).

**Concluding Remarks**

This volume concludes the second part of the *Brāh-  
masphuṭasiddhānta*, covering an extraordinary range of  
mathematical topics. The sections included in this por-  
tion demonstrate the depth and sophistication of Brah-  
magupta’s mathematical thought.

1

1. **Commercial Arithmetic:** Partnership problems (*bhāga*), compound interest (*miśra dhana*), barter exchange (*bhāṇḍa-pratibhāṇḍaka*), and the Rule of Three and its extensions.
2. **The Algebra of Zero:** Brahmagupta's revolutionary rules for arithmetic with zero, including the controversial but pioneering treatment of division by zero as yielding the quantity *khahara*.
3. **Series:** Comprehensive treatment of arithmetic and geometric progressions, sums of natural numbers, squares, and cubes.
4. **Plane Geometry:** Areas of triangles, quadrilaterals, and circles, with theorems on cyclic quadrilaterals and circumscribed circles.
5. **Solid Geometry:** Volumes of excavations, wells, and grain heaps.
6. **Spherical Geometry:** The beginning of the chapter on spheres (*gītādhyaṃya*), dealing with the geometry of spherical segments.

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*ds as one of the most influential  
tory of world mathematics."*

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#### Note on Sources:

The present edition is based on the Chowkhamba Sanskrit Series publication, with the extensive Hindi commentary (*Hindī bhāṣā*) of Paṇḍita Sudhākara Dvivedin. Frequent references are made to the *Līlāvatī* of Bhāskarācārya II, whose rules often parallel or expand upon those of Brahmagupta.

The work concludes with the transition into the *Gītādhyaṃya*, the chapter on spherical geometry, which will be continued in the third and final part of this edition.

# Brāhmasphuṭasiddhānta

*of*

Brahmagupta

(c. 598–668 CE)

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## Volume III — Chunk I

Pages 1677–1816

comprising Chapters XII–XXI

(Final Volume: Concluding Chapters & Appendices)

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### BILINGUAL EDITION

Sanskrit (Devanagari) || English Translation with IAST

Edited with Sanskrit Commentary, Hindi Translation,  
Mathematical Proofs and Astronomical Notes

by

**Dr. Ram Swarup Sharma**

Published by  
**Indian Institute of Astronomical and Sanskrit Research**  
New Delhi, India

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## English Translation

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This final volume of the *Brāhmasphuṭasiddhānta* completes the monumental work begun in the first two volumes. Pages 1677–1816 cover the concluding chapters of Brahmagupta’s masterpiece, including the completion of the *Gaṇitādhyāya*, extraction of roots of non-square numbers, time computations, lunar eclipses, equations of centre and anomaly, daily motion, and the final colophons. The text preserves Brahmagupta’s original verses in authentic Sanskrit form, accompanied by detailed commentary and mathematical proofs (*upapatti*).

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## cumference & Shadow

The present section begins with the final verses of the twelfth chapter (*Gaṇitādhyāya*), treating the calculation of circumference (*paridhi*), proportion (*anupapatti*), and the computation of shadows (*chāyā vyavahāra*).

*paridhau kramaśaḥ prāgvat svaguāt bhavet phalam |*  
*śrīpatyuktamidam cācāryairanukrameāvabodhita ||17||*

(BSS 12.37 / Vol. II, p. 891) This verse states that the circumference is obtained by multiplying the diameter successively by the prescribed factors, as taught by previous teachers including Śrīpati. Brahmagupta, Bhāskarācārya, and Līlāvātī have all taught this in their respective works.

*dviveda sacitrabhāge kaniṭhāntu paridhiḥ phalam |*  
*mithyantabodako’śiḥ svaguābhijam tat ||171||*

||171|| The true circumference is that obtained by the rules taught by the two Dvivedis; the circumference otherwise (from crude approximation) is erroneous.

## 1.2 Mathematical Proof

-

*Upapatti* (Proof): The incorrect circumference associated with a quantity of grains is proportional to the true circumference. The inclined-angle quantity of grains is one-quarter of the true circumference; the base-angle quantity is one-sixth. The circumference- triple measure, multiplied successively by the Dvivedi-sacitrabhāga factors, yields the true circumference. The previous penetration result, divided successively by the Dvivedi-sacitrabhāga factors, gives the true result — thus the teacher's statement is proved. ||171||

## 1.3 End of Arithmetic Chapter

Here ends the *Rāśi Vyavahāra* (Arithmetic Operations). Now, under division, the quantities of grains corresponding to the inclined angle and base angle are stated for the derivation of measures.

## 1.4 Shadow Calculations

BSS 12.41–42: First, for finding the time from a given shadow, or the shadow at a given time. The *nara* (sine of latitude) multiplied by the shadow, divided by the *śaṅku* (gnomon), gives the *śuddha* (mean) value; considering the eastern or western direction, this is divided by the *dinagata-doṣaṁsa* (ascensional difference) to obtain the accurate shadow. ||42||

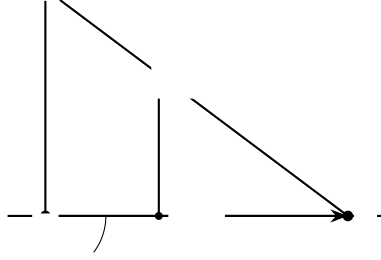
*Sūtra-bhāya*: The *nara* is the *śaṅku* multiplied by eight — the *nara* of the shadow divided by the *śaṅku* is that in the *saṅkrama*; this is the *śuddha*; divided by the eastern or western semi-diurnal arc, it becomes the corrected value.

The *Chāyā Vyavahāra* (shadow computations) section continues with detailed derivations. The *nara* is the sine of latitude, and the *śaṅku* is the gnomon (12 aṅgulas). The shadow is computed from the relationship between these quantities.



## 1.5 Geometric Construction of Shadows

The accompanying diagrams illustrate the geometric relationships: the gnomon (vertical pole) casts a shadow on the ground, forming an angle with the sun's ray. The *karna* (hypotenuse) equals the distance from the lamp's height. Similar triangles yield the proportions.



*Shadow* (Chāyā), *śāna* = base. *Karna* = lamp-height. From similarity of triangles, the *karna*–base proportion gives: base = *chāyā-prāntara*; then by proportional reasoning,  $12 \times \text{base/shadow} = \text{lamp-height}$ . Similarly for the other triangle pair, the same result follows. Thus the teacher's statement is proved. (BSS 12.41–42 commentary)

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## 2 Algebraic Operations —

BSS 12.46: Now (the author) teaches the method of finding the square root of a non-square number. The square of the *sakala* (integer part) is subtracted from the number; the remainder, multiplied by the square of the denominator, becomes the *vikalavarga* (square of the non-square part). This is to be added to the square of the *sakala*, and the result gives the square root approximately. ||46||

(BSS 12.46) Where a quantity has both integer (*kala*) and fractional (*vikala*) parts, the integer part is called *sakala* and the fractional part *vikala*; these are then processed by the rule to obtain the square root.

*Upapatti*: Let the integer part be  $k$  and the *vikala* part  $vi$ . Then the quantity is  $k + \frac{vi}{60}$ . The square is:

$$= k^2 + \frac{2k \cdot vi}{60} + \frac{vi^2}{60^2}$$

*Vṛti-bhāṣya*: When a quantity has both *kala* and *vikala* parts, to find its square, the integer part is called *sakala* and the fractional part *vikala*; then the rule states the method. Note: means ‘square’ in Sanskrit. (BSS 12.46 c - mentary)

*chedeneṭayutone-nārdhaṃ bhājayapa-daṭṭatigunaṃ syāt |*  
*prakṛtiratachchedahṛtaṃ labdhā yutaṃ hainika-*  
*managhaṃ ||47||*

(BSS 12.47) This verse gives the method for division with fractions. The dividend, combined with the divisor (in the manner of adding fractions), is divided by half the divisor multiplied by the denominator of the divisor; the result, added to the *prakṛti*, gives the quotient. (BSS 12.47 commentary)

*idāñim bhāgahāre viropamāha |*  
*chedeneṭayutone-nārdhaṃ bhājayapa-daṭṭatigunaṃ syāt |*  
*prakṛtiratachchedahṛtaṃ labdhā yutaṃ hainika-*  
*managhaṃ ||49||*

(BSS 12.49) The section on division with fractions (*bhāgaḥāra*) is explained. The rule involves the manipulation of fractional quantities through their denominators. (BSS 12.49 commentary)

### 3 Astronomical Problems —

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The following sections deal with astronomical computations, including the determination of time, planetary positions, and the computation of various astronomical quantities.

#### 3.1 Mixed Astronomical Problems

*idāñim madhyān praśnānāha |  
vyatīpātaṃ vaidhṛtibṛhaspativarṣayoścārcanīcari-  
vastīna |  
dviprahayogāṃśa yuge yo veti sa kālatatva-  
jñah ||6||*

**(BSS 13.6)** He who knows the *vyatīpāta*, *vaidhṛti*, *bṛhaspati-varṣa*, and the motion of the nodes in a *yuga* — he is the knower of the essence of time. (BSS 13.6 commentary)

*sāvanamāsāvadhipraharekk'aniṭamadhyasaṃyogāt |  
iṭadvayu saṃyu tonaniṭāñam yo veti grahyakah  
sah ||10||*

**(BSS 13.10)** He who knows the sum of the *avama* (omitted tithis) and the *praghara* (excess days), combined with the mean (longitudes) at the desired time — he is the knower of the *grahyaka* (the quantity to be known). (BSS 13.10 commentary)

#### 3.2 Third Set of Problems

*idāñim tṛtīya praś nasamūha (chāyāḍiparitodivasa  
vāre veṭityasya) uttaramāha |  
kalparkśadinasaptakabadhāt kalpagatāhar-  
gaāñam tacchēat |  
saptahr addinavārah śeṣah sāyyādiko bha-  
vati ||40||*

**(BSS 13.40)** From the product of the elapsed days of the *kalpa* and seven, divided by seven — the remainder gives the day of the week (counting from Sunday).

*Upapatti:* From the *kalpa* elapsed days multiplied by seven, the remainder from division by seven gives the day of the week starting from Sunday. (BSS 13.40 commentary)

### 3.3 Lunar Calculations

yadiṭa grahayugmabhagaāṃ sarobha maga-  
 ādytaṃ labhyate  
 tadā yuḡiyapātamaḡāṃ candrabhagāyogayogi  
 kiṃ phalaṃ  
 bhagādikamiṭahṛt rāśyādimadhyamasapātaca-  
 draṃ yathāgataṃ  
 deśāntaramandaphalāḡin yathāgataṃ sarobha  
 sapātacandro  
 bhavet ||31||

## 4 Planetary Theory — &

### 4.1 Epicyclic Model

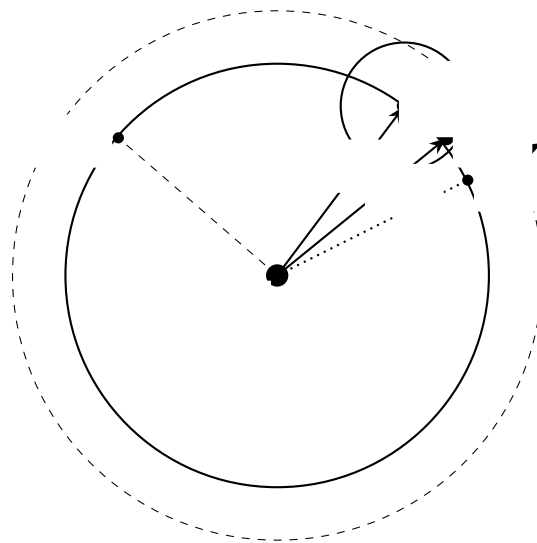
The computation of planetary positions involves the *mandasphuṭa* (equation of centre) and *śighraphala* (equation of anomaly, or second inequality). Brahmagupta gives the following method:

mandasphuṭārkendrāmśonāyām      grahasyandam  
kendram |  
eva yadi tat spaṭam trikoṃ tadāñiṃ viṣeṣam  
nodyet |  
tadyāt spaṭam trikoṃ tasya spaṭam dvitīyam  
kendram  
krameā kendrāntaram phalamiti | vā jñānaman-  
dam spaṭam  
prāpyaccḥissācāryyodinā      śīghra-karttavyam  
kāryamiti sarvaṃ śuddham || 378–581 ||

*Vṛti-bhāṣya*: The *phala* obtained by multiplying by the *bhuja* (sine of anomaly) and dividing by the *karna* — its arc gives the difference between the first and second true positions; that is the true position. (BSS 378–581 commentary)

## 4.2 Proof of Epicyclic Correction

*Upapatti:* *Ja* = geocentric planet; *Ma* = mean planet via manda; *Ma.3* = *śīghrakendra*; *Mar* = *śīghrakendra-jyā* = geocentric sine; *Sp.* = true planet; *Sp.3* = true *kendra*. (BSS *upapatti*)



*Bhum* = geocentric planet. *Spaṭa* = true *kendra-jyā*. *Ma.3. Ma.* = mean *kendra*. *Mū.spaṭa* = *śīghraphala*. From similarity of triangles: geocentric sine/*śīghrakarna* = true *kendra-jyā*. Alternatively, true *kendra* = difference between true planet and apogee. The manda-true *kendra-jyā* meets the epicycle; by arc-correspondence, manda-true *kendra-rāśi* = difference between manda-true and equation. (BSS *upapatti*)

The *mandasphuṭa* (true planet via equation of centre) is computed as:

1. Compute the *mandakendra* (mean anomaly): difference between mean longitude and apogee
2. Find the *mandaphala* (equation of centre) from the sine table
3. Apply to the mean longitude to get *mandasphuṭa*
4. Compute *śīghrakendra* from *mandasphuṭa*
5. Find *śīghraphala* and apply to get the true longitude

## 5 Advanced Astronomical Computations — &

*ītajyāṃ sagūṭitāḥ paścakarmāṃ karmac candram |*  
*ītajyāpādyutthyāsarvaṃ vibhājitā labdham | 25 |*  
*navati chate proha pade navate manorathaṃ śeama-*  
*agakalā |*  
*evaṃ dhanuriṭadayā bhavati jyayā vinā jyārd-*  
*hiḥ | 26 |*

(BSS 21.25–26) The *ītajyā* (desired sine) is multiplied by the results of the five operations and divided by the radius; the remainder, when 90 is subtracted from the *nava* position, gives the *śeṣa*; the *manoratha* (desired arc) is obtained. The arc is thus found from the sine without the *jyārdha* (half-chord).

*Sūtra-bhāya*: The desired sine is divided by the semi-diameter, together with the two-foot arc-difference; the remainder is clear. (BSS 21.25–26 commentary)

*Upapatti*: From previous proofs:  $jy = \frac{(180-ca) \cdot ca \cdot jra}{10125 - \frac{(180-ca)ca}{4}}$ . Let  $y = (180 - ca) \cdot ca$ . Then  $jy = \frac{jy \cdot y}{10125 - y/4}$ . (BSS 21.25–26 *upapatti*)

Let  $y = (180 - ca) \cdot ca$ . By algebraic manipulation:  $jy \times 40500 = 4 \cdot jy \cdot y$ , so  $y = 10125 \cdot jy / (jy + 6)$ . Let  $l = (180 - ca)ca = 180ca - ca^2$ . By equating:  $ca^2 - 180ca + l = 0$ . Therefore  $ca = 90 \pm \sqrt{90^2 - l}$ . Taking the smaller arc as the teacher taught:  $ca = 90 - \sqrt{90^2 - l}$ . Thus all is proved. || 25–26 ||

*Bhogakalā* (daily motion in arc-minutes): For six *bhogakalās*, one = 9 *caga*; for another six, one = 3 *caga*; for fifteen, one = 15 *caga*. Total *yogakalā* = 27 *caga*. Subtracted from *cakrakalā*, the resulting *sarvayogakalā* gives the daily motion = *cak* - 27 *caga*. The *kalpa* elapsed days multiplied by the desired *cakra-kalā* yield the result; in the *kalpa*, <sup>3</sup>the revolutions = *ku* — 27 *kbhaṃ*. The remainder is derived from *mṛta*, *nakṣatra*, and *ayana* differences. || 50–52 ||



(BSS 21.50–52) These verses compute the *bhogakalā* (daily motion in minutes of arc). For the Sun, Moon, and planets, the daily motion is found by taking the difference in longitude between successive days and converting to *kalās*.

*Vṛti-bhāya*: The mean daily motion in *kalās* of the Sun (or Moon), multiplied by the respective factor, gives the *bhogakalās* of the *nakṣātras*. (BSS 21.50–52 commentary)

The *bhogakalā* is the daily motion of a planet in terms of *kalās* (arc-minutes). The computation involves:

- Finding the difference in longitude between successive days
- Converting this difference into minutes (*kalās*)
- For the Sun: about 1° per day (approximately 60 *kalās*)
- For the Moon: about 13° per day

## 6 Final Chapters — Eclipses and Colophons

The concluding portion of the *Brāhmasphuṭasiddhānta* contains extensive rules for eclipse calculations, including the computation of the *sparsa* (first contact) and *mokṣa* (last contact), the magnitude of the eclipse, and the *akṣavalana* (deflection due to latitude).

*mandasphuṭārṅkendrāṁśonāyāṁ grahasyandaṁ  
kendram |  
eva yadi tat spaṭaṁ trikoṁ tadāñiṁ viśeṣaṁ  
nodyet |  
tadyāt spaṭaṁ trikoṁ tasya spaṭaṁ dvitīyaṁ  
kendram  
krameā kendrāntaraṁ phalamiti | vā jñānaman-  
daṁ spaṭaṁ  
prāpyacchissācāryyodinā śīghra-karttavyaṁ  
kāryamiti sarvaṁ śuddham || 578–581 ||*

## Colophons —

*kalāmāyo yāvatā māñāṁ bhogakalāḥ sṛḍhāstāvad  
gaāni |  
śēaḥ kha likha vartamānakātrasya gatakalāstās-  
tadugkalāmaya |  
śuddhā gamya kalā bhavati | tato grahadvayako di-  
vasastadā  
gatagamyakalābhiḥ kimitvanupātena gaā divamā  
bhavati |*

*Here ends the Brāhmasphuṭasiddhānta*

Thus concludes the *Brāhmasphuṭasiddhānta*. This treatise contains 24 chapters in all, comprising 12 chapters of mathematics (*Gaṇitādhyāya*) and 12 chapters of astronomy (*Golādhyāya*). The teacher Brahmagupta composed this work in Śaṃvat 628 (CE 666).

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3 | Bilingual Edition

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**[Introductory Note.]** Now [the Ācārya] speaks of the computation of the motion (bhoga) of the starry nakṣatra and the computation of the nakṣatra as freed from the planets (graha-mukta-nakṣatrānayaṇa).

**[Mathematical Rule.]** The fifteenth [operation]: The sum of the degrees of conjunction-motion (yoga-bhoga-kalā) =  $\text{Caṃ} \times 15$ . The sum of all =  $1 \text{ Caṃ} + 3 \text{ Caṃ} + 15 \text{ Caṃ} = 27 \text{ Caṃ}$  = the total motion of all the nakṣatras (sarvāna-kṣatra-bhogaḥ). Here  $\text{Caṃ}$  (cakra) denotes a complete revolution.

**[Dvivedi's Commentary.]** The pure revolutions (cakkala) are the numbers [representing] the total motion of all the nakṣatras; then the degrees of motion of the starry nakṣatra (śrambijñud bhoga-kalā) [are obtained]; its daily motion = revolution —  $27 \text{ Caṃ}$ . From that, the kalpa-born revolutions [are obtained] as: (revolution —  $27 \text{ Caṃ}$ ) civil days (kudina) = cakkala.

In 27 civil days, the kalpa revolutions [are obtained] by this Ācārya; [this is] well established (tumūpapannam). In the *Siddhāntaśekhara*: “cakkāriyā vā śāsvata-sativadhana-āhatāni śodhayin sudina-laya-dvaroṣa-cakra”. [The meaning is that] the daily motion of the kalpa (kalikā-gatiḥ) which is the Vaivasvata [value] at mid-[period] is to be known (mṛtiḥ). The degrees of nakṣatra motion (nakṣatra-bhoga-kalikā), the times of the prahara, etc., the remaining degrees to be traversed (bhogya-kalikā) — from the excess, the gatagamyā [values] become the days traversed and remaining to be traversed (gata-gamyā-dināni). Thus [it is said] by Śrīpati in the fifteenth [chapter].

“The sum of the motions of all [nakṣatras] multiplied by the revolutions [gives] the Vaivasvata [value]; this is the yoga of the starry [nakṣatra],” and so on — thus spoke Bhāskara; and all teachers [hold] this same meaning.

**[Verses 50–51]**

| *Upapatti* (Proof/Demonstration)

- ×

**[Mathematical Derivation.]** The sum of the six doubled (ṣaṭ-śraddha) motion-degrees of the nakṣatras =  $\frac{3 \text{ Cam}}{2} \times 6 = 3 \text{ Cam} \times 3 = 6 \text{ Cam}$ . The sum of the six [times] single-motion degrees of the nakṣatras =  $\frac{\text{Cam}}{2} \times 6 = \text{Cam} \times 3$ . The sum of the fifteen [times] single motion (sama-bhoga) degrees of the nakṣatras =  $\text{Cam} \times 15$ . The sum of all = total nakṣatra motion degrees =  $6 \text{ Cam} + 3 \text{ Cam} + [15] \text{ Cam} = 27 \text{ Cam}$ .

**[Hindi Bhāṣya — English Summary.]** When the mean daily motion of the Moon (candra-madhyagati-kalā) is multiplied by [certain] quantities, the motion degrees of the six, the śarva, and the equal nakṣatras are obtained. The remaining [degrees], when subtracted from the revolution degrees (bhagaṇa-kalā), give the motion of the starry nakṣatra (śrambijñuta-bhoga).

Alternatively: The kalpa revolutions of the starry nakṣatra are obtained by multiplying the kalpa revolutions of the Moon by eighty-seven and subtracting from the kalpa civil days. From these revolutions, the daily motion of the starry nakṣatra is obtained by proportion (anupāta): if in kalpa civil days such-and-such kalpa revolutions [are obtained], then in one day [what]? — thus the daily motion in time-units.

When the nakṣatra motion degrees are subtracted from the planet-time aggregate (grahakāla-samūha), the nakṣatras are obtained. That is, in the planet-freed degrees (grahamukta-kalā), as many pure motion degrees of the nakṣatras as there are, so many are the traversed nakṣatras (gata-nakṣatra). The remaining degrees are the current degrees of the present nakṣatra; subtracting these from the nakṣatra motion degrees yields the remaining degrees to be traversed (gamyā-kalā). Then, if in one day [a certain value] is obtained in planetary motion degrees, by proportion from the gata-gamyā degrees, the days traversed and days remaining are obtained.

**[Verses 50–52]**

| **Upapatti (Proof).** Here the Ācārya has postulated (kalpita) the previously stated time-quantity (kāla-jñā) as the desired [value] for the second [process]. Then śrada, sama-śāim, kudūdit — these three quantities (muñjā) produce one śramba field (kṣetra); and śrada-jñā, labdha-jñā, tijñā — these three quantities produce a second śramba field. These two śramba fields are homogeneous (sajātiya); therefore, proportion (anupāta) is applied [between them].

$$\frac{\text{labdhajñā} \times \text{iṣṭa}}{\text{tijñā}} = \text{samaśāim-kud}$$

And the time-quantity (kārajñā) = [the result of this proportion]. The three quantities (trayaṃ muñjānām) generate the proportional relationship between the two śramba fields.

| **Atrôpapattiḥ** (Demonstration thereof)

[Śloka on Shadow Computation.] In the shadow-hypotenuse circle (châyā-karṇe gole), how the equinoctial shadow (palabhā) and the gnomon (śaṅku) become equal in nature (tulya-tattvaṃ) — this is demonstrated thus: if the palabhā is obtained as the base (bhuja) in the twelve-aṅgula sign (dvāda-śāṅgula-rāśi), then what is the gnomon-like [quantity]? The gnomon-equivalent (śaṅku-tulya) =  $\frac{\text{pama} \times \text{śaṅku}}{12}$ ; but śaṅku =  $\frac{12 \times \text{tri}}{\text{chāka}}$ . By elevation (utthāpanena), the gnomon-equivalent =  $\frac{\text{pama} \times 12 \times \text{tri}}{12 \times \text{chāka}}$ .

[Dvivedī's Explanation.] Here, for computing the palabhā at a time different from midday (dvāda/dinārka), [the Ācārya] speaks.

[Hindi Bhāṣya — Summary.] At a time different from midday, multiplying the Sun's shadow by the shadow-hypotenuse (châyā-karṇa) and dividing by the tijñā yields the circular shadow of the hypotenuse-circle. The difference between the root of the gnomon (śaṅku-mūla) and the east-west line is the base (bhuja). In the northern hemisphere (uttara-gola), subtracting the northern bhuja from the karṇa-dvitiya and adding the southern bhuja yields the palabhā. In the southern hemisphere (dakṣiṇa-gola), merely subtracting the second shadow from the bhuja yields the palabhā.

[Verses 46–50]

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[Introduction.] Now [the Ācārya] speaks of the questions of the śrambya [type].

[Hindi Bhāṣya — Summary.] Those who know the sama-śaku [from] the declination (krānti), who know the sama-śutta [from] the latitude (lamba-āṃśa), who know the koṇa-śaku, who know the koṇa-śaku-chāyā, and who know the ghaṭī [from] the koṇa-dvitāja entrance — they are the scholars (paṇḍita) of jyotiṣa-siddhānta. Here there are five questions.

[Verse 7]

| *Idānīm anyān praśnān āha* (Now he states other questions.)

*śaṅku-tala-prācyaparāntara-dvayaṃ bodhyaṃ yo vijānāti |  
viśuvacchāyām ekaṃ hastvā 'āsanna-dityaṃ ca gaṇyakaḥ  
saḥ || 7 ||*

He who, having known the two differences of the east-west [line] at the gnomon-base (śaṅku-tala-prācyaparāntara), understands [also] the equinoctial shadow (viśuvac-chāyā) and the near-adjacent sun (āsanna-āditya) — he is a calculator (gaṇyaka).

[Śloka 7]

[Sūtra-bhāṣya.] The difference of the east-west [line] from the gnomon-base (śaṅku-tala-prācyaparāntara) is the base (bhuja). He who, having understood the two bhuja-values, knows the equinoctial shadow (viśuvac-chāyā) [i.e.,] the palabhā, or who knows one bhuja [together with] the equinoctial shadow and the sun — he is a gaṇyaka. Thus here there are three questions.

[Verse 8]

[Vivaraṇa-bhāṣya.] The difference of the east-west line [measured] from the gnomon-base (śaṅku-tala-pūrvāpararekha) is declared to be the bhuja. He who, having known the two bhuja-values, knows the palabhā, or having known one bhuja, knows the palabhā and the Sun (ravi) — he is a gaṇyaka. Thus here there are three questions.

[Verse 7]

[Dvivedī's Introduction.] Now [the Ācārya] speaks of the śrambya questions.

[Hindi Bhāṣya — Summary.] The difference between the gnomon-base (śaṅku-tala) and the east-west line is the bhuja. The person who, having known the two bhuja-values, knows the palabhā, and likewise [he who], having known one bhuja, knows the palabhā and the Sun (ravi) — they are the paṇḍitas of jyotiḥ-śāstra. Here there are three questions.

[Verse 7]

| *Idānīm anyān praśnān āha* (Now he states other questions.)

*pātāla-śaṅkum uddhyeste vā hasta-jñāṃ raviṃ vijānāti |  
iṣṭa-pātāla-gaṇakoḥ pṛthag tale vā sa tantra-jñāḥ || 8 ||*

He who knows the pātāla-śaṅku [and] the uddeśa-śaṅku, or [who knows] the sun having known the hasta; [or who knows] the iṣṭa-pātāla separately, each at its own base (tale) — he is a knower of the tantra (tantra-jñā).

[Śloka 8]

[Sūtra-bhāṣya.] He who knows the pātāla-śaṅku of the Sun [and] the kumatha-śaṅku, or who knows the Sun having known the uddheśa, or [who knows] the iṣṭa-śaṅku [and] the kudivo-iṣṭa-śaṅku [as] the pātāla-gaṇaḥ and pātāla-śaṅku [respectively], the ravi-ratha-śaṅku — of these two, separately, each at its own base (pṛthag tale) or at the gnomon-base (śaṅku-tale), he who knows — he alone is the tantra-jñā. Thus here there are four questions.

[Verse 9]

[Dvivedī's Introduction.] Now [the Ācārya] speaks of the śrambya questions.

[Hindi Bhāṣya — Summary.] The person who knows the Rāhu-path (bhūmā-mārga) in an eclipse, who knows that path, and who knows the koṇa-śaku-chāyā — here there are five questions.

[Verse 7]



**[Eclipse Triangle Computation.]**

*koṇaduta-stha-ravi-gataṃ bhuva-protkuṭaṃ bhuvādri-  
viparyaṃ bhuvyacāpāṃśaṃ ekaṃ bhujaḥ |*

*bhuvālpasatikāvadhi lamba-aṃśa dvitīyo bhujaḥ | koṇadu-  
taṃ va svainikādri-viparyaṃ nanāṃśāstu-tīyo bhujaḥ |*

In this triangle: the Sun's passage at the koṇaduta, the earth's protuberance (protkuṭa), the earth-mountain reversal — the degrees of the earth-curve (bhuvyacāpāṃśa) — these form one bhuja. The latitude-degrees (lamba-aṃśa) up to the earth's alpaka-satika [form] the second bhuja. The koṇaduta and the svainika-mountain reversal [form] the third bhuja.

In this triangle, the Sun's passage being [known] as the earth's protuberance (bhuva-protkuṭa), the angle [is] the natakala (nata-kālaḥ). The southern koṇaduta-nata-amṛtyanna-koṇa = 85; by the rule of proportion (anupātaḥ): if by the natakāla-jyā the hasta-jyā is obtained, then by the arc-jyā of 85 (jyā 85), what is obtained?

$$\text{The bhuja-jyā} = \frac{\text{hasta-jyā} \times \text{jyā } 85}{\text{natakāla-jyā}} = \frac{\text{hasta-jyā} \times \text{tri} \times \text{jyā } 85}{\text{natakāla-jyā} \times \text{tri}} = \frac{\text{hasta-jyā} \times \text{tri} \times 285}{2495 \times \text{natakāla-jyā}}.$$

| *Upapattiḥ* (Proof/Demonstration)

[Dvivedi's Upapatti — Summary.] Here, since the longitude difference (deśāntara) is not known, one should compute the conjunction-time (saṃmilana-kāla) and disjunction-time (unmilana-kāla) of the total eclipse by the method of lunar eclipse, from the true Sun (sphuṭa-ravi) and true Moon (sphuṭa-candra), devoid of deśāntara correction. On that day, the second saṃmilana-kāla should [also] be computed; if it exceeds the computational saṃmilana-kāla, then one should know whether [the place] is situated in the eastern or western part from the second line.

Because to the east of the line, midday (madhyāhna-kāla) occurs first in one's own country and later in the line-country; therefore, the saṃmilana-kāla of one's own country exceeds the saṃmilana-kāla of the line-country. In the west, the opposite occurs. Thus, from the unmilana-kāla, the iṣṭa-prāsa-kāla, the sparśa-kāla, or the mokṣa-kāla should be examined.

*uttara-gole yāmye viṣuvacchāyāyuktas taram hīnam |  
evaṃ viṣuvacchāyā yuktavihīnāstareṣābhā || 50 ||*

In the northern hemisphere (uttara-gola), [the latitude] being joined with the equinoctial shadow (viṣuvac-chāyā) [in the] south (yāmya) is diminished (hīnam). Thus, the equinoctial shadow [is] joined with or diminished from the latitudes (tareṣā) [depending on hemisphere].

[Śloka 50]

[Sūtra-bhāṣya.] The first half of the earlier [section] of the Tri-praśna-adhyāya [is to be] placed as the fourth subsidiary section (upayupardha-saṃ nyasyāt). All else of the Tri-praśna [verses] 58–60, the extent (āyāmi) [is] clear.

[Verses 49–50]

[Vivaraṇa-bhāṣya.] At the half-day time (dinārdha-dina-kāle), the shadow multiplied by the shadow-hypotenuse (chāyā-karṇa-guṇā), divided by the tijñā, then the circular shadow of the hypotenuse-circle (chāyā-karṇa-goliyā) would be [obtained] as the ravi-value. The difference between the root of the gnomon (śaṅku-mūla) and the east-west line is the bhuja. By that bhuja, diminished by the karṇa-dvitiya-prācyā, [or] joined with it — in the northern hemisphere, the equinoctial shadow (viṣuvac-chāyā), i.e., the palabhā, results. In the south (i.e., the southern hemisphere), when that is diminished by the difference from the karṇa-dvitiya-prācyā (bhuja-māna), then the palabhā results. Thus, [the shadow] joined with or diminished from the difference by the bhuja, or diminished by the southern bhuja, then the palabhā results.

[Verses 49–50]

| *Atrôpapattiḥ* (Demonstration thereof)

[Proof of Gnomon-Equivalence.]

How the palabhā in the shadow-hypotenuse circle (chāyā-karṇa-gola) and the gnomon (śaṅku) become equal in nature (tulya-tattvaṃ) is demonstrated thus: if the palabhā is obtained as the bhuja in the twelve-aṅgula sign, then what is the gnomon-equivalent (śaṅku-tulya)?

$$\begin{aligned}\text{śaṅku-tulya} &= \frac{\text{pama} \times \text{śaṅku}}{12}, \quad \text{but } \text{śaṅku} = \frac{12 \times \text{tri}}{\text{chāka}}, \text{ by elevation} \\ \text{śaṅku-tulya} &= \frac{\text{pama} \times 12 \times \text{tri}}{12 \times \text{chāka}} = \text{pama}\end{aligned}$$

Therefore, it is proved (siddhaṃ) that in the karṇa-gola, the gnomon-equivalent (śaṅku-tulya) equals the palabhā. The karṇa-dvitiya-jyā becomes the inverse circle (vyasta-golā). The palabhā is the north-south [quantity]. Thus, by the correction of the chāyā-pravi-parasūna, the bhuja is spoken of; by the addition-subtraction (kṣepa-riyena) with the karṇa-dvitiya-jyā, by the correction of the bhuja, the palabhā results.

In the *Siddhāntaśekhara*: “Now, the difference of the natapūrva-pāścima (east-west natal difference), the śakibuta-jñā-makā. The dakṣiṇōttara-bhujā, joined with and diminished, the Sun dwelling in the north (saumya-vartin) [gives] the palabhā. In the southern [hemisphere], again, at the inanāyām, thus it is always diminished by that difference. Thus, joined with and diminished by the difference, the shadow becomes fixed [in] aṅgula-ścakā.” This [is] said by Śrīpati; [it is] in accordance with the Ācārya alone.

[Verses 49–50]

*valanasya muce taimyaḥ prthag grahak khaṇḍakena | vṛt-  
tiḥ kṛtiḥ sparśavimarśanimadhyagrataḥ krameṇôr bimha-  
vagamayaḥ ||*

Of the deflection (valana), the release (muca), the taimya, separately; by the section of the planet (graha-khaṇḍakena); the vṛtti, the kṛti, the contact and release (sparśa-vimarśa), the centrality of the middle (nimadhyagrataḥ) — in sequence, the understanding of the eclipse-disc (ur-bimha) [is obtained].

By Bhāskarācārya, the very method of Śrīpati [was] propounded (pratipāditaḥ) in the *Viśvadarpaṇa*.

[Verses 14–15]

[Dvivedī's Commentary on Eclipse Drawing.]

The answers to the two questions beginning “viśvadarpa-  
maṇḍala-dikā” etc., have been explained in the eclipse  
chapter (grahaṇādhikāra). The latter half of that [is called]  
śrambe.

Now the answers to the question beginning “sāmparkaṃ  
maṇḍalai” etc. are spoken.

[Hindi Bhāṣya — Summary.] In the prāhlādūta, the arc of  
the instantaneous declination (tātkālīka krānti) is of a sim-  
ple form with slight difference; [this is] the balana-sūtra.  
From the balana-jyā, the divya-jyā should be computed.  
In the māna-cakra-bāhṛta [circle], above the balana-sūtra,  
the contact-sphāra (sparśa-kālīka sāra) and release-sphāra  
(mokṣa-kālīka sāra) of the Sun should be given in the direc-  
tion of their [respective] sāra. In the madhya-balana-sūtra,  
the madhya should be given from the centre of the prāhla-  
disc. From the moon's horns (śarāghra), by the prāhla-  
pramāṇa-vyāsa [diameter], having drawn quickly, the con-  
tact, release, and mid-line prāsa should be computed. And  
by drawing the parilekha [projection] on the earth, the  
contact-release śrīra-prāsa is obtained.

| *Upapattiḥ* (Proof/Demonstration)

[Dvivedī's Proof — Summary.] When in the māna-cakra-bāhṛta circle there is the centre of the prāhlā-kṛtā, then the prāhla-bimbagrānta and prāhla-bimagrānta remain contiguous (saṃlagna). Therefore, in the known māna-cakra-bāhṛta circle, the direction should be fixed (śrīṣṭa). The common endpoint of the sama-maṇḍala and the māna-cakra-bāhṛta circle is the eastern mark (prācī cihna) in the māna-cakra-bāhṛta circle.

From there, giving the balana-jyā, the line from the centre [through] the balana-jyā-pragata is the krānti-tṛt-prācī. One should give the jyāvat-khādana from the balana-sūtra. Because the sāra remains south and north from the krānti-vṛt-prācī. Thus it happens in the sparśa and mokṣa. The madhya-balana [is] in the dakṣiṇōttara direction from the tātkālīka-krānti-vṛt-prācī; therefore, from the centre it should be given in the balana-sūtra. In the śarāghra, the prāhla-vṛtta has its centre; therefore, from there, by the tātkālīka-prāhlābhi-pramāṇa, from the constructed circles, the sparśa-mokṣa-madhyā [is obtained].

In the *Siddhāntaśekhara*, from the verse “mānacakrabāhṛtalaye tatpariyāminivi” etc., written in the Sanskrit proof, Śrīpati has spoken in accordance with the Ācāryas alone. In the *Siddhāntaśiromaṇi*, from “prāhlākhasūtreya vidhāya vṛttaṃ mānacakrabāhṛtaṃ ca sādhitāṃśam” etc., Bhāskarācārya has propounded in the form of deliberation (vicāra-rūpa) the very method of Śrīpati.

[Verses 14–15]

*idānī yaḥ parilikhatīṣṭadyāsamilyādi praśnasyōttarām āha*  
/  
*paścāt pragrahayo prāgamoṣe ravi-vibbalo bāhuḥ |*  
*sva-balana-siddha-jñā-diśi viparītaḥ śītakara-madhyāt || 16*  
//  
*bhānumato bāhu-guṇyākhāyā diśaṃ koṭir atyathā śavinaḥ*  
/  
*ravi-śāsti-vamadhyāt karaṇāstinyakam, karaṇāṅgu-*  
*koṭiyuteḥ || 17 ||*

Now he who draws the answer to the question of iṣṭa-dyā-samilyādi: the bāhu [is] the western and eastern pragraha at the release (moṣa) of the Sun's vibbalo [force]. Opposite in direction to the svabalana-siddha-jyā, from the centre of the Moon (śītakara).

Of the Sun (bhānumat), the direction of the bāhu multiplied (bāhu-guṇyākhāyā diśaṃ); the koṭi [is] otherwise (atyathā) of Saturn (śanaiścara/śavina). From the centre of the Sun's śāsti [force], karaṇa-astinyakam; from the karaṇāṅgu-koṭi-yuti [is obtained the result].

[Śloka 16–17]

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**[Introduction.]** In the Sūrya-siddhānta, the following verbal description of instruments (yantra-vivaraṇa) is given —

*tuṅgârkasamâyuktaṃ gola-yantraṃ prasādhayet |  
gopayeetat-prakāśoktaṃ sarvagamyam bhaved dhi ||*

One should prepare the spherical instrument (gola-yantra) endowed with the exalted Sun (tuṅgârka-samâyuktaṃ); having concealed it, [as] spoken of in the Prakāśa, it becomes indeed universally accessible (sarvagamyam).

*kāla-saṃsāadhanārthaṃ tathā yantrārīṇā sādhyet |  
ekākī yojayed bījaṃ yantraṃ vismaya-kārīṇi ||*

For the purpose of time-determination (kāla-saṃsādhana), one should thus accomplish [it] by the yantrārīṇā; the solitary one should apply the seed (bījaṃ) [to] the instrument, causing wonder.

*śaṅku-kuyaṣṭi dhanuś-cakreṣa-chāyā-yantrair naikathā |  
gupta-deśāṅka-jñeyam kāla-jñānam atandritaḥ ||*

By gnomon-staff (śaṅku-kuyaṣṭi), bow-wheel (dhanu-cakra), and shadow-instruments (chāyā-yantraiḥ) in manifold ways; the knowledge of time (kāla-jñāna) is to be known from the hidden place-marker (gupta-deśāṅka) [by one who is] vigilant.

*toya-yantra-kapālāṅka-gharmmy-udanatravānaraiḥ |  
sa-sūtra-rekhu-gamaiś ca samaya-kālam prasādhayet ||*

By water-instrument (toya-yantra), bowl (kapāla), aṅka of heat (gharma), water-stream (udanatra), and monkey [devices?] (vānaraiḥ), and with cord-line-rekhuga (sa-sūtra-rekhu-gamaiḥ), one should determine the proper time (samaya-kālam).

*paradārā-manuṣyāṇām gulvataila-jalāni ca |  
bijāni pāṃsavas teṣu prayogās te 'pi durlabhāḥ ||*

The wives of others (paradāra) of men, oil (taila), and water — [these are] seeds (bijāni) [in] the sand (pāṃsavaḥ) of these; their applications are also difficult to obtain (durlabhāḥ).

*tā prapañcamadhiśchadd nyastaṃ kuṇḍe smalābhamis |  
dviṭṭha-meja-laya-horātraṃ sphuṭaṃ yantraṃ kapālakam ||*

That covering of the world (prapañcam-adhiśchadd), placed in the bowl (kuṇḍe), smalābhamis [?]; the double-positioned (dviṭṭha) meja-laya day-night (horātra) — the clear instrument [is] the kapālaka.

*na yantraṃ tathā sādhu divā ca vimale ravau |  
chāyā saṃsāadhanaiḥ proktaṃ kāla-sādhanaṃ uttamam ||*

Not the instrument thus [alone is] good; by day, when the Sun is pure (vimale ravau), the shadow [is] declared by the saṃsādhana-methods [to be] the best time-determination (kāla-sādhana).

| *Mānādhyāya* (Chapter on Measures)

*mānāni saura-cāndrārṅkṣa-sāvanāni grahāṇayanam aibhiḥ |  
mānaiḥ prṭhak caturbhiḥ saṃkhyabahāroḍjña lokasya ||*

The measures (mānāni): solar (saura), lunar (cāndra), stellar (nakṣa), and civil (sāvana) — by these, the computation of planets (grahāṇayanam). By these four measures separately, the multitudinous arrangements (saṃkhyabahāroḍjña) of the world [are made].

**[Dvivedī's Hindi Commentary — Summary.]**

From this, the solar measure (saura-māna), lunar measure (cāndra-māna), stellar measure (nākṣatra-māna), and civil measure (sāvana-māna) — these four types of measures are spoken. By these very four measures, all the affairs of people are conducted. Which objects are grasped by which measure — all this is propounded (pratipādita).

Brāhma, divya, paitṛ, prājāpatya, bāhṛspatya, saura, sāvana, cāndra, nākṣa — these are nine measures. Of these measures, in the world of humans, only the saura, cāndra, sāvana, and nākṣa have primacy, because by these measures alone all human affairs are accomplished. In the Sūrya-siddhānta, Siddhāntaśekhara, Siddhāntaśiromaṇi, and all other treatises, the subject of measures is spoken of in the same manner. In this chapter of the Brāhmasphuṭasiddhānta, the methods of viewing the earth (bhūma-darśana) are also given.



[Introduction.] In the Smṛtis and in the Purāṇas, the doctrine of Rāhu-grasta eclipse (rāhugrasta-grahaṇa) is present (pratipādana). Therefore, reconciling the two doctrines, the Ācārya has said —

*rāhus tac chadayati praviśati yac chuklapacchadayante |  
chāyā tamasi-dvairaghra-dānām kamala-yoniḥ ||  
candro 'smuyojaḥ syo yadinaimaya-bhāskarasya māsāt |  
chādayati śarmi-tatāpo rāhuś chadayati tat savituḥ ||*

Rāhu covers that [Moon], entering [it] which the bright and dark [portions] cover. The shadow (chāyā) [is] the tamasi [force] of the enemy of the lotus-born [i.e., Rāhu, enemy of Viṣṇu?].

The Moon (candra), the asmu-yojaḥ, the syo [?], from the māsa of the sun-of-the-yadinaimaya [?] — [this] Rāhu covers the scorching-heat (śarmi-tatāpaḥ) [of] that Sun (savituḥ).

In the Golādhyāya of the Siddhāntaśiromaṇi also, the unspoken (apokṛta) statement of Bhāskara [is as follows] —

*dvi-deśa-kāla-varayāṇi medānām chādako rāhur iti sa  
varṇita |  
yan māninaḥ kevala-golavidhās tat-sahitā vedy  
araṇābāhyām ||*

Rāhu, the eclipser (chādakaḥ) of the vara of the two--place-time (dvi-deśa-kāla-varayāṇi medānām), thus he is described. Those who know only the spherical method (kevala-golavidhāḥ), together with that [knowledge], should know the araṇā-bāhyām [?].

*rāhuḥ kṣubhā-maṇḍalagaḥ śaśāṅkaḥ śaśāṅkaś chādayatīva  
bimbam |*

*tamo-mayaḥ śamyuva-pradānāt sarvagamānām avirud-  
dham etat ||*

Rāhu, situated in the agitated orb (kṣubhā-maṇḍalagaḥ), [eclipses] the Moon (śaśāṅkaḥ); the Moon appears to cover the disc (bimbam). [Rāhu is] composed of darkness (tamo-mayaḥ); from the bestowal of śamyuva [?], this is not contradictory (aviruddham) for all who go [i.e., all observers].

[Dvivedī's Commentary.] This reconciliation (samanvaya) has been made. In the Siddhāntaśekhara, for the refutation (khaṇḍanārtha) of the Rāhu-grasta eclipse, a chapter named “Rāhu-nirākaraṇādhyāya” has been placed. In this, Śrīpati also has made the reconciliation with the following śloka —

*viṣṇu-lūna-viśirasaḥ kila pañcad ettavān varam imaṃ  
parameṣṭhī |  
hema-dāna-vidhinā tavatupatis tam-arati-mahas tad  
uparāge ||*

The Supreme Lord (parameṣṭhī) indeed gave this excellent boon (varam imaṃ) to the headless Viṣṇu-lūna [i.e., the decapitated Rāhu], [that] by the method of donating gold (hema-dāna-vidhinā), at that eclipse (tad uparāge), [there is] the great darkness (tam-arati-mahaḥ) [for him?].

*bhūmecchāyā praviṣṭaḥ sthagayati śaśinaṃ śuklapakṣā-  
vasāne |*

*rāhur bhu-prasādāt samavigatavastu-tāmo vyasutulyaḥ ||*  
Entering the earth's shadow (bhūmecchāyā praviṣṭaḥ), [Rāhu] obscures the Moon (sthagayati śaśinaṃ) at the end of the bright fortnight (śuklapakṣāvasāne). Rāhu, by the grace of the earth (bhu-prasādāt), [being] equal to the vyasu [?], the darkness [is] the real state of things (samavigatavastu-tāmaḥ).

*urvyāṃ sthaṃ bhānu-bimbaṃ salila-maya-tanor  
yathaivōktaṃ bimbaṃ |*

*saṃsūyaivaṃ ca māsavy uparati-samaye svasya sāhitya-  
hetoḥ ||*

Standing on the earth (urvyāṃ sthaṃ), the solar disc (bhānu-bimbaṃ), [this] disc as described [earlier], of the water-bodied one [i.e., the Moon] — thus indeed at the time of lunar cessation (māsavy-uparati-samaye), for the sake of its conjunction (svasya sāhitya-hetoḥ).

**[Dvivedī's Commentary on Spherical Construction.]**

In the Golabandhâdhikāra, having constructed the great circles (mahad-vṛttāḥ: pūrvâpara-vṛtta, yāmyôttara-vṛtta, sthitija-vṛtta, etc.) and the small circles (laghu-vṛttāḥ: the outer royal circles of the twelve signs from Meṣa, etc.), the Ācārya has demonstrated the nature (svarūpa) of the maximum equation (paramalambana-nati) and computed the knowledge of the circles (vṛttôpāyanam).

As the Ācārya has spoken of the golabandha in the Brāhmasphuṭasiddhānta, so also has Śrīpati in the Siddhāntaśekhara, and Bhāskarācārya in the Golādhyāya of the Siddhāntaśiromaṇi. In the graha-gola and nakṣatra-gola, five fixed circles (sthira-vṛtta) are spoken: pūrvâpara-vṛtta, sthitija-vṛtta, yāmyôttara-vṛtta, unmaṇḍala, and viṣuvad-vṛtta. All these are equal to the kalā-maṇḍala.

And the moving circles (calavṛtta) of the planets: mandanīcôcavṛtta = 7, the śighranīcôcavṛtta of the manda-etc. planets = 5. The manda-prativṛtta = 7, the śighra-prativṛtta = 7. The hamaṇḍala and hṛd-kṣepa of the seven planets [are also given].

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$$\frac{(\phi) \cdot (\delta)}{(\alpha)} \Big)$$